Basic Mathematics

Mathematics

Lecture Notes

A friendly path through algebra, geometry, and calculus.

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Chapter 1

Introduction

1.1 Purpose and Philosophy

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognises the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, formulate precise questions, distinguish an object from its notation, notice structure, and build simple but meaningful chains of reasoning. A solved problem is valuable not only because it produces an answer, but also because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course is organised into three connected parts: linear algebra, analytic geometry, and calculus. In the first part, we focus on matrix methods: matrices, determinants, inverse matrices, and systems of linear equations. The second part develops the language of coordinates and vectors and uses it to describe points, lines, planes, directions, distances, and angles. The third part introduces sequences and functions, limits, continuity, derivatives, integrals, change, and accumulation. These parts are not isolated collections of techniques. Together they form a basic mathematical language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorise, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as an answer to a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some habits of thought will deliberately return in different places. The same question of what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. A mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognise patterns of reasoning rather than only patterns of exercises.

The number of answers produced is not, by itself, a measure of progress. What matters is the quality of the work. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each important step was valid? Did we interpret or verify the result? These questions are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The

goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as the art of asking good questions and giving well-reasoned answers.

1.2 How the Course Works

This book is part of a weekly learning process rather than a collection of material to be read only before an examination. Each of the twelve thematic chapters is designed to support one week of study. A chapter provides the mathematical structure for the lecture, a complete narrative to which the student can return, and a starting point for individual problem solving and classroom discussion.

In a fifteen-week semester, the twelve thematic weeks establish the main rhythm of the course. The remaining time can be used to introduce the method of work, connect topics, revise important ideas, and present more substantial solutions.

1.2.1 The Weekly Learning Cycle

A student should not leave a topic behind when the lecture ends. A typical week consists of the following connected stages:

1. **Attend the lecture.** The lecture introduces the topic, motivates the central questions, and demonstrates how the relevant mathematical objects and methods should be read.

2. **Return to the chapter.** Read the corresponding chapter carefully after the lecture. The PDF is not merely a summary of slides. It is an organised narrative containing definitions, explanations, examples, interpretations, and mathematical arguments that can be examined more than once.

3. **Ask questions and clarify the material.** Mark every place where a definition, transition, calculation, or conclusion is not yet clear. Use the text, your own examples, classroom discussion, and appropriate digital tools to resolve these questions.

4. **Solve the complete assigned problem set.** Move from following an explanation to constructing arguments independently. Work on every assigned problem and prepare a separate, self-contained note for each one.

5. **Prepare professional mathematical notes.** Use precise notation, explanatory sentences, justified transitions, and a clear conclusion. The note should show not only the answer, but also how it follows from the given information, definitions, and earlier results.

6. **Present and discuss your work.** During exercise classes, be prepared to explain a solution, answer questions about its steps, and describe what you learned while preparing it.

These stages form one continuous process. Each stage prepares the next, and the presentation and feedback lead back to further reading, revision, and improved work.

1.2.2 Exercise Classes: Presentation, Feedback, and Revision

Presenting mathematics is part of learning mathematics. It reveals whether an argument has genuinely been understood and whether the student can communicate it clearly. During a presentation, the student should be able to read the notation, explain the role of each important formula, justify the method, and respond to questions without relying on AI to speak on the student's behalf.

Feedback given during exercise classes is intended for the whole group, not only for the student whose work is currently being presented. Everyone should follow the argument and

listen carefully to the discussion. By comparing different examples, students learn what a complete solution looks like, which explanations make an argument clear, which steps require justification, and which weaknesses require revision.

Students should not focus only on comments addressed directly to their own work. Listening to feedback on the work of others provides a broader set of examples and develops an operational understanding of the course standards: what makes mathematical work clear, justified, and complete, and what still needs improvement.

The feedback concerns the mathematical work and its results, not the personal worth of the student. What is examined is the quality of the note, the understanding demonstrated in the argument, and the clarity of the presentation. Feedback should identify what already works and what should be changed in order to produce a stronger solution.

The first submitted version does not always have to be the final version. Students should learn to receive comments, reconsider their choices, correct errors, add missing explanations, and improve both the written note and its presentation. Revision is not evidence of failure. It is a normal part of serious intellectual work.

This process is important throughout university study and later in professional life. Students and professionals are given tasks, asked to solve problems, expected to present results, and required to respond to feedback and critical review. Learning to improve an initial solution is therefore not an additional classroom exercise. It is a fundamental habit of responsible academic and professional work.

1.2.3 Progress and Assessment

Progress is demonstrated by the quality and increasing independence of the student's work. By the end of a weekly cycle, the student should be able to identify the main mathematical objects, state the essential definitions, solve representative problems, justify the principal steps, verify or interpret the result, and communicate the argument clearly.

Assessment in the exercise component focuses primarily on how the student works throughout the semester. Evidence of serious engagement includes:

- regular preparation and completion of the assigned problem sets,
- documented attempts to solve problems rather than bare answers,
- active participation in presentations and classroom discussion,
- separate, self-contained notes for individual problems,
- careful revision of work after feedback,
- increasing clarity, independence, and mathematical precision.

A student is not defined by the quality of the first solution or presentation. Some students may initially find mathematical notation, formal writing, or public explanation difficult. A weak beginning does not prevent a very good final grade for the exercise component. A student who works conscientiously, responds to feedback, corrects earlier weaknesses, and demonstrates substantial development may finish the exercise component with a very good grade.

This approach does not lower the standard of mathematical work. It recognises that reaching the standard through sustained effort and revision is itself a meaningful achievement. The exercise-component grade reflects a process of learning, not merely a snapshot taken at the beginning of the semester.

Systematic work according to these guidelines provides the basis for earning a passing grade for the exercise component. The examination is assessed separately and tests the student's mathematical knowledge and ability to use it independently. Regular work does not replace the examination, but it develops the understanding and skills that substantially increase the likelihood of successfully completing the course.

1.3 How to Prepare Mathematical Notes

A mathematical note is not merely a place where an answer is recorded. Its purpose is to show how a problem is understood, how relevant information is selected, how a method is chosen, and how a conclusion follows from definitions, assumptions, and earlier results.

Mathematical expressions are compact. A single formula may contain several ideas at once: objects, operations, conditions, and logical relationships. Reading a formula therefore means more than pronouncing its symbols. A student should be able to explain what the objects represent, what operation is being performed, why the expression is meaningful, and what information the formula provides.

The same principle applies to writing. A sequence of formulas without explanatory sentences rarely communicates a complete mathematical argument. Formulas should be connected by words that explain their role. A good solution tells the reader why a calculation is being performed, which definition or theorem is being used, whether its assumptions are satisfied, and how the result answers the original question.

1.3.1 Make the Note Self-Contained

Prepare one separate, self-contained note for each assigned problem, including all of its subparts. Begin by writing the problem or a complete and faithful formulation of it. The reader should not need to search through another document to discover what is being solved. After several weeks, you should still be able to open the note and understand both the question and the solution.

Identify what is given and what must be found, proved, or explained. Introduce the relevant notation and determine what kinds of mathematical objects appear in the problem. Before calculating, ask what information is available and what would count as a satisfactory answer.

1.3.2 Make the Reasoning Visible

Present the principal idea or method before, or while, carrying out the calculation. Important transitions should be explained with complete sentences. For example:

- Because the determinant is non-zero, the matrix is invertible.
- We may therefore multiply both sides by the inverse matrix.
- By the definition of the derivative, we consider the following limit.
- Since the direction vector is non-zero, it determines a line.
- Substituting the result into the original equation verifies the solution.

Such sentences are not decoration around the mathematics. They are part of the mathematics because they show the logical relationship between successive formulas.

It is not necessary to describe every elementary arithmetic operation. The amount of explanation should be proportional to the problem. A simple exercise may require only a few sentences, while a more substantial problem may require a longer argument. The goal is not to make every note long. The goal is to make every essential step understandable and justified.

> Write as briefly as possible, but as completely as necessary.

1.3.3 A Complete Solution

A complete note should make the following five stages visible:

1. **Understand the problem.** Record the problem, identify the given information, and state clearly what must be found, proved, or explained.

2. **Choose a method.** Select the definitions, theorems, or techniques that connect the available information with the goal.

3. **Construct the argument.** Carry out the necessary steps and connect formulas with sentences that explain why each important transition is valid.

4. **Verify the result.** Check calculations, assumptions, domain restrictions, or geometric meaning whenever appropriate.

5. **State the conclusion.** Answer the original question clearly and in its proper context.

A result without its context is not a complete solution. A bare final answer, an unexplained sequence of formulas, or a few unconnected calculations will be treated as incomplete work and should be revised.

Quality cannot be replaced by quantity, but quality does not remove the obligation to complete the assigned work. Students are expected to prepare separate, complete solutions for the entire assigned problem set. Producing many unexplained answers is insufficient, but preparing one polished solution while ignoring the remaining problems is also insufficient.

1.3.4 Working with AI

We are studying in an age in which AI tools are widely available. The free versions of these tools are sufficient for the kinds of support expected in this course: asking questions, clarifying notation, generating simpler examples, checking reasoning, identifying missing steps, and improving the organisation of mathematical writing. Access to expensive software is not required.

AI should be used as an individual learning assistant. It may help transform rough calculations into clear mathematical prose, but it does not replace the student's mathematical work. Receiving a well-written answer is not the same as understanding it. The student remains responsible for checking every definition, calculation, assumption, conclusion, and formula included in the submitted note.

Use AI with a focused context. Work on one problem at a time and prepare one complete note for that problem. Do not paste an entire problem set into a single conversation and request all answers at once. Handling many unrelated tasks together encourages brief, generic, and mathematically weak notes. Focused work makes it easier to question each step, identify missing justifications, and produce a solution that can be understood and presented.

The one-problem-at-a-time principle does not mean that only one problem from the set should be prepared. Students are expected to repeat this focused process for every problem in the complete assigned set.

A polished AI-generated note is insufficient if the student cannot read its formulas, explain its terminology, justify its transitions, or present the argument independently. When using AI, ask not only for a solution, but also questions that deepen understanding:

- What does this formula mean?
- What mathematical object does each symbol represent?
- Why is this step valid?
- Which definition or theorem is being used?
- Are the assumptions of the theorem satisfied?
- Is there a simpler way to explain this transition?
- How can the result be checked?

- What should I be able to explain during class?

Preparing notes in this way is part of learning mathematics. It develops the ability to read notation, recognise mathematical objects, organise information, construct arguments, and communicate ideas precisely. Repeating this process for the complete problem set builds the habits of systematic and conscientious work that should accompany the student throughout university study and later professional life.

1.3.5 Student GitHub Workspace

The current instructions, problem templates, and technical workflow for student work are maintained in the public repository `dchorazkiewicz/Math_Problems_Repo`. Students should create a public fork of this repository and commit their notes to their own fork. The course repository is the common template; it is not the place where individual student solutions should be submitted.

The repository contains a short map of the working process:

- `README.md` explains the purpose and structure of the workspace and provides the weekly index;
- `SOURCE_MATERIAL.md` links the current lecture PDF and maps each week to the corresponding LaTeX problem source used by an AI assistant;
- `NOTE_GUIDELINES.md` defines the standard of a complete mathematical note;
- `AI_WORKFLOW.md` describes the one-problem-at-a-time workflow with AI, review, commits, presentation, feedback, and revision;
- `AGENTS.md` gives a connected AI assistant the operational rules for working safely in the student's fork;
- the `docs/` directory contains twelve weekly folders and one Markdown file for each problem.

The workspace uses Markdown for student notes, Git and GitHub for version history and public forks, MathJax for mathematical typesetting, MkDocs for the web presentation, and GitHub Actions for automatic validation and deployment. These instructions and templates may be updated independently of this PDF, so students should consult the repository for the current working procedure. The repository links back to this PDF and to its LaTeX sources. Students use the PDF as the readable course material containing the theory, definitions, examples, and problem sets. A connected AI assistant uses the current mapped LaTeX file as the exact machine-readable source of a selected problem statement.

Chapter 2

Matrices

2.1 What matrices are

A matrix is a rectangular arrangement of mathematical objects, usually numbers. This description is correct, but it is not yet very informative. If we stop at this description, a matrix looks like a table, and matrix algebra looks like a collection of rules for moving numbers around. In this chapter we will try to see more than that. A matrix is a way of organizing information. It can describe data, relations between quantities, coefficients of equations, or a rule that transforms one vector into another.

The first skill is therefore not calculation, but reading. When we see a matrix, we should ask what its shape is, what its entries represent, and how its rows and columns are meant to be interpreted. The same array of numbers may be only a table in one problem, but in another problem it may describe a transformation, a system of equations, or a geometric operation.

Definition

A matrix with m rows and n columns is called an $m \times n$ matrix. It has the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The entry a_{ij} is located in row i and column j.

This notation is simple, but it contains an important discipline. The first index tells us where to look vertically, and the second index tells us where to look horizontally. A student who confuses these indices will often make mistakes later when multiplying matrices or describing systems of equations.

Example

Consider the matrix

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}.$$

This is a 2×3 matrix. The entry m_{11} is 4, the entry m_{12} is -2, and the entry m_{23} is 5. We are not doing any operation yet. We are only learning to read the object.

The size of a matrix is part of its identity. A row matrix, a column matrix, and a square matrix may contain similar numbers, but they play different roles. A column matrix can represent a vector. A square matrix can sometimes represent a transformation from a space to itself. A rectangular matrix may describe a relation between two collections of quantities of different sizes.

Definition

A matrix is called square if it has the same number of rows and columns. An $n \times n$ matrix is a square matrix of order n.

We will often begin with square matrices because many basic ideas are easiest to see there. However, rectangular matrices are just as important. The point is not to memorize a list of shapes, but to develop the habit of asking: what kind of object am I holding, and what operations make sense for this object?

2.2 Basic matrix operations

Once we know how to read a matrix, we can begin to operate on it. The simplest operations are addition, subtraction, and multiplication by a number. These operations are deliberately simple because they preserve the shape of the matrix. They do not change a table into a different kind of object; they change its entries while keeping the same positions.

Definition

Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same size $m \times n$. Their sum is the matrix

$$A + B = (a_{ij} + b_{ij}),$$

and their difference is the matrix

$$A - B = (a_{ij} - b_{ij}).$$

If λ is a number, then

$$\lambda A = (\lambda a_{ij}).$$

The phrase "of the same size" is not a technical detail. It is the reason the operation is meaningful. When two matrices have the same size, each entry of one matrix has a partner in the other matrix. Addition and subtraction then happen position by position.

Example

Let

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 2+5 & -1+4 \\ 0+(-2) & 3+1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix}.$$

Also,

$$A - B = \begin{pmatrix} 2-5 & -1-4 \\ 0-(-2) & 3-1 \end{pmatrix} = \begin{pmatrix} -3 & -5 \\ 2 & 2 \end{pmatrix}.$$

Finally,

$$-3A = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}.$$

The calculation is not difficult. The important thing is the discipline of positions. We do not add rows to columns, and we do not rearrange entries because it looks convenient. We respect the shape of the object.

These operations satisfy the familiar rules one expects from ordinary arithmetic. For matrices of the same size,

$$A + B = B + A, \qquad (A + B) + C = A + (B + C).$$

This should not be surprising: addition is performed entry by entry, and ordinary addition of numbers is commutative and associative.

Remark

Whenever a matrix operation is defined entry by entry, many properties of ordinary arithmetic pass directly to matrices. But this principle must be used carefully. Matrix multiplication will not be entry-by-entry multiplication, and therefore it will behave differently.

2.3 Matrix multiplication

Matrix multiplication is the first operation in this chapter that is not simply performed position by position. This is why it deserves careful attention. Many mistakes with matrices come from treating multiplication as if it were only another version of addition. It is not. Addition compares matching positions. Multiplication combines rows of the first matrix with columns of the second matrix.

Definition

Let $A = (a_{ij})$ be an $m \times n$ matrix and let $B = (b_{ij})$ be an $n \times p$ matrix. The product AB is the $m \times p$ matrix $C = (c_{ij})$, where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

In words, the entry c_{ij} is obtained by multiplying row i of A by column j of B.

The condition on sizes is built into the definition. If A has size $m \times n$, then each row of A has n entries. To multiply this row by a column of B, each column of B must also have n entries. Therefore B must have n rows. We summarize this by writing

$$(m \times n)(n \times p) = m \times p.$$

The inner dimensions must match; the outer dimensions describe the size of the result.

Example

Let

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}.$$

The first entry of AB comes from the first row of A and the first column of B:

$$1 \cdot 2 + 3 \cdot 5 = 17.$$

The first row and second column give

$$1 \cdot 0 + 3 \cdot (-1) = -3.$$

Continuing in the same way,

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}.$$

The example is small, but the point is general. Each entry of the product answers one local question: how does a chosen row of the first matrix interact with a chosen column of the second matrix?

Why the order matters

For ordinary numbers, $ab = ba$. For matrices this is not generally true. The reason is conceptual, not only computational. The product AB asks rows of A to interact with columns of B. The product BA asks a different question.

Using the matrices from the previous example,

$$BA = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}.$$

Thus $AB \neq BA$. The matrices have not changed, but the order has changed, and the question asked by the multiplication has changed.

Remark

The statement $AB \neq BA$ does not mean that two matrices can never commute. Some matrices do commute. The point is that commutativity is not automatic. If it matters, it must be checked or justified from the structure of the matrices.

Matrix multiplication is associative:

$$(AB)C = A(BC),$$

whenever the products are defined. This means that if several compatible matrices are multiplied in a fixed order, parentheses do not change the result. But the order of the matrices themselves still matters. Associativity tells us how to group operations. It does not allow us to swap them.

2.4 Special matrices and visible structure

Some matrices are important because their structure is easy to see. They help us learn a useful lesson: a matrix is not only a collection of entries. The position of the entries matters, and certain patterns can make an operation simple or meaningful.

The zero matrix and the identity matrix

The zero matrix is the matrix whose entries are all zero. It behaves like zero for matrix addition. If A is an $m \times n$ matrix and 0 is the zero matrix of the same size, then

$$A + 0 = A.$$

An identity matrix is always square. The size needed in a product is determined by the dimensions of the other matrix.

Definition

The $n \times n$ identity matrix is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

If $A \in \mathbb{R}^{m\times n}$, then

$$I_m A = A, \qquad A I_n = A.$$

The two identity matrices in these products need not have the same size. The left identity must have m rows and columns, while the right identity must have n rows and columns.

The identity matrix is important because it represents doing nothing. This phrase is informal, but helpful. Multiplying by the correctly sized identity preserves the matrix, just as multiplying a number by 1 preserves the number.

Transpose

The transpose interchanges rows and columns. If

$$A = (a_{ij}) \in \mathbb{R}^{m\times n},$$

then its transpose is the $n \times m$ matrix

$$A^{\mathsf{T}} = (a_{ji}).$$

Definition

For

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

the transpose is

$$A^{\mathsf{T}} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}.$$

For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^{\mathsf{T}} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

A square matrix is called symmetric if

$$A^{\mathsf{T}} = A.$$

This means that its entries are mirrored across the main diagonal.

Diagonal matrices

A diagonal matrix is a square matrix whose non-zero entries, if any, lie only on the main diagonal.

Definition

A square matrix $D = (d_{ij})$ is diagonal if

$$d_{ij} = 0 \quad \text{whenever } i \neq j.$$

We often write

$$D = \operatorname{diag}(d_1, d_2, \ldots, d_n).$$

Diagonal matrices are useful because they act independently on different positions. There is no mixing between coordinates. This is why many calculations with diagonal matrices are transparent.

Example

Let

$$D = \operatorname{diag}(3, -1, 2).$$

Then

$$D^2 = \operatorname{diag}(9, 1, 4), \qquad D^3 = \operatorname{diag}(27, -1, 8).$$

The power is computed by taking powers of the diagonal entries. This works because all off-diagonal entries are zero, so the coordinates do not mix.

If two diagonal matrices have the same size, then they commute. Indeed,

$$\operatorname{diag}(a_1, \ldots, a_n) \operatorname{diag}(b_1, \ldots, b_n) = \operatorname{diag}(a_1 b_1, \ldots, a_n b_n),$$

and the same result is obtained in the opposite order because ordinary numbers commute. This is a good example of structure explaining an algebraic fact.

Patterns worth noticing

When a matrix appears in a problem, it is useful to ask whether it has visible structure. Is it diagonal? Does it contain many zeros? Are two rows equal? Are rows or columns multiples of one another? Is it symmetric? Such observations can simplify computations, but more importantly they help us understand the object.

Remark

A calculation often becomes easier after we notice structure. But noticing structure is not a trick; it is part of mathematical understanding. We learn not only to compute with objects, but also to see them.

2.5 Row operations and problem solving

Matrices are often used because they allow us to transform a problem without rewriting every sentence of the problem. One of the most important examples is the use of row operations. Row operations will become essential when we solve systems of linear equations, but it is useful to meet them already here.

A row operation changes a matrix in a controlled way. We do not change entries randomly. We perform an operation that has a clear description and a clear purpose.

Definition

The basic elementary row operations are:

- swapping two rows,
- multiplying a row by a non-zero number,
- adding a multiple of one row to another row.

The value of row operations is not that they make matrices look different. Their value is that they help reveal structure. For example, they can help us simplify a matrix, detect whether

equations are dependent, or prepare a system for solving.

Example

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}.$$

Suppose we replace the second row by the second row minus twice the first row. Then

$$(2, 4, 3) - 2(1, 2, -1) = (0, 0, 5).$$

The new matrix is

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}.$$

This operation did not merely produce new numbers. It revealed that part of the second row was already explained by the first row, except for the last entry.

This example should be read slowly. The operation has three layers. First, there is arithmetic. Second, there is a transformation of the row. Third, there is interpretation: the new row tells us something about dependence between rows.

Column vectors as matrices

A column vector may be viewed as a matrix with one column. This viewpoint will be used constantly in linear algebra. It allows us to write vector equations and matrix equations in the same language.

For instance,

$$u = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

is a 3×1 matrix. Its transpose is the row matrix

$$u^\top = \begin{pmatrix} 2 & -1 & 4 \end{pmatrix}.$$

The distinction between a row and a column is not cosmetic. The products $u^\top u$ and $uu^\top$ have different sizes and different meanings:

$$u^\top u$$

is a 1×1 matrix, while

$$uu^\top$$

is a 3×3 matrix.

This is another example of a recurring principle: before calculating, read the shapes.

How to approach matrix exercises

When solving matrix exercises, the aim is not to rush toward the answer. The answer is only the last line of a process. A good solution should make the process visible.

Summary

When working with matrices, ask:

- What is the size of each matrix?
- Which operations are defined?
- Am I adding entries position by position, or multiplying rows by columns?
- Does the order of multiplication matter here?
- Is there a visible structure that simplifies the problem?
- What does the result tell me about the original objects?

A student who asks these questions is not merely applying a recipe. They are beginning to use matrices as mathematical objects. That is the real purpose of this chapter.

Chapter 3

Determinants

3.1 Why determinants matter

In the previous chapter we learned how to read matrices and how to perform basic operations on them. We can add matrices, multiply them by numbers, multiply compatible matrices, and recognize simple structures such as the identity matrix or a diagonal matrix. But after learning these operations, a natural question appears.

Are all square matrices equally good?

This question is intentionally vague at first. Mathematics often begins in this way: we notice that some objects behave differently from others, and only later do we find the right words and the right definitions. Consider the following matrices:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

The first matrix is the identity matrix. It preserves everything under multiplication. The second matrix stretches the first coordinate by a factor of 2 and the second coordinate by a factor of 3. The third matrix is different. Its second row is exactly twice its first row:

$$(2, 4) = 2(1, 2).$$

So the two rows do not contain two independent pieces of information. One row is only a repetition of the other, scaled by a number.

This observation suggests a deeper question. Can a matrix lose information? Can a matrix collapse different inputs into outputs that are no longer distinguishable? Can we detect this collapse by assigning one carefully chosen number to the matrix?

The determinant is such a number.

This statement should not be read as a definition yet. At this stage it is only a promise. We are looking for a number attached to a square matrix, but not just any number. We want a number that reacts to the structure of the matrix. It should notice when rows or columns become dependent. It should distinguish matrices that preserve two-dimensional or three-dimensional structure from matrices that collapse it. It should also help us decide, later, whether a matrix can be inverted.

Remark

A determinant is not a summary of all entries of a matrix in the sense of an average or a sum. It is a number designed to detect structural behaviour. In particular, the value 0 will become extremely important: it will mean that the matrix has lost a certain kind of independence.

There is also a geometric way to think about determinants. A 2×2 matrix can transform the plane. Some matrices stretch areas, some reverse orientation, and some flatten the plane onto a line. The determinant measures this behaviour. In dimension 2, its absolute value describes an area scale factor. In dimension 3, its absolute value describes a volume scale factor. A determinant equal to zero means that area or volume has been flattened completely.

For the purposes of this course, however, we will not begin from geometry alone. We will move between several viewpoints:

- determinants as numbers computed from square matrices,
- determinants as tests for structural collapse,
- determinants as tools for recognizing invertibility,
- determinants as quantities affected in predictable ways by row operations.

The chapter will therefore follow a path of discovery. First we study small matrices, where the formulas can be seen directly. Then we introduce a systematic method for larger matrices: minors, cofactors, and Laplace expansion. After that we collect the properties that make determinants useful in actual calculation. Finally we connect determinants with singular matrices and prepare the transition to the next chapter, where invertibility becomes the central topic.

3.2 Determinants of small matrices

The determinant is defined only for square matrices. This is already meaningful: we are not attaching this number to every rectangular table. We are studying matrices that can be interpreted as transforming a space into itself, or as square coefficient matrices in systems of equations.

We begin with small cases. This is not only because small cases are easier. Small cases allow us to see the idea before it becomes hidden behind notation.

3.2.1 The determinant of a 1×1 matrix

The simplest square matrix has only one entry:

$$A = \begin{pmatrix} a \end{pmatrix}.$$

There is no structure to compare, no rows to exchange, and no area or volume to measure. The determinant is simply the entry itself:

$$\det A = a.$$

This case looks trivial, but it is useful because larger definitions will reduce determinants to smaller determinants. Eventually, a long computation ends at 1×1 determinants.

3.2.2 The determinant of a 2×2 matrix

Now consider a general 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The determinant is defined by

$$\det A = ad - bc.$$

The formula should be read carefully. We multiply along the main diagonal and subtract the product along the other diagonal:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

At first this may look like an arbitrary rule. It is not arbitrary. It is a rule that reacts very strongly to dependence between rows or columns.

Example

For

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$$

we get

$$\det A = 3 \cdot 5 - 1 \cdot 2 = 15 - 2 = 13.$$

The determinant is non-zero. This suggests that the two rows are not repetitions of each other and that the matrix has not collapsed its two-dimensional structure.

Example

For

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$$

we get

$$\det B = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0.$$

The second row is three times the first row:

$$(3, 6) = 3(1, 2).$$

Thus the determinant detects that the two rows are dependent.

This example gives us the first important interpretation: when the determinant of a 2×2 matrix is zero, the two rows or the two columns fail to provide two independent directions of information.

3.2.3 What changes the determinant?

Let us observe how the formula reacts to simple operations. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

we have

$$\det A = ad - bc.$$

If we exchange the two rows, we obtain

$$A' = \begin{pmatrix} c & d \\ a & b \end{pmatrix}.$$

Then

$$\det A' = cb - da = -(ad - bc) = -\det A.$$

So exchanging two rows changes the sign of the determinant.

If the two rows are equal,

$$A = \begin{pmatrix} a & b \\ a & b \end{pmatrix},$$

then

$$\det A = ab - ba = 0.$$

This is exactly what we should expect: two equal rows certainly do not provide two independent directions.

Remark

The determinant is sensitive to order and independence. It changes sign when rows are swapped, and it becomes zero when rows become dependent. These facts will later become general properties for determinants of all sizes.

3.2.4 The determinant of a 3×3 matrix and Sarrus' rule

For a 3×3 matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

one useful formula is

$$\det A = aei + bfg + cdh - ceg - bdi - afh.$$

This formula can be remembered using Sarrus' rule. Repeat the first two columns after the matrix:

$$\begin{array}{ccc|cc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{array}$$

Then add the products along the three downward diagonals and subtract the products along the three upward diagonals:

$$\det A = (aei + bfg + cdh) - (ceg + bdi + afh).$$

Example

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}.$$

Using Sarrus' rule,

$$\det A = (1 \cdot 3 \cdot 5 + 2 \cdot 4 \cdot 2 + 0 \cdot (-1) \cdot 0) - (0 \cdot 3 \cdot 2 + 2 \cdot (-1) \cdot 5 + 1 \cdot 4 \cdot 0).$$

Hence

$$\det A = (15 + 16 + 0) - (0 - 10 + 0) = 31 - (-10) = 41.$$

The determinant is non-zero.

The computation above is mechanical, but it is still worth reading. A non-zero determinant tells us that the rows and columns of the matrix do not collapse into a lower-dimensional structure. Later this will imply that the matrix is invertible.

Remark

Sarrus' rule is convenient, but it is special. It works for 3×3 matrices only. It is not a method for 4×4 or larger matrices. For larger matrices we need a more general idea.

3.2.5 A warning about calculation

When computing determinants, students often try to memorize many separate rules. A better habit is to ask three questions:

- What is the size of the matrix?

- Which method is appropriate for this size and this structure?
- Is there visible structure that can simplify the calculation?

For 2×2 matrices, the formula $ad-bc$ is usually fastest. For 3×3 matrices, Sarrus' rule may be useful. But as soon as the matrix becomes larger, or as soon as many zeros appear, the method of expansion by minors and cofactors becomes much more important.

3.3 Minors, cofactors, and Laplace expansion

Sarrus' rule gives a quick method for 3×3 matrices, but it does not generalize to larger matrices. If we want a method that works for every square matrix, we need a different idea. The key idea is reduction: we compute a large determinant by reducing it to smaller determinants.

This is a recurring mathematical pattern. When an object is too large to understand at once, we try to break it into smaller objects of the same kind. For determinants, the smaller objects are called minors.

3.3.1 Minors

Let $A=(a_{ij})$ be an $n\times n$ matrix. Choose an entry a_{ij}. Remove row i and column j. The determinant of the remaining $(n-1)\times(n-1)$ matrix is called the minor of a_{ij}, and is denoted by M_{ij}.

Example

Let

$$A=\begin{pmatrix}2 & 1 & 3\\ 0 & -1 & 4\\ 5 & 2 & 1\end{pmatrix}.$$

To find M_{12}, we remove row 1 and column 2. The remaining matrix is

$$\begin{pmatrix}0 & 4\\ 5 & 1\end{pmatrix}.$$

Therefore

$$M_{12}=\det\begin{pmatrix}0 & 4\\ 5 & 1\end{pmatrix}=0\cdot 1-4\cdot 5=-20.$$

A minor is not an entry of the original matrix. It is a determinant obtained after deleting a row and a column. This distinction matters: the entry a_{ij} is a number already present in the matrix; the minor M_{ij} is computed from what remains after removing the row and column of that entry.

3.3.2 Cofactors

The minor still needs a sign. The signs follow a checkerboard pattern:

$$\begin{pmatrix}+ & - & + & - & \cdots\\ - & + & - & + & \cdots\\ + & - & + & - & \cdots\\ - & + & - & + & \cdots\\ \vdots & \vdots & \vdots & \vdots & \ddots\end{pmatrix}.$$

This pattern is encoded by the factor $(-1)^{i+j}$.

Definition

The cofactor of the entry a_{ij} is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Example

Using the matrix from the previous example, we found

$$M_{12} = -20.$$

Since $(-1)^{1+2} = (-1)^3 = -1$, the cofactor is

$$C_{12} = (-1)M_{12} = 20.$$

For the entry in position $(1,1)$, the sign is positive. Removing row 1 and column 1, we get

$$M_{11} = \det \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} = (-1) \cdot 1 - 4 \cdot 2 = -9.$$

Thus

$$C_{11} = M_{11} = -9.$$

The alternating signs are not decoration. Without them, the determinant would not have the correct behaviour under row exchange, and it would not detect orientation properly.

3.3.3 Laplace expansion along a row

Laplace expansion expresses a determinant as a sum of entries multiplied by their cofactors. If we expand along row i, then

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}.$$

Equivalently,

$$\det A = \sum_{j=1}^{n} a_{ij}C_{ij}.$$

Example

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix}.$$

Expand along the first row. We need the cofactors C_{11}, C_{12}, and C_{13}.
First,

$$C_{11} = +\det \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} = (-1) \cdot 1 - 4 \cdot 2 = -9.$$

Second,

$$C_{12} = -\det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} = -(0 \cdot 1 - 4 \cdot 5) = 20.$$

Third,

$$C_{13} = +\det \begin{pmatrix} 0 & -1 \\ 5 & 2 \end{pmatrix} = 0 \cdot 2 - (-1) \cdot 5 = 5.$$

Therefore

$$\det A = 2C_{11} + 1C_{12} + 3C_{13} = 2(-9) + 1(20) + 3(5).$$

Hence

$$\det A = -18 + 20 + 15 = 17.$$

Notice how the computation works. A 3×3 determinant was reduced to three 2×2 determinants. For a 4×4 determinant, Laplace expansion reduces the problem to 3×3 determinants, and so on. The method is recursive.

3.3.4 Laplace expansion along a column

We may also expand along a column. If we expand along column j, then

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

Equivalently,

$$\det A = \sum_{i=1}^{n} a_{ij}C_{ij}.$$

The freedom to choose a row or a column is important. A wise choice can make a determinant much easier to compute.

Example

Consider

$$B = \begin{pmatrix} 4 & 0 & 1 & 2 \\ 0 & 3 & 0 & -1 \\ 2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 6 \end{pmatrix}.$$

The second column contains three zeros:

$$\begin{pmatrix} 0 \\ 3 \\ 0 \\ 0 \end{pmatrix}.$$

So we expand along the second column. Only one term survives:

$$\det B = 3C_{22}.$$

Since $(-1)^{2+2} = 1$,

$$C_{22} = \det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix}.$$

Now compute the remaining 3×3 determinant. Using Sarrus' rule,

$$\det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix} = (4 \cdot 5 \cdot 6 + 1 \cdot 0 \cdot 1 + 2 \cdot 2 \cdot 0) - (2 \cdot 5 \cdot 1 + 1 \cdot 2 \cdot 6 + 4 \cdot 0 \cdot 0).$$

Thus

$$C_{22} = 120 - (10 + 12) = 98.$$

Therefore

$$\det B = 3 \cdot 98 = 294.$$

This example shows why Laplace expansion is not merely a theoretical formula. It gives a strategy. Before computing, inspect the matrix. A row or column with many zeros is usually the best place to expand.

3.3.5 A method, not a ritual

Laplace expansion can be used along any row or any column. The result is always the same determinant. But the amount of work may be very different. The mathematical question is not only "Can I compute it?" but also "How can I compute it intelligently?"

Summary

When using Laplace expansion, follow this order.

- Inspect the matrix before calculating.
- Look for a row or column with many zeros.
- Write the sign pattern carefully.
- Compute the minors as smaller determinants.
- Keep the structure visible; do not compress too many steps at once.

3.4 Properties of determinants

At this point we know how to compute determinants of small matrices, and we know a general method: Laplace expansion. But if we use Laplace expansion blindly, determinant calculations can become long and unpleasant. Mathematics is not only about having a method. It is also about knowing when a method can be simplified.

The determinant has several important properties. These properties are not just facts to memorize. They are tools for reading a matrix before calculating.

3.4.1 Triangular matrices

A square matrix is upper triangular if all entries below the main diagonal are zero. It is lower triangular if all entries above the main diagonal are zero.

For example,

$$U = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{pmatrix}$$

is upper triangular.

Theorem

The determinant of a triangular matrix is the product of its diagonal entries.

$$\det U = u_{11}u_{22}\cdots u_{nn}.$$

For the matrix above,

$$\det U = 2 \cdot 3 \cdot (-2) = -12.$$

This is much faster than expanding the determinant directly.

The reason is visible from Laplace expansion. If we expand a triangular matrix along a row or column with many zeros, the determinant keeps reducing to smaller triangular matrices.

Eventually only diagonal entries remain.

3.4.2 Swapping two rows

If two rows of a matrix are exchanged, the determinant changes sign.

Theorem

If B is obtained from A by swapping two rows, then

$$\det B = -\det A.$$

The same is true for swapping two columns.

For a 2×2 matrix this can be checked directly:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc,$$

while

$$\det \begin{pmatrix} c & d \\ a & b \end{pmatrix} = cb - da = -(ad - bc).$$

The general case follows from the same alternating nature of the determinant.

This property explains why the sign pattern in cofactors matters. The determinant remembers orientation. Exchanging two rows reverses that orientation.

3.4.3 Equal rows and proportional rows

If two rows of a matrix are equal, then the determinant is zero.

Indeed, suppose two rows are equal. If we swap these two rows, the matrix does not change. Therefore the determinant should remain the same. But swapping two rows changes the sign of the determinant. Hence

$$\det A = -\det A,$$

so

$$2 \det A = 0,$$

and therefore

$$\det A = 0.$$

This is a good example of a short argument. We did not compute the determinant. We used a property to force its value.

If two rows are proportional, the determinant is also zero. For example,

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & -4 \\ 0 & 1 & 5 \end{pmatrix}$$

has the second row equal to twice the first row. The determinant is zero because the first two rows do not contain independent information.

Remark

A determinant equal to zero often means that some hidden dependence is present. Repeated rows and proportional rows are the easiest visible cases of this phenomenon.

3.4.4 Multiplying a row by a number

If one row of a matrix is multiplied by a number λ, then the determinant is also multiplied by λ.

Theorem

If B is obtained from A by multiplying one row by λ, then

$$\det B = \lambda \det A.$$

The same is true for multiplying one column by λ.

For example,

$$\det \begin{pmatrix} 2 & 4 \\ 3 & 5 \end{pmatrix} = 2 \cdot 5 - 4 \cdot 3 = 10 - 12 = -2.$$

If we multiply the first row by 3, we get

$$\det \begin{pmatrix} 6 & 12 \\ 3 & 5 \end{pmatrix} = 6 \cdot 5 - 12 \cdot 3 = 30 - 36 = -6.$$

The new determinant is three times the old one:

$$-6 = 3(-2).$$

This property is often used when simplifying determinants or when factoring common numbers out of rows or columns.

3.4.5 Adding a multiple of one row to another

One of the most useful properties is the following.

Theorem

Adding a multiple of one row to another row does not change the determinant.

This property is extremely important because it allows us to use row operations to simplify a determinant without changing its value. It is the determinant version of a powerful idea: transform the object into a simpler one while preserving the quantity we care about.

Example

Compute

$$\det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix}.$$

We simplify the matrix using row operations that do not change the determinant. Replace

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

Then

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & 3 \end{pmatrix}.$$

Now replace

$$R_3 \leftarrow R_3 + 2R_2.$$

We get

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{pmatrix}.$$

This matrix is upper triangular, so its determinant is the product of the diagonal entries:

$$\det A = 1 \cdot 1 \cdot 5 = 5.$$

This calculation is much more efficient than expanding immediately. It also demonstrates how determinant properties support intelligent computation.

3.4.6 The determinant of a product

Another fundamental property is

$$\det(AB) = \det(A)\det(B),$$

for square matrices A and B of the same size.

This formula says that determinants multiply under matrix multiplication. If one matrix scales area or volume by one factor and another matrix scales it by another factor, then doing both transformations multiplies the scale factors.

One important consequence is immediate. If A has determinant zero, then for every compatible square matrix B,

$$\det(AB) = 0.$$

A matrix that has already collapsed structure cannot be repaired by multiplying by another matrix on one side.

3.4.7 How to use the properties

The properties of determinants are not separate tricks. They form a strategy.

Summary

Before computing a determinant, ask:

- Is the matrix triangular?
- Are two rows or columns equal or proportional?
- Can I create zeros using row operations that preserve the determinant?
- If I swap rows, did I change the sign?
- If I multiply a row by a number, did I multiply the determinant by that number?
- Is Laplace expansion now easier after simplification?

A determinant should not be expanded mechanically unless there is no better path. Often the best computation begins with looking at the matrix and asking what structure is already visible.

3.5 Singularity and parameters

The determinant is especially important because of what happens when it is equal to zero. A zero determinant is not just a numerical accident. It tells us that the matrix has lost a certain

kind of independence.

Definition

A square matrix A is called singular if

$$\det A = 0.$$

It is called non-singular if

$$\det A \neq 0.$$

The word "singular" means exceptional or special. In this context, it means that the matrix does not behave like a fully independent square matrix. Something has collapsed. Rows or columns may be dependent, the associated transformation may flatten space, or a system of equations may fail to have a unique solution.

3.5.1 Reading singularity from rows

Consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & -1 \end{pmatrix}.$$

The second row is twice the first row:

$$(2, 4, 6) = 2(1, 2, 3).$$

Therefore the rows are dependent, and the determinant is zero:

$$\det A = 0.$$

We do not need to expand the determinant to know this. The structure is visible.

This is an important habit: before calculating, inspect. Sometimes the answer is already present in the shape of the object.

3.5.2 A determinant as a test for invertibility

A central theorem, which will become essential in the next chapter, is the following.

Theorem

A square matrix A is invertible if and only if

$$\det A \neq 0.$$

Equivalently, A is non-invertible if and only if

$$\det A = 0.$$

This theorem is one of the reasons determinants matter. It says that a single number can answer a structural question: can the matrix be reversed? If the determinant is non-zero, the matrix has not collapsed its structure, and an inverse matrix exists. If the determinant is zero, the matrix has lost information, and no inverse matrix can recover it.

We will study inverse matrices in the next chapter. Here we should understand the direction of thought:

$$\det A \neq 0 \quad \Longrightarrow \quad \text{the matrix can be inverted,}$$

while

$$\det A = 0 \quad \Longrightarrow \quad \text{the matrix cannot be inverted.}$$

3.5.3 Matrices depending on a parameter

Determinants are often used when a matrix contains a parameter. Then the determinant becomes an expression depending on that parameter. Solving

$$\det A = 0$$

tells us for which parameter values the matrix becomes singular.

Example

Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

Then

$$\det A(t) = t \cdot t - 1 \cdot 4 = t^2 - 4.$$

The matrix is singular when

$$t^2 - 4 = 0.$$

Therefore

$$t = 2 \quad \text{or} \quad t = -2.$$

For all other values of t, the matrix is non-singular.

This example is simple, but it shows the main idea. A parameter does not merely make the computation longer. It changes the question. We are no longer asking for one determinant value. We are asking when the structure of the matrix changes.

Example

Let

$$B(a) = \begin{pmatrix} 1 & a & 0 \\ 0 & 1 & a \\ a & 0 & 1 \end{pmatrix}.$$

Using Sarrus' rule,

$$\det B(a) = (1 \cdot 1 \cdot 1 + a \cdot a \cdot a + 0 \cdot 0 \cdot 0) - (0 \cdot 1 \cdot a + a \cdot 0 \cdot 1 + 1 \cdot a \cdot 0).$$

Thus

$$\det B(a) = 1 + a^3.$$

The matrix is singular when

$$1 + a^3 = 0,$$

so

$$a = -1.$$

Thus $B(a)$ is singular only for $a = -1$, and non-singular for all other real values of a.

The determinant turns a structural question about a matrix into an algebraic equation. This is one of the most useful aspects of the concept.

3.5.4 What should be concluded?

When a problem asks for parameter values for which a matrix is singular, the answer is not just a list of numbers. The reasoning has a structure:

1. compute $\det A$ as a function of the parameter,
2. solve the equation $\det A = 0$,
3. interpret the roots as values where the matrix loses invertibility.

Summary

The determinant separates square matrices into two broad classes.

- If $\det A \neq 0$, the matrix is non-singular and invertible.
- If $\det A = 0$, the matrix is singular and not invertible.
- If A depends on a parameter, the equation $\det A = 0$ identifies the exceptional parameter values.

This prepares the next chapter. Once we know how to detect whether a matrix is invertible, we can ask a new question: if the inverse exists, how do we find it?

3.6 Determinant patterns

A student who has just learned determinants often tries to compute every determinant by direct expansion. This is understandable, but it is not the best habit. Determinants reward attention to structure. Before expanding, we should ask what kind of matrix we are looking at.

3.6.1 Zeros are invitations

A row or column with many zeros usually tells us where to expand.

Example

For

$$A = \begin{pmatrix} 2 & 0 & 1 & 3 \\ 0 & 0 & 5 & 1 \\ 4 & 0 & -1 & 2 \\ 1 & 7 & 0 & 6 \end{pmatrix},$$

the second column has only one non-zero entry. Expansion along that column gives

$$\det A = 7 \det \begin{pmatrix} 2 & 1 & 3 \\ 0 & 5 & 1 \\ 4 & -1 & 2 \end{pmatrix}.$$

The matrix itself suggests an efficient first move.

3.6.2 Triangular form is a goal

Triangular matrices have easy determinants, so determinant-preserving row operations can be used to create triangular form.

Example

Let

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 3 & 5 & 12 \end{pmatrix}.$$

Apply

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - 3R_1,$$

and then

$$R_3 \leftarrow R_3 - 2R_2.$$

This produces

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Therefore $\det A = 0$, and the zero row reveals linear dependence.

3.6.3 Common factors

A common factor in one row or column may be taken outside the determinant. For example,

$$\det \begin{pmatrix} 3x & 6x \\ 2 & 5 \end{pmatrix} = 3x \det \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} = 3x.$$

Such a factor may reveal parameter values for which the determinant vanishes.

3.6.4 Vandermonde-type structure

A famous structured example is the 3×3 Vandermonde determinant

$$V = \begin{pmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{pmatrix}.$$

Its factorization follows from determinant-preserving column operations. Apply

$$C_2 \leftarrow C_2 - C_1, \qquad C_3 \leftarrow C_3 - C_1.$$

Then

$$\det V = \det \begin{pmatrix} 1 & 0 & 0 \\ x & y - x & z - x \\ x^2 & y^2 - x^2 & z^2 - x^2 \end{pmatrix}.$$

Expanding along the first row gives

$$\det V = \det \begin{pmatrix} y - x & z - x \\ y^2 - x^2 & z^2 - x^2 \end{pmatrix}.$$

Using

$$y^2 - x^2 = (y - x)(y + x), \qquad z^2 - x^2 = (z - x)(z + x),$$

we obtain

$$\det V = (y - x)(z - x) \det \begin{pmatrix} 1 & 1 \\ y + x & z + x \end{pmatrix}.$$

The remaining determinant is $z - y$. Hence

$$\boxed{\det V = (y - x)(z - x)(z - y)}.$$

The factors describe exactly when two columns become equal. For example, if $x = y$, then the first two columns coincide and the factor $y - x$ is zero.

3.6.5 Looking before expanding

Summary

Before computing a determinant, ask:

- Is the matrix triangular or nearly triangular?
- Is there a row or column with many zeros?
- Are two rows or columns equal or proportional?
- Can determinant-preserving row or column operations create zeros?
- Is there a common factor in a row or column?
- If parameters are present, when could dependence occur?

What to remember

A determinant is not merely a computational ornament. It detects structural information, becomes zero when independence is lost, and changes predictably under row and column operations. Efficient calculations begin by reading the matrix before expanding it.

Chapter 4

Matrix Inversion

4.1 The meaning of an inverse matrix

In the previous chapter we learned that the determinant can distinguish between singular and non-singular square matrices. The condition

$$\det A \neq 0$$

means that the matrix has not collapsed its structure. This raises a natural question: if a matrix has not collapsed information, can we reverse its action?

This question leads to the idea of an inverse matrix.

For ordinary numbers, the inverse of a non-zero number a is the number a^{-1} such that

$$aa^{-1} = 1.$$

For example, the inverse of 5 is $\frac{1}{5}$, because

$$5 \cdot \frac{1}{5} = 1.$$

The number 1 is special because multiplication by 1 changes nothing. In matrix algebra, the identity matrix plays the same role.

Definition

Let A be a square matrix of order n. A matrix B is called an inverse of A if

$$AB = I_n \quad \text{and} \quad BA = I_n.$$

If such a matrix exists, it is denoted by A^{-1}, and we write

$$AA^{-1} = I_n, \qquad A^{-1}A = I_n.$$

The two equations are both important. Matrix multiplication is not generally commutative, so the statements $AB = I$ and $BA = I$ are not automatically the same piece of information. The inverse must undo the matrix on both sides.

Remark

The notation A^{-1} does not mean $\frac{1}{A}$. Matrices are not numbers, and matrix division is not defined in the same way as numerical division. The symbol A^{-1} means: the matrix which reverses the action of A, if such a matrix exists.

4.1.1 Undoing an operation

The idea of an inverse is the idea of undoing. If a matrix A transforms a vector x into a vector y, then

$$Ax = y.$$

If A^{-1} exists, we can apply it to both sides:

$$A^{-1}Ax = A^{-1}y.$$

Since $A^{-1}A = I$, this becomes

$$x = A^{-1}y.$$

Thus the inverse matrix recovers the original vector from the transformed vector.

This explains why singular matrices cannot have inverses. If a matrix collapses different inputs into the same output, then there is no way to recover the original input uniquely. Information has been lost.

Example

Consider the diagonal matrix

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Multiplication by D doubles the first coordinate and triples the second coordinate:

$$D\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 3y \end{pmatrix}.$$

To reverse this action, we should divide the first coordinate by 2 and the second coordinate by 3. Therefore

$$D^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}.$$

Indeed,

$$DD^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2.$$

This example is simple because the matrix is diagonal. Each coordinate is changed independently. More general matrices mix coordinates, so their inverses are harder to find. But the meaning stays the same: an inverse matrix reverses the action of the original matrix.

4.1.2 When does an inverse exist?

The determinant gives the fundamental test.

Theorem

A square matrix A is invertible if and only if

$$\det A \neq 0.$$

If $\det A = 0$, then A is singular and has no inverse.

This theorem connects the present chapter with the previous one. Determinants do not merely produce numbers. They answer the structural question of whether a matrix can be reversed.

Summary

The inverse matrix should be understood through three questions:

- What operation does the original matrix perform?

- Has this operation lost information?
- If no information was lost, what matrix reverses the operation?

The rest of this chapter develops methods for finding inverse matrices and for using them correctly.

4.2 Inverses of 2×2 matrices

The first non-trivial case is a 2×2 matrix. It is small enough that the inverse can be written explicitly, but rich enough to show the main ideas: the determinant appears, the order of entries matters, and the inverse exists only under a clear condition.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We want to find a matrix

$$A^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

such that

$$AA^{-1} = I_2.$$

Multiplying, we get

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}.$$

For this to be the identity matrix, we need

$$\begin{cases} ap + br = 1, \\ cp + dr = 0, \end{cases} \qquad \begin{cases} aq + bs = 0, \\ cq + ds = 1. \end{cases}$$

Thus finding the inverse is already a problem about solving systems of linear equations. Instead of solving these systems every time, we use the known formula.

Theorem

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

If

$$\det A = ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $ad - bc = 0$, then A has no inverse.

The formula has two visible parts. First, the determinant appears in the denominator. This is not accidental: if the determinant is zero, the inverse cannot exist. Second, the matrix

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

is obtained by switching the diagonal entries a, d and changing signs of the off-diagonal entries b, c.

Example

Find the inverse of

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}.$$

First compute the determinant:

$$\det A = 2 \cdot 3 - 1 \cdot 5 = 6 - 5 = 1.$$

Since the determinant is non-zero, the inverse exists. Therefore

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

We should verify the result:

$$AA^{-1} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Verification is not a decorative step. It helps us remember the meaning of the inverse. We are not only producing a matrix; we are producing a matrix that must multiply with the original one to give the identity.

Example

Consider

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}.$$

The determinant is

$$\det B = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0.$$

The formula cannot be used because it would require division by zero. More importantly, the inverse does not exist. The second row is three times the first row, so the matrix has lost independent information.

This example shows why the determinant test is not just a technical condition. A zero determinant indicates structural collapse. There is no inverse because there is no unique way to undo the action of the matrix.

4.2.1 A formula should still be read

It is tempting to memorize the 2×2 inverse formula mechanically. But it should be read as a statement with meaning:

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The determinant controls whether the inverse exists and how strongly the inverse scales the result. The second matrix rearranges the entries in the only way that makes multiplication return the identity.

Summary

For a 2×2 inverse:

- compute $\det A = ad - bc$,
- if $\det A = 0$, stop: the matrix is not invertible,
- if $\det A \neq 0$, use the inverse formula,
- verify the result by multiplying with the original matrix.

4.3 The Gauss–Jordan method

The formula for a 2×2 inverse is useful, but it does not give a practical method for larger matrices. For a 3×3 matrix, and especially for larger matrices, we need a systematic procedure. The most important procedure is the Gauss–Jordan method.

The idea is simple and powerful. If we want to find A^{-1}, we want to transform A into the identity matrix. Every row operation we apply to A must also be applied to the identity matrix beside it. When the left side becomes I, the right side becomes A^{-1}.

We write this as an augmented matrix:

$$[A \mid I].$$

The goal is

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

Definition

An elementary matrix is a matrix obtained from an identity matrix by performing exactly one elementary row operation. If E represents that operation, then multiplying EA applies the same row operation to A.

Remark

The method does not work by magic. Each row operation corresponds to multiplication by an elementary matrix. If

$$E_k \cdots E_2 E_1 A = I,$$

then

$$E_k \cdots E_2 E_1 = A^{-1}.$$

Applying the same operations to the identity block therefore constructs the matrix that reverses A.

4.3.1 A complete 2×2 example

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}.$$

We form the augmented matrix

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array}\right].$$

Our goal is to turn the left block into the identity matrix.

First eliminate the entry below the first pivot:

$$R_2 \leftarrow R_2 - 3R_1.$$

Then

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

Now eliminate the entry above the second pivot:

$$R_1 \leftarrow R_1 - 2R_2.$$

This gives

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

The left side is now I_2, so

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

We verify:

$$\begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}\begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 7-6 & -2+2 \\ 21-21 & -6+7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

4.3.2 A complete 3×3 example

Now consider

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

We start with

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$$

Use the first row to eliminate the entry below the first pivot:

$$R_3 \leftarrow R_3 - R_1.$$

Then

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right].$$

Use the second row to eliminate the entry below the second pivot:

$$R_3 \leftarrow R_3 + R_2.$$

This gives

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array}\right].$$

Normalize the third row:

$$R_3 \leftarrow \frac{1}{2}R_3.$$

Thus

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

Now eliminate the entry above the third pivot in row 2:

$$R_2 \leftarrow R_2 - R_3.$$

We get

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

Finally eliminate the entry above the second pivot in row 1:

$$R_1 \leftarrow R_1 - R_2.$$

Therefore

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

So

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

This example is longer than the 2×2 case, but the logic is the same. We create pivots, eliminate entries below and above them, and keep track of every operation on the identity matrix.

4.3.3 What if the method fails?

The Gauss–Jordan method may fail to produce the identity matrix on the left. This happens precisely when the matrix is singular.

Example

Let

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Start with

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1.$$

We obtain

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right].$$

The left side cannot become the identity matrix, because the second row of the left block is zero. Thus B has no inverse.

This is not a failure of the method. It is information. The method reveals that the matrix is singular.

Summary

The Gauss–Jordan method for finding A^{-1}:

1. Form the augmented matrix $[A \mid I]$.
2. Use elementary row operations to reduce the left block.
3. If the left block becomes I, the right block is A^{-1}.
4. If the left block cannot become I, then A is not invertible.

4.4 The adjugate method

The Gauss–Jordan method is often the most practical method for finding inverse matrices by hand. There is also another method, built from determinants and cofactors. This method is called the adjugate method.

The adjugate method is important for two reasons. First, it gives an explicit formula for the inverse of a matrix. Second, it shows a deep connection between determinants, cofactors, and invertibility.

4.4.1 The cofactor matrix and the adjugate matrix

Let $A = (a_{ij})$ be an $n \times n$ matrix. For each entry a_{ij}, we can compute its cofactor

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} is the minor obtained by deleting row i and column j.

The matrix of cofactors is

$$C = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{pmatrix}.$$

The adjugate matrix is the transpose of the cofactor matrix.

Definition

The adjugate of a square matrix A, denoted by $\operatorname{adj}(A)$, is defined as

$$\operatorname{adj}(A) = C^{\top},$$

where C is the cofactor matrix of A.

The transpose is essential. The adjugate is not simply the cofactor matrix; it is the cofactor matrix with rows and columns exchanged.

4.4.2 The inverse formula

The central formula is the following.

Theorem

If A is a square matrix and $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A}\operatorname{adj}(A).$$

This formula should be read slowly. It says that the inverse exists only when $\det A \neq 0$. It also says that the inverse can be built from cofactors. Thus the same objects used to compute determinants also describe the inverse matrix.

A compact way to express the reason behind the formula is

$$A\operatorname{adj}(A) = (\det A)I.$$

If $\det A \neq 0$, we divide by $\det A$ and obtain

$$A\left(\frac{1}{\det A}\operatorname{adj}(A)\right) = I.$$

So

$$\frac{1}{\det A}\operatorname{adj}(A)$$

is the inverse matrix.

4.4.3 A 3×3 example

Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}.$$

First compute the determinant. Expanding along the first row,

$$\det A = 1 \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} + 2 \det \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}.$$

The first minor gives

$$3 \cdot 1 - 1 \cdot 2 = 1,$$

and the second minor gives

$$(-1) \cdot 2 - 3 \cdot 0 = -2.$$

Therefore

$$\det A = 1 + 2(-2) = -3.$$

Since $\det A \neq 0$, the inverse exists.

Now compute the cofactors. We list them carefully:

$$C_{11} = \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} = 1,$$

$$C_{12} = -\det \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} = 1,$$

$$C_{13} = \det \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix} = -2,$$

$$C_{21} = -\det \begin{pmatrix} 0 & 2 \\ 2 & 1 \end{pmatrix} = 4,$$

$$C_{22} = \det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1,$$

$$C_{23} = -\det \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = -2,$$

$$C_{31} = \det \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix} = -6,$$

$$C_{32} = -\det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} = -3,$$

$$C_{33} = \det \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix} = 3.$$

Thus the cofactor matrix is

$$C = \begin{pmatrix} 1 & 1 & -2 \\ 4 & 1 & -2 \\ -6 & -3 & 3 \end{pmatrix}.$$

Therefore

$$\operatorname{adj}(A) = C^{\top} = \begin{pmatrix} 1 & 4 & -6 \\ 1 & 1 & -3 \\ -2 & -2 & 3 \end{pmatrix}.$$

Finally,

$$A^{-1} = \frac{1}{-3} \begin{pmatrix} 1 & 4 & -6 \\ 1 & 1 & -3 \\ -2 & -2 & 3 \end{pmatrix}.$$

So

$$A^{-1} = \begin{pmatrix} -\frac{1}{3} & -\frac{4}{3} & 2 \\ -\frac{1}{3} & -\frac{1}{3} & 1 \\ \frac{2}{3} & \frac{2}{3} & -1 \end{pmatrix}.$$

This method is longer than Gauss–Jordan for this example, but it shows exactly how determinants and cofactors build the inverse.

4.4.4 When should this method be used?

For hand calculations, the adjugate method is convenient for 2×2 matrices and sometimes manageable for 3×3 matrices. For larger matrices it becomes computationally heavy because many cofactors must be computed.

Its conceptual value is very high: it explains why the determinant appears in inverse formulas and why zero determinant prevents invertibility.

Summary

The adjugate method:

1. Compute $\det A$.
2. If $\det A = 0$, the inverse does not exist.
3. Compute all cofactors C_{ij}.
4. Form the cofactor matrix C.
5. Transpose it to obtain $\operatorname{adj}(A)$.

6. Use $A^{-1} = \frac{1}{\det A}\operatorname{adj}(A)$.

4.5 Invertibility and structure

It is possible to treat invertibility as a purely computational question: compute the determinant, or try Gauss–Jordan elimination, and see what happens. This works, but it is not the only way to think. Often the structure of a matrix already tells us something about whether an inverse should exist.

The central rule remains the same:

$$A \text{ is invertible} \iff \det A \neq 0.$$

But before computing the determinant, we should learn to look at the matrix.

4.5.1 Diagonal matrices

A diagonal matrix is one of the clearest cases. Let

$$D = \operatorname{diag}(d_1, d_2, \ldots, d_n).$$

Then

$$\det D = d_1 d_2 \cdots d_n.$$

Therefore D is invertible exactly when none of the diagonal entries is zero.

If every $d_i \neq 0$, then

$$D^{-1} = \operatorname{diag}\left(\frac{1}{d_1}, \frac{1}{d_2}, \ldots, \frac{1}{d_n}\right).$$

Example

Let

$$D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Then

$$D^{-1} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{5} \end{pmatrix}.$$

This inverse is easy to understand: each coordinate was scaled independently, so the inverse simply rescales each coordinate back.

If a diagonal entry is zero, one coordinate is collapsed to zero. That information cannot be recovered.

4.5.2 Triangular matrices

Triangular matrices are also easy to test. If T is triangular, then

$$\det T = \text{product of diagonal entries.}$$

Thus T is invertible if and only if every diagonal entry is non-zero.

Example

Let

$$T = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 6 \end{pmatrix}.$$

Since

$$\det T = 1 \cdot (-1) \cdot 6 = -6 \neq 0,$$

the matrix is invertible.
On the other hand,

$$S = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 5 \\ 0 & 0 & 3 \end{pmatrix}$$

has determinant

$$\det S = 2 \cdot 0 \cdot 3 = 0.$$

So S is not invertible.

This is not merely a shortcut. A zero on the diagonal of a triangular matrix means that one pivot is missing. Without all pivots, the matrix cannot be reduced to the identity matrix.

4.5.3 Repeated or dependent rows

If two rows of a matrix are equal, the determinant is zero. If one row is a multiple of another, the determinant is also zero. More generally, if one row can be built from the others, the determinant is zero.

Example

Consider

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 0 & 1 & 4 \end{pmatrix}.$$

The second row is three times the first row. Therefore the rows are dependent, and

$$\det A = 0.$$

So A is not invertible.

The interpretation is important. If one row repeats information already contained in another row, then the matrix does not contain enough independent information to be reversed.

4.5.4 Products of matrices

The determinant rule

$$\det(AB) = \det A \det B$$

has a direct consequence for invertibility.

Theorem

If A and B are invertible square matrices of the same size, then AB is invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The order in this formula is essential. To undo the product AB, we must undo the last action first.

Suppose a vector x is transformed by B and then by A:

$$x \longmapsto Bx \longmapsto A(Bx) = ABx.$$

To recover x, first undo A, and then undo B:

$$ABx \longmapsto Bx \longmapsto x.$$

Thus the inverse is

$$B^{-1}A^{-1}.$$

We can verify algebraically:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

4.5.5 Matrices satisfying special identities

Sometimes a matrix is invertible because it satisfies an identity that already reveals its inverse.

Example

Suppose a square matrix A satisfies

$$A^2 = I.$$

Then A is its own inverse, because

$$AA = I.$$

Therefore

$$A^{-1} = A.$$

Such a matrix is called an involution.

Example

Suppose a square matrix A satisfies

$$A^2 - 3A + 2I = 0.$$

Then

$$A^2 - 3A = -2I.$$

Factor the left side:

$$A(A - 3I) = -2I.$$

Multiplying by $-\frac{1}{2}$, we get

$$A\left(-\frac{1}{2}(A - 3I)\right) = I.$$

Therefore

$$A^{-1} = -\frac{1}{2}(A - 3I) = \frac{1}{2}(3I - A).$$

This argument finds the inverse without performing Gauss–Jordan elimination.

These examples show that invertibility is not only a computational topic. It is also a structural topic. Sometimes the matrix tells us how it can be undone.

Summary

To recognize invertibility, look for structure:

- diagonal matrices are invertible when no diagonal entry is zero,
- triangular matrices are invertible when all diagonal entries are non-zero,
- repeated or dependent rows imply non-invertibility,
- products of invertible matrices are invertible,
- special identities may reveal the inverse directly.

4.6 Matrix equations

One of the main reasons for studying inverse matrices is that they allow us to solve matrix equations. The idea is similar to solving ordinary equations, but there is one crucial difference: matrix multiplication is not commutative. Therefore the side on which we multiply by an inverse matters.

4.6.1 Equations of the form $AX = B$

Suppose

$$AX = B,$$

where A is a known square matrix, B is a known matrix, and X is unknown. If A is invertible, we multiply on the left by A^{-1}:

$$A^{-1}AX = A^{-1}B.$$

Since $A^{-1}A = I$, this gives

$$X = A^{-1}B.$$

The inverse must be placed on the left because A is on the left of X. Writing XA^{-1} would not undo AX.

Example

Solve

$$AX = B,$$

where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

From an earlier example,

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

Therefore

$$X = A^{-1}B = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

Compute the product:

$$X = \begin{pmatrix} 28 - 10 & 7 - 4 \\ -12 + 5 & -3 + 2 \end{pmatrix} = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix}.$$

Thus

$$X = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix}.$$

4.6.2 Equations of the form $XA = B$

Now suppose

$$XA = B.$$

Here A is on the right of X. To undo multiplication by A, we must multiply on the right by A^{-1}:

$$XAA^{-1} = BA^{-1}.$$

Since $AA^{-1} = I$, we get

$$X = BA^{-1}.$$

Example

Solve

$$XA = B,$$

where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

Again,

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

But now

$$X = BA^{-1} = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

Thus

$$X = \begin{pmatrix} 28-3 & -8+1 \\ 35-6 & -10+2 \end{pmatrix} = \begin{pmatrix} 25 & -7 \\ 29 & -8 \end{pmatrix}.$$

This is not the same answer as in the previous example. The equation is different because the unknown matrix is multiplied by A on a different side.

4.6.3 Equations with matrices on both sides

Some equations have the unknown matrix between two known matrices:

$$AXB = C.$$

If A and B are invertible, then we undo A on the left and B on the right:

$$A^{-1}(AXB)B^{-1} = A^{-1}CB^{-1}.$$

Therefore

$$X = A^{-1}CB^{-1}.$$

The order is not optional. Each inverse must be placed on the side where it can cancel the corresponding matrix.

4.6.4 Solving vector equations

The equation

$$Ax = b$$

where x and b are column vectors is a special case of $AX = B$. If A is invertible, then

$$x = A^{-1}b.$$

This gives a conceptual method for solving linear systems. In the next chapter we will study systems of linear equations in detail, and we will compare different methods: elimination, inverse matrices, and Cramer's rule.

Example

Solve

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The determinant is

$$\det A = 2 \cdot 1 - 1 \cdot 1 = 1,$$

so

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 - 3 \\ -5 + 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Thus

$$x = 2, \qquad y = 1.$$

4.6.5 Common mistakes

The most common mistake is to move matrices as if they were numbers. For example, from

$$AX = B$$

one may be tempted to write

$$X = BA^{-1}.$$

This is wrong in general. Multiplying by A^{-1} on the right does not cancel the A on the left of X. The correct step is

$$A^{-1}AX = A^{-1}B,$$

so

$$X = A^{-1}B.$$

Summary

When solving matrix equations:

- identify where the unknown matrix is placed,
- multiply by inverses on the side where cancellation is valid,

- preserve the order of multiplication,
- check that the matrices involved are invertible,
- verify the final answer by substitution when possible.

Inverse matrices are powerful because they allow equations to be solved by undoing operations. But the power comes with a responsibility: the order of multiplication must be respected at every step.

Chapter 5

Systems of Linear Equations

5.1 What systems of linear equations describe

A system of linear equations is not only a collection of equations to solve. It is a collection of conditions that must be satisfied at the same time. Each equation gives one condition. A solution is an assignment of values to the unknowns that satisfies all conditions simultaneously.

This point is important. When we solve a system, we are not solving the equations one after another as separate problems. We are looking for values that make the whole system true.

Example

Consider the system

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

The pair $(2, 3)$ satisfies the first equation because

$$2 + 3 = 5.$$

It also satisfies the second equation because

$$2 \cdot 2 - 3 = 1.$$

Therefore $(2, 3)$ is a solution of the system.
The pair $(1, 4)$ satisfies the first equation, since $1 + 4 = 5$, but it does not satisfy the second equation, since $2 \cdot 1 - 4 = -2$. Therefore $(1, 4)$ is not a solution of the system.

The example should be read slowly. A solution of a system must pass every test in the system. Passing one equation is not enough.

Unknowns, coefficients, and constants

A linear equation in the unknowns $x_1, x_2, \ldots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b.$$

The numbers $a_1, a_2, \ldots, a_n$ are coefficients. The number b is the constant term or right-hand side.

A system of linear equations is made of several such equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \qquad\qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

Here m is the number of equations and n is the number of unknowns.

Remark

The coefficients describe how the unknowns enter the equations. The right-hand side describes what values the linear combinations must equal. A system compares these two pieces of information.

Solution sets

The solution set of a system is the set of all solutions. Sometimes this set contains one element. Sometimes it is empty. Sometimes it contains infinitely many elements.

These are the three basic possibilities for a linear system:

- exactly one solution,
- no solution,
- infinitely many solutions.

The goal of solving a system is not only to find numbers. The goal is to determine which of these cases occurs and to describe the solution set accurately.

Matrix form

Systems of linear equations are naturally connected with matrices. The coefficients form a matrix, the unknowns form a column vector, and the right-hand side forms another column vector.

For example, the system

$$\begin{cases} 2x + y = 5, \\ x - y = 1 \end{cases}$$

can be written as

$$\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

Using the notation

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \qquad x = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 1 \end{pmatrix},$$

we write the system compactly as

$$Ax = b.$$

This notation is not merely shorter. It reveals the structure of the problem. The matrix A transforms the unknown vector x into the right-hand side vector b. Solving the system means finding which input vector x, if any, is sent to b.

The augmented matrix

When solving a system, it is often convenient to write the coefficients and the right-hand side together in an augmented matrix:

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array}\right].$$

The vertical line reminds us that the last column is not another column of coefficients. It is the right-hand side.

The augmented matrix is a compact record of the system. Row operations on this matrix correspond to controlled transformations of the equations.

Summary

When reading a system of linear equations, ask:

- What are the unknowns?
- What are the equations, and what condition does each one impose?
- What are the coefficients and right-hand sides?
- What would it mean for one assignment to satisfy all equations at once?
- Can the system be written as $Ax = b$?
- What does the augmented matrix record?

A system of equations should first be read as a structured mathematical object. Only then should we choose a method for solving it.

5.2 Geometric meaning of solutions

Linear systems have an algebraic form, but they also have a geometric meaning. This geometric interpretation is often the easiest way to understand why a system may have one solution, no solution, or infinitely many solutions.

The central idea is simple: each equation describes a set of points. The solution set of the system is the intersection of all these sets.

Two variables: lines in the plane

In two variables, a linear equation

$$ax + by = c$$

usually describes a line in the plane. A system of two linear equations describes two lines, and a solution is a point that lies on both lines.

Example

Consider

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

The first equation describes one line. The second equation describes another line. The pair $(2, 3)$ is a solution because the point $(2, 3)$ lies on both lines.

This is the geometric meaning of solving the system: find the common point of the two lines.

One solution

If two lines in the plane intersect in one point, the system has exactly one solution.

Example

The system

$$\begin{cases} x + y = 5, \\ 2x - y = 1 \end{cases}$$

has exactly one solution. Solving gives

$$x = 2, \qquad y = 3.$$

Geometrically, the two lines meet at the point $(2, 3)$.

When a system has exactly one solution, the conditions are independent enough to determine the unknowns completely.

No solution

If two lines are parallel and distinct, they never meet. Then the system has no solution.

Example

Consider

$$\begin{cases} x + y = 1, \\ x + y = 4. \end{cases}$$

The left-hand sides are the same, but the right-hand sides are different. The same expression $x + y$ cannot equal both 1 and 4 at the same time.
Geometrically, these are two parallel distinct lines. Therefore the system has no solution.

A system with no solution is called inconsistent. The equations impose conditions that cannot all be satisfied simultaneously.

Infinitely many solutions

If two equations describe the same line, then every point on that line is a solution. The system has infinitely many solutions.

Example

Consider

$$\begin{cases} x + y = 3, \\ 2x + 2y = 6. \end{cases}$$

The second equation is just twice the first equation. It gives the same condition, not a new independent condition.
Therefore the solution set is the whole line

$$x + y = 3.$$

There are infinitely many solutions, for example $(0, 3)$, $(1, 2)$, and $(3, 0)$.

This case is not a failure of mathematics. It means that the equations do not contain enough independent information to determine one point.

Three variables: planes in space

In three variables, a linear equation

$$ax + by + cz = d$$

usually describes a plane in space. A system of linear equations describes the common points of several planes.

Several things may happen:

- three planes may meet at one point,
- they may have no common point,
- they may share a line,
- they may all be the same plane.

The algebraic classification remains the same: one solution, no solution, or infinitely many solutions. The geometry becomes richer because planes can intersect in more ways than lines in the plane.

Reading geometry from algebra

The algebraic form of a system often reveals the geometric situation. For example, proportional left-hand sides may indicate parallel or identical lines. A contradiction such as

$$0 = 5$$

means that the conditions cannot all be satisfied. A row of zeros such as

$$0 = 0$$

means that one equation became redundant.

These patterns will appear naturally during Gaussian elimination.

Summary

When interpreting a system geometrically, ask:

- What geometric object does each equation describe?
- Is the solution set an intersection of lines, planes, or other linear objects?
- Do the objects meet in one point?
- Are they parallel or inconsistent?
- Is one condition a repetition of another condition?
- Does the geometry suggest one solution, no solution, or infinitely many solutions?

The geometric viewpoint does not replace algebraic methods. It explains what those methods are detecting.

5.3 Gaussian elimination

Gaussian elimination is a method for simplifying a system of linear equations without changing its solution set. The method uses elementary row operations on the augmented matrix. These operations are not random changes of notation. They correspond to replacing equations by equivalent equations.

The aim is to transform a complicated system into a readable one.

Equivalent systems

Two systems are equivalent if they have the same solution set. Gaussian elimination works because elementary row operations produce equivalent systems.

The allowed row operations are:

- swapping two rows,
- multiplying a row by a non-zero number,
- adding a multiple of one row to another row.

These are the same operations introduced earlier for matrices. Here their meaning is sharper: they change the written form of the equations, but not the set of solutions.

Remark

The purpose of row operations is not to make the matrix look different. The purpose is to reveal the solution set by simplifying the conditions.

A first example

Consider the system

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

Its augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 1 \end{array}\right].$$

We use the first row to eliminate the x-coefficient in the second row:

$$R_2 \leftarrow R_2 - 2R_1.$$

This gives

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -9 \end{array}\right].$$

The second row now means

$$-3y = -9,$$

so

$$y = 3.$$

Substitute into the first equation:

$$x + 3 = 5,$$

so

$$x = 2.$$

Therefore the solution is

$$(x, y) = (2, 3).$$

This example shows the basic strategy. We simplify the system until the unknowns can be read one by one.

Pivots

A pivot is an entry used to eliminate other entries in its column. In the previous example, the first pivot was the 1 in the upper-left corner. It allowed us to remove the 2 below it.

Pivots are not only computational markers. They identify independent information in the system. Each pivot usually corresponds to a variable that can be solved from the system.

Example

Consider

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 2 & 6 \end{array}\right].$$

The pivot positions are in the first, second, and third variable columns. The last row gives

$$2z = 6,$$

so $z = 3$. The second row then gives

$$y + 4z = 5,$$

so $y + 12 = 5$, hence $y = -7$. The first row gives

$$x + 2y - z = 3,$$

so $x + 2(-7) - 3 = 3$, and therefore $x = 20$.

This process is called back substitution. We solve from the bottom upward after the system has been simplified.

Row echelon form

A matrix is in row echelon form when the leading non-zero entries move to the right as we move downward, and zero rows, if any, are placed at the bottom. Row echelon form is useful because it makes the structure of the system visible.

For example,

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 4 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

is in row echelon form. The last row represents the equation

$$0 = 0,$$

which gives no new restriction.

Detecting no solution

During elimination, a row may become

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 5 \end{array}\right].$$

This row represents

$$0 = 5.$$

That is impossible. Therefore the system has no solution.

Example

Consider

$$\begin{cases} x + y = 1, \\ 2x + 2y = 5. \end{cases}$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 5 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1.$$

We get

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 3 \end{array}\right].$$

The second row says

$$0 = 3,$$

which is impossible. The system has no solution.

The contradiction is not a technical detail. It is the algebraic sign that the conditions are inconsistent.

Free variables and infinitely many solutions

If a variable column has no pivot, that variable is free. A free variable can be chosen as a parameter. The remaining variables are expressed in terms of it.

Example

Consider the row echelon form

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right].$$

This represents

$$\begin{cases} x + 2y - z = 3, \\ z = 4. \end{cases}$$

The variable y has no pivot, so let

$$y = t.$$

Then $z = 4$, and the first equation gives

$$x + 2t - 4 = 3,$$

so

$$x = 7 - 2t.$$

Thus the solution set is

$$(x, y, z) = (7 - 2t, t, 4), \qquad t \in \mathbb{R}.$$

There are infinitely many solutions.

A parameter is not a decoration. It records the freedom that remains after all independent conditions have been used.

What elimination tells us

Gaussian elimination does more than compute answers. It classifies the system.

Summary

After row reduction, look for these patterns:

- A pivot in every variable column and no contradiction: exactly one solution.
- A row of the form $0 = \text{non-zero}$: no solution.
- At least one free variable and no contradiction: infinitely many solutions.

Gaussian elimination is the most flexible method in this chapter. It works for square and non-square systems, and it reveals not only the solution but also the structure of the solution set.

5.4 Cramer's rule

Cramer's rule is a method for solving certain square systems of linear equations using determinants. It is not the most efficient method for large systems, but it is conceptually important because it shows a direct connection between determinants and solvability.

The method applies to systems where the number of equations equals the number of unknowns and the coefficient matrix has non-zero determinant.

The two by two case

Consider the system

$$\begin{cases} ax + by = e, \\ cx + dy = f. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and its determinant is

$$\det A = ad - bc.$$

If $\det A \neq 0$, the system has exactly one solution.

Cramer's rule says that the solution can be computed by replacing one column of the coefficient matrix at a time by the right-hand side column.

Define

$$A_x = \begin{pmatrix} e & b \\ f & d \end{pmatrix}, \qquad A_y = \begin{pmatrix} a & e \\ c & f \end{pmatrix}.$$

Then

$$x = \frac{\det A_x}{\det A}, \qquad y = \frac{\det A_y}{\det A}.$$

Remark

The denominator $\det A$ is not a technical detail. If $\det A = 0$, the coefficient matrix is singular, and Cramer's rule does not apply.

A worked example

Example

Solve the system

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its determinant is

$$\det A = 2 \cdot (-1) - 1 \cdot 1 = -2 - 1 = -3.$$

Since $\det A \neq 0$, Cramer's rule applies.
For x, replace the first column by the right-hand side:

$$A_x = \begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$\det A_x = 5 \cdot (-1) - 1 \cdot 1 = -5 - 1 = -6.$$

So

$$x = \frac{\det A_x}{\det A} = \frac{-6}{-3} = 2.$$

For y, replace the second column:

$$A_y = \begin{pmatrix} 2 & 5 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\det A_y = 2 \cdot 1 - 5 \cdot 1 = 2 - 5 = -3.$$

So

$$y = \frac{\det A_y}{\det A} = \frac{-3}{-3} = 1.$$

Thus

$$(x, y) = (2, 1).$$

This result should be checked by substitution. We have

$$2 \cdot 2 + 1 = 5, \qquad 2 - 1 = 1.$$

Both equations are satisfied.

The general square case

For a square system

$$Ax = b,$$

where A is an $n \times n$ matrix and b is a column vector, Cramer's rule applies if

$$\det A \neq 0.$$

For each unknown x_i, form the matrix A_i by replacing the i-th column of A with b. Then

$$x_i = \frac{\det A_i}{\det A}.$$

The rule is elegant, but the number of determinant computations grows quickly. For this reason, Gaussian elimination is usually more practical for larger systems.

Why the determinant condition matters

The condition

$$\det A \neq 0$$

means that the coefficient matrix is invertible. Geometrically and structurally, the equations contain enough independent information to determine one solution.

If

$$\det A = 0,$$

then the matrix is singular. This does not automatically mean that the system has no solution. It means that the system cannot have a unique solution determined by Cramer's rule. The system must be analyzed further: it may have no solution or infinitely many solutions.

Example

Consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

and

$$\det A = 1 \cdot 2 - 1 \cdot 2 = 0.$$

Cramer's rule does not apply. But the system is not inconsistent. The second equation is twice the first, so there are infinitely many solutions.

This example is important. A zero determinant means loss of uniqueness, not automatically loss of existence.

Summary

When using Cramer's rule, ask:

- Is the system square?
- What is the coefficient matrix A?
- Is $\det A \neq 0$?
- Which column must be replaced to compute each unknown?
- Does the final answer satisfy the original equations?
- If $\det A = 0$, what further classification is needed?

Cramer's rule shows how determinants can solve systems, but it must be used only when its structural condition is satisfied.

5.5 The inverse matrix method

Some systems of linear equations can be solved by using the inverse of the coefficient matrix. This method is based on the matrix equation

$$Ax = b.$$

Here A is the coefficient matrix, x is the column vector of unknowns, and b is the right-hand side vector.

The method is simple in form, but its meaning should be understood carefully. If A is invertible, then multiplying by A^{-1} undoes the action of A.

Solving $Ax = b$

Suppose

$$Ax = b$$

and A is invertible. Multiply both sides on the left by A^{-1}:

$$A^{-1}Ax = A^{-1}b.$$

Since

$$A^{-1}A = I,$$

we get

$$Ix = A^{-1}b.$$

Therefore

$$x = A^{-1}b.$$

Theorem

If A is an invertible square matrix, then the system

$$Ax = b$$

has the unique solution

$$x = A^{-1}b.$$

The word "invertible" is essential. If A^{-1} does not exist, this method cannot be used.

A worked example

Example

Solve

$$\begin{cases} 2x + y = 5, \\ x + y = 3. \end{cases}$$

Write the system as

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Thus

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The determinant is

$$\det A = 2 \cdot 1 - 1 \cdot 1 = 1,$$

so A is invertible. For a 2×2 matrix,

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Compute the product:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5-3 \\ -5+6 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Thus

$$x = 2, \qquad y = 1.$$

The inverse matrix method gives the same result as other correct methods. The difference is the language: here solving means applying the inverse transformation.

Why the method works

The equation

$$Ax = b$$

can be read as follows: the matrix A acts on the unknown vector x and produces the vector b. If A is invertible, then this action can be reversed. Applying A^{-1} to b recovers the original vector x.

This interpretation is important. The formula

$$x = A^{-1}b$$

is not magic. It is the matrix version of undoing an operation.

Connection with determinants

For a square matrix A, invertibility is equivalent to

$$\det A \neq 0.$$

Therefore the inverse matrix method applies exactly when the coefficient matrix has non-zero determinant.

If

$$\det A = 0,$$

then A^{-1} does not exist. The system may have no solution or infinitely many solutions, but it cannot be solved by multiplying by A^{-1}.

Example

Consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

and

$$\det A = 0.$$

The inverse matrix method cannot be used. However, the system does have solutions: every pair (x, y) satisfying $x + y = 2$ is a solution. There are infinitely many such pairs.

Again, the failure of invertibility means loss of uniqueness, not automatically inconsistency.

When this method is useful

The inverse matrix method is conceptually clear, but it is not always the most practical computational method. Finding A^{-1} can be more work than solving the system directly by elimination.

The method is especially useful when:

- the inverse matrix is already known,
- several systems have the same coefficient matrix A but different right-hand sides,
- we want to understand the system as a matrix equation.

For routine computation, Gaussian elimination is usually more flexible.

Summary

When using the inverse matrix method, ask:

- Can the system be written as $Ax = b$?
- Is A square?
- Is A invertible, equivalently $\det A \neq 0$?
- Have I multiplied on the correct side by A^{-1}?
- Does the vector $A^{-1}b$ satisfy the original system?

The inverse matrix method is powerful because it connects solving systems with the idea of reversing a linear transformation.

5.6 Parametric systems and classification

Some systems contain a parameter. Then the question is not only “What is the solution?” but also “For which parameter values does the structure of the system change?”

This is where determinants, row reduction, and geometric interpretation come together.

The three possible outcomes

Every linear system has one of the following outcomes:

- exactly one solution,
- no solution,
- infinitely many solutions.

For a square system

$$Ax = b,$$

one important test is the determinant of A. If

$$\det A \neq 0,$$

then A is invertible and the system has exactly one solution. If

$$\det A = 0,$$

then the system does not have a unique solution obtained by inversion. But we must still decide whether it has no solution or infinitely many solutions.

Remark

The condition $\det A = 0$ is a warning sign, not a complete answer. It says that uniqueness is lost. It does not by itself decide whether solutions exist.

A simple parameter example

Consider the system

$$\begin{cases} x + y = 1, \\ 2x + 2y = a. \end{cases}$$

The left-hand side of the second equation is twice the left-hand side of the first equation. Therefore consistency depends on the right-hand side.

If we multiply the first equation by 2, we get

$$2x + 2y = 2.$$

So for the second equation to describe the same condition, we need

$$a = 2.$$

If $a = 2$, the two equations describe the same line, and there are infinitely many solutions:

$$x + y = 1.$$

If $a \neq 2$, the equations describe parallel distinct lines, and there is no solution.

Summary

For

$$\begin{cases} x + y = 1, \\ 2x + 2y = a, \end{cases}$$

we have:

- if $a = 2$, infinitely many solutions,
- if $a \neq 2$, no solution.

This example shows that a parameter may affect only the right-hand side, but that can still change the solution set completely.

Using determinants to find exceptional values

Now consider a parameter in the coefficient matrix. Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

The determinant is

$$\det A(t) = t^2 - 4.$$

The exceptional values occur when

$$t^2 - 4 = 0,$$

so

$$t = 2 \qquad \text{or} \qquad t = -2.$$

For all other values of t, the matrix is invertible. Therefore any system

$$A(t)x = b$$

has exactly one solution for $t \neq 2$ and $t \neq -2$.

The exceptional values $t = 2$ and $t = -2$ require separate analysis. At those values, the determinant test tells us that the system is not uniquely solvable, but not whether it is inconsistent or has infinitely many solutions.

A full classification example

Example

Classify the system depending on the parameter a:

$$\begin{cases} x + y = 3, \\ 2x + ay = 6. \end{cases}$$

The coefficient matrix is

$$A(a) = \begin{pmatrix} 1 & 1 \\ 2 & a \end{pmatrix}.$$

Its determinant is

$$\det A(a) = a - 2.$$

If

$$a \neq 2,$$

then $\det A(a) \neq 0$, so the system has exactly one solution.
Now consider the exceptional value $a = 2$. The system becomes

$$\begin{cases} x + y = 3, \\ 2x + 2y = 6. \end{cases}$$

The second equation is twice the first. Therefore the system has infinitely many solutions.
Thus:

- if $a \neq 2$, the system has exactly one solution,
- if $a = 2$, the system has infinitely many solutions.

The determinant found the value where the structure changes. Row interpretation then classified what happens at that value.

Example

Classify the system depending on a:

$$\begin{cases} x + y = 3, \\ 2x + 2y = a. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

so

$$\det A = 0.$$

There is no value of a that makes this coefficient matrix invertible. We must compare the equations.
Twice the first equation gives

$$2x + 2y = 6.$$

Therefore:

- if $a = 6$, the two equations describe the same line, so there are infinitely many solutions,
- if $a \neq 6$, the two equations are inconsistent, so there is no solution.

This example shows that a singular coefficient matrix may still lead to different outcomes depending on the right-hand side.

Parameters and row reduction

For larger systems, row reduction is often the safest method. It allows us to see contradictions, zero rows, and free variables directly.

A row such as

$$0 = 0$$

indicates redundancy. A row such as

$$0 = 5$$

indicates inconsistency. A missing pivot indicates a free variable. These patterns remain valid even when parameters are present.

The difficulty is that some row operations may require dividing by an expression containing a parameter. Before dividing by such an expression, we must consider when it might be zero.

Remark

When a parameter appears, do not divide by an expression such as $a - 2$ without asking whether $a - 2 = 0$ is possible. Parameter values that make a denominator zero are usually exceptional cases and must be treated separately.

A classification strategy

A good solution to a parameter system should not only list parameter values. It should explain why the classification changes.

Summary

For a square parameter system $A(t)x = b(t)$, a useful strategy is:

1. Compute $\det A(t)$.
2. Solve $\det A(t) = 0$ to find exceptional parameter values.
3. For non-exceptional values, conclude that there is exactly one solution.
4. For exceptional values, analyze the system separately using row reduction or equation comparison.
5. Classify each case as exactly one solution, no solution, or infinitely many solutions.

The determinant identifies where uniqueness can fail. Elimination and interpretation decide what happens there.

What the chapter has built

Systems of linear equations bring together the earlier algebraic tools.

Matrices organize the coefficients. Row operations simplify the system. Determinants detect whether a square coefficient matrix is invertible. Inverse matrices give the formula $x = A^{-1}b$ when the inverse exists. Cramer's rule gives a determinant-based expression for the solution when $\det A \neq 0$.

These methods are not isolated tricks. They are different ways of reading the same object: a system of linear conditions.

Summary

At the end of this chapter, remember:

- A system is a set of conditions to be satisfied simultaneously.
- The augmented matrix records the coefficients and the right-hand side.
- Gaussian elimination is the most flexible method for solving and classifying systems.
- If $\det A \neq 0$, a square system $Ax = b$ has exactly one solution.
- If $\det A = 0$, uniqueness is lost, but existence must still be checked.
- A linear system may have exactly one solution, no solution, or infinitely many solutions.

The goal is not only to solve systems. The goal is to understand what the equations say about structure, dependence, and compatibility.

Chapter 6

Vectors and Coordinates

6.1 Points and coordinates

Analytic geometry begins with a simple but powerful idea: geometric objects can be described by numbers. A point on a line can be described by one number. A point in a plane can be described by two numbers. A point in space can be described by three numbers. This numerical description allows us to use algebra to answer geometric questions.

The important word here is description. A point is not the same thing as the symbols used to describe it. The symbols depend on a coordinate system. Once a coordinate system is chosen, the symbols become a precise way to say where the point is.

Definition

In the plane, a point is usually written as an ordered pair

$$P = (x, y).$$

The first coordinate x describes horizontal position, and the second coordinate y describes vertical position. In three-dimensional space, a point is usually written as an ordered triple

$$P = (x, y, z).$$

The order of the coordinates matters. The points $(2, 5)$ and $(5, 2)$ are generally different points. The numbers are the same, but their roles are not the same. The first number answers one question, and the second number answers another question.

Example

The point

$$P = (3, -2)$$

means the following: starting from the origin, move 3 units in the positive horizontal direction and 2 units in the negative vertical direction.
The point

$$Q = (-2, 3)$$

uses the same two numbers, but in different positions. It describes a different point.

This is a small example of a general habit that will appear throughout analytic geometry. Before doing a calculation, we should read the object. A pair of numbers is not just a pair of numbers. It is an ordered description of position.

The origin and the coordinate axes

The origin is the point from which positions are measured. In the plane it is

$$O = (0, 0),$$

and in space it is

$$O = (0, 0, 0).$$

The coordinate axes give directions of measurement. In the plane, the x-axis records horizontal displacement from the origin, while the y-axis records vertical displacement. In space, the z-axis adds a third independent direction.

A coordinate system therefore does two things at once. It gives a point of reference, and it gives directions in which numbers are measured. Without this structure, a coordinate tuple has no geometric meaning.

Remark

Coordinates are not only labels. They are labels chosen relative to a system of measurement. Changing the coordinate system may change the coordinates of a point, even though the point itself has not moved.

Coordinates as variables

Sometimes coordinates are fixed numbers. For example,

$$A = (1, 4), \qquad B = (-3, 2)$$

are particular points. But often coordinates are variables. When we write

$$P = (x, y),$$

we may be describing an arbitrary point in the plane. The letters x and y are not yet known numbers. They are placeholders for possible positions.

This distinction is essential. A fixed point is one object. A point with variable coordinates can represent many possible points. Equations will later use this idea. For instance, an equation in x and y may describe all points whose coordinates satisfy a certain condition.

Example

The expression

$$P = (x, 2)$$

does not describe one fixed point unless x is known. It describes all points whose second coordinate is 2. Geometrically, these points lie on a horizontal line.
Similarly,

$$P = (1, y)$$

describes all points whose first coordinate is 1. These points lie on a vertical line.

This example already hints at the main idea of analytic geometry: an algebraic condition can describe a geometric object. The condition $y = 2$ describes a horizontal line. The condition $x = 1$ describes a vertical line. Later we will use more complicated conditions to describe more complicated objects.

Reading before calculating

The first task in coordinate geometry is not to apply a formula. It is to understand what kind of object is being described.

Summary

When reading coordinates, ask:

- Am I looking at a fixed point or a point with variable coordinates?

- Which coordinate describes which direction?
- Is the order of the coordinates important in this problem?
- Is an equation describing one point or a whole set of points?
- What geometric information is encoded in the algebraic notation?

The aim is to develop a controlled language. Coordinates allow us to speak about geometry with numbers, but the numbers must be read with care.

6.2 Vectors as displacements

A point describes a position. A vector describes a displacement. This difference is one of the most important distinctions in analytic geometry.

If we say that a point is $(3, 2)$, we are saying where something is. If we say that a vector is $(3, 2)$, we are saying how something moves: three units in one coordinate direction and two units in another coordinate direction. The same pair of numbers may appear in both situations, but the meaning is different.

Definition

Let

$$A = (a_1, a_2), \qquad B = (b_1, b_2)$$

be points in the plane. The vector from A to B is

$$\overrightarrow{AB} = B - A = (b_1 - a_1,\, b_2 - a_2).$$

In space, if

$$A = (a_1, a_2, a_3), \qquad B = (b_1, b_2, b_3),$$

then

$$\overrightarrow{AB} = (b_1 - a_1,\, b_2 - a_2,\, b_3 - a_3).$$

The formula is simple, but it should be read carefully. We subtract the starting coordinates from the ending coordinates. A displacement is not only about where we end. It is about how we get from one position to another.

Example

Let

$$A = (1, 4), \qquad B = (6, 2).$$

Then

$$\overrightarrow{AB} = B - A = (6 - 1,\, 2 - 4) = (5, -2).$$

This means that to move from A to B, we move 5 units in the positive x-direction and 2 units in the negative y-direction.

The vector $(5, -2)$ does not remember the absolute position of the starting point. It remembers the displacement. The same displacement could begin somewhere else.

Example

Let

$$C = (-3, 1), \qquad D = (2, -1).$$

Then

$$\overrightarrow{CD} = D - C = (2 - (-3),\ -1 - 1) = (5, -2).$$

Thus

$$\overrightarrow{AB} = \overrightarrow{CD},$$

even though the points A, B, C, D are not the same. The arrows begin in different places, but they describe the same displacement.

This is why vectors are often called free objects in elementary geometry. A vector can be represented by an arrow, but the arrow may be moved parallel to itself without changing the vector. What matters is direction, length, and orientation, not the particular location of the drawing.

Position vectors

There is one special way in which points and vectors are connected. If a vector begins at the origin and ends at a point P, then the coordinates of the vector and the coordinates of the point are the same.

If

$$P = (p_1, p_2),$$

then

$$\overrightarrow{OP} = (p_1, p_2),$$

where $O = (0, 0)$.

Remark

The equality of coordinates does not mean that points and vectors are the same kind of object. It means that a point can be represented by the vector from the origin to that point. This representation is useful, but the distinction should not be forgotten.

This distinction will help us later. A line can be described by starting from one point and moving in a vector direction. A plane can be described by a point and two independent directions, or by a point and one normal vector. In every case, we must know whether an expression describes a position, a displacement, or a direction.

Opposite displacements

The vector from B to A is the opposite of the vector from A to B:

$$\overrightarrow{BA} = A - B = -(B - A) = -\overrightarrow{AB}.$$

This is not a new formula to memorize. It simply says that going back reverses the displacement.

Example

For the points

$$A = (1, 4), \qquad B = (6, 2),$$

we found

$$\overrightarrow{AB} = (5, -2).$$

In the opposite direction,

$$\overrightarrow{BA} = A - B = (1 - 6,\ 4 - 2) = (-5, 2).$$

Thus

$$\overrightarrow{BA} = -\overrightarrow{AB}.$$

The sign of a vector therefore carries geometric information. It tells us orientation. The vectors $(5, -2)$ and $(-5, 2)$ are parallel and lie along the same supporting line, but they point in opposite directions.

Summary

When working with vectors as displacements, ask:

- Which point is the starting point?
- Which point is the ending point?
- Am I describing a position or a displacement?
- Would the same displacement be obtained if the arrow were moved elsewhere?
- What changes when the direction of the arrow is reversed?

A vector is not merely a pair or triple of numbers. It is a controlled way of recording movement from one position to another.

6.3 Vector operations

Vectors can be added, subtracted, and multiplied by numbers. These operations look algebraic, but each of them has a geometric meaning. If we treat them only as coordinate formulas, we lose much of their value.

The guiding idea is this: vectors describe displacements, and vector operations describe how displacements are combined or changed.

Adding vectors

Suppose a first displacement moves an object by u, and a second displacement moves it by v. The total displacement is $u+v$. In coordinates, addition is performed component by component.

Definition

For vectors

$$u = (u_1, u_2), \qquad v = (v_1, v_2),$$

their sum is

$$u + v = (u_1 + v_1,\ u_2 + v_2).$$

In space, for

$$u = (u_1, u_2, u_3), \qquad v = (v_1, v_2, v_3),$$

we define

$$u + v = (u_1 + v_1,\ u_2 + v_2,\ u_3 + v_3).$$

The formula says that horizontal changes add to horizontal changes, vertical changes add to

vertical changes, and third-coordinate changes add to third-coordinate changes. We do not mix the roles of the coordinates.

Example

Let

$$u = (3, -1), \qquad v = (-2, 5).$$

Then

$$u + v = (3 + (-2),\ -1 + 5) = (1, 4).$$

Geometrically, moving by u and then by v produces the same final displacement as moving once by $(1, 4)$.

This interpretation explains why vector addition is commutative:

$$u + v = v + u.$$

If the same two displacements are applied, the final total displacement is the same, even though the path used to draw the two arrows may look different.

Subtracting vectors

Subtraction measures the difference between displacements. If u and v are two vectors, then

$$u - v = u + (-v).$$

Here $-v$ is the vector with the same length and opposite orientation.

Definition

For vectors

$$u = (u_1, u_2), \qquad v = (v_1, v_2),$$

their difference is

$$u - v = (u_1 - v_1,\ u_2 - v_2).$$

Example

Let

$$u = (4, 2), \qquad v = (1, 5).$$

Then

$$u - v = (4 - 1,\ 2 - 5) = (3, -3).$$

This vector records how the displacement v must be changed to become the displacement u.

Subtraction should be read with care. The vectors $u - v$ and $v - u$ are opposites:

$$v - u = -(u - v).$$

Changing the order changes the orientation.

Multiplying a vector by a number

Multiplying a vector by a number changes its scale. If the number is positive, the orientation is preserved. If the number is negative, the orientation is reversed. If the number is zero, the vector collapses to the zero vector.

Definition

If

$$u = (u_1, u_2),$$

and λ is a real number, then

$$\lambda u = (\lambda u_1,\ \lambda u_2).$$

In space,

$$\lambda(u_1, u_2, u_3) = (\lambda u_1,\ \lambda u_2,\ \lambda u_3).$$

Example

Let

$$u = (2, -3).$$

Then

$$3u = (6, -9), \qquad -2u = (-4, 6), \qquad 0u = (0, 0).$$

The vector $3u$ points in the same direction as u, but is three times as long. The vector $-2u$ lies on the same direction line, but points in the opposite direction.

Parallel vectors

Scalar multiplication gives a simple way to recognize parallel vectors. Two non-zero vectors are parallel when one is a scalar multiple of the other.

Definition

Non-zero vectors u and v are parallel if there exists a real number $\lambda \neq 0$ such that

$$v = \lambda u.$$

Example

The vectors

$$u = (2, 3), \qquad v = (6, 9)$$

are parallel because

$$v = 3u.$$

The vector

$$w = (-4, -6)$$

is also parallel to u, because

$$w = -2u.$$

The negative scalar means that w points in the opposite orientation along the same direction line.

This idea will be central in the next chapter. A line can be described by a point and a direction vector. All vectors parallel to that direction vector describe the same direction line, although they may have different lengths or orientations.

The zero vector

The zero vector is the displacement that changes nothing:

$$0 = (0, 0)$$

in the plane, and

$$0 = (0, 0, 0)$$

in space.

It is useful, but it must be handled carefully. The zero vector has no direction. For this reason, when we discuss directions or parallelism, we usually require direction vectors to be non-zero.

Remark

A zero displacement is meaningful: it says that the starting point and ending point coincide. But it should not be used as a direction vector for a line or a plane, because it does not point anywhere.

Summary

When using vector operations, ask:

- Am I combining displacements or comparing them?
- Does scalar multiplication preserve or reverse orientation?
- Is a vector being used as a direction?
- If so, is it non-zero?
- Are two vectors parallel because one is a scalar multiple of the other?

The purpose of vector operations is not only to produce new coordinates. The purpose is to control displacements, directions, and geometric relationships.

6.4 Length, distance, and midpoints

Coordinates allow us to measure geometric quantities. Once a displacement is written as a vector, its length can be computed. Once two points are written in coordinates, the distance between them can be computed. The formulas are familiar, but they should not be treated as isolated rules. They come from the Pythagorean theorem applied to coordinate differences.

Length of a vector

A vector tells us how far and in which direction to move. Its length measures only the size of the displacement, not its orientation.

Definition

For a vector

$$v = (v_1, v_2)$$

in the plane, its length is

$$\|v\| = \sqrt{v_1^2 + v_2^2}.$$

For a vector

$$v = (v_1, v_2, v_3)$$

in space, its length is

$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

The squares appear because coordinate directions are perpendicular. A displacement in the horizontal direction and a displacement in the vertical direction form the legs of a right triangle. The vector is the diagonal.

Example

Let

$$v = (3, 4).$$

Then

$$\|v\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

The vector has length 5. The coordinates $(3, 4)$ describe how the displacement is decomposed into coordinate directions; the length 5 describes the total size of the displacement.

A vector and its opposite have the same length:

$$\|-v\| = \|v\|.$$

This is natural. Reversing a displacement changes its orientation, but not how long it is.

Distance between two points

The distance between two points is the length of the vector joining them. If

$$A = (a_1, a_2), \qquad B = (b_1, b_2),$$

then

$$\overrightarrow{AB} = (b_1 - a_1,\ b_2 - a_2).$$

Therefore the distance between A and B is

$$d(A, B) = \|\overrightarrow{AB}\|.$$

Definition

For points

$$A = (a_1, a_2), \qquad B = (b_1, b_2)$$

in the plane,

$$d(A, B) = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2}.$$

For points in space,

$$A = (a_1, a_2, a_3), \qquad B = (b_1, b_2, b_3),$$

we have

$$d(A, B) = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2}.$$

The formula is not new information. It is the length formula applied to the displacement from A to B.

Example

Let

$$A = (1, -2), \qquad B = (5, 1).$$

Then

$$\overrightarrow{AB} = (5 - 1,\ 1 - (-2)) = (4, 3).$$

Hence

$$d(A,B) = \|\overrightarrow{AB}\| = \sqrt{4^2 + 3^2} = 5.$$

This calculation should be read in two layers. Algebraically, we subtract coordinates and take a square root. Geometrically, we first form the displacement from one point to the other, and then measure its length.

Midpoints

A midpoint is the point halfway between two given points. In coordinates, it is obtained by averaging corresponding coordinates.

Definition

For points

$$A = (a_1, a_2), \qquad B = (b_1, b_2),$$

the midpoint of the segment AB is

$$M = \left(\frac{a_1 + b_1}{2}, \frac{a_2 + b_2}{2}\right).$$

In space, for

$$A = (a_1, a_2, a_3), \qquad B = (b_1, b_2, b_3),$$

the midpoint is

$$M = \left(\frac{a_1 + b_1}{2}, \frac{a_2 + b_2}{2}, \frac{a_3 + b_3}{2}\right).$$

The formula is easy to remember, but again we should understand it. Each coordinate is placed halfway between the corresponding coordinates of A and B.

Example

Let

$$A = (-1, 4), \qquad B = (5, 2).$$

The midpoint is

$$M = \left(\frac{-1 + 5}{2}, \frac{4 + 2}{2}\right) = (2, 3).$$

The first coordinate 2 is halfway between -1 and 5. The second coordinate 3 is halfway between 4 and 2.

There is also a vector way to read the midpoint. Starting from A, the full displacement to B is $B - A$. Half of that displacement is $\frac{1}{2}(B - A)$. Therefore

$$M = A + \frac{1}{2}(B - A).$$

After simplifying coordinates, this gives the same midpoint formula. This form is often more meaningful: the midpoint is reached by starting at A and moving halfway toward B.

What is being measured?

Length, distance, and midpoint calculations are common, but they answer different questions.

A length belongs to a vector. It asks how large a displacement is. A distance belongs to a pair of points. It asks how far apart two positions are. A midpoint is itself a point. It asks

where the halfway position lies.

Summary

When measuring with coordinates, ask:

- Am I measuring a vector or comparing two points?
- Have I first formed the correct displacement?
- Does the result represent a number, such as a distance, or a point, such as a midpoint?
- Can the calculation be interpreted geometrically after it is done?

Analytic geometry becomes clearer when every formula is connected to the object it measures.

6.5 Dot product and angles

Some geometric questions are not only about length. We often want to know whether two directions are similar, opposite, or perpendicular. We may want to measure the angle between two vectors. For this purpose, one of the most important operations is the dot product.

The dot product takes two vectors and produces a number. This number is not a new vector. It is a measurement of how the two vectors interact directionally.

Definition

For vectors

$$u = (u_1, u_2), \qquad v = (v_1, v_2),$$

the dot product is

$$u \cdot v = u_1 v_1 + u_2 v_2.$$

In space, for

$$u = (u_1, u_2, u_3), \qquad v = (v_1, v_2, v_3),$$

we define

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

The formula multiplies corresponding coordinates and then adds the results. But the meaning is not merely arithmetic. The dot product measures how much two vectors point in compatible directions.

Example

Let

$$u = (2, 3), \qquad v = (4, -1).$$

Then

$$u \cdot v = 2 \cdot 4 + 3 \cdot (-1) = 8 - 3 = 5.$$

The result is a number. It is not a vector, and it should not be interpreted as a point or a displacement.

The dot product and length

The dot product of a vector with itself gives the square of its length:

$$v \cdot v = \|v\|^2.$$

Indeed, if $v = (v_1, v_2)$, then

$$v \cdot v = v_1^2 + v_2^2,$$

while

$$\|v\| = \sqrt{v_1^2 + v_2^2}.$$

Therefore

$$\|v\| = \sqrt{v \cdot v}.$$

This connection is important. It shows that the dot product contains metric information: it knows something about lengths.

Example

For

$$v = (-2, 5),$$

we have

$$v \cdot v = (-2)^2 + 5^2 = 4 + 25 = 29.$$

Therefore

$$\|v\| = \sqrt{29}.$$

The dot product and angles

The deeper geometric meaning of the dot product is expressed by the formula

$$u \cdot v = \|u\|\|v\| \cos\theta,$$

where θ is the angle between the non-zero vectors u and v.

This formula should be read as an interpretation. The dot product is large and positive when the vectors point in similar directions. It is negative when they point more against each other. It is zero when the vectors are perpendicular.

Theorem

For non-zero vectors u and v,

$$\cos\theta = \frac{u \cdot v}{\|u\|\|v\|},$$

where θ is the angle between the vectors.

The denominator $\|u\|\|v\|$ removes the effect of the lengths. What remains is information about direction.

Example

Let

$$u = (1, 0), \qquad v = (0, 1).$$

Then

$$u \cdot v = 1 \cdot 0 + 0 \cdot 1 = 0.$$

Both vectors have length 1. Hence

$$\cos\theta = 0,$$

so

$$\theta = 90^\circ.$$

The vectors are perpendicular.

Perpendicular vectors

The most important special case is orthogonality. Two non-zero vectors are perpendicular exactly when their dot product is zero.

Theorem

Non-zero vectors u and v are perpendicular if and only if

$$u \cdot v = 0.$$

This criterion is one of the main reasons the dot product is so useful in analytic geometry. It turns a geometric condition, perpendicularity, into an algebraic equation.

Example

Let

$$u = (3, 2), \qquad v = (4, -6).$$

Then

$$u \cdot v = 3 \cdot 4 + 2 \cdot (-6) = 12 - 12 = 0.$$

Therefore the two vectors are perpendicular.

This is not only a convenient test. It is a model for much of analytic geometry: translate a geometric relation into an algebraic condition, solve or simplify the condition, and then interpret the result geometrically.

The sign of the dot product

The sign of the dot product gives qualitative information about the angle.

- If $u \cdot v > 0$, then the angle between u and v is acute.
- If $u \cdot v = 0$, then the angle is right.
- If $u \cdot v < 0$, then the angle is obtuse.

This follows from the sign of $\cos\theta$. Positive cosine corresponds to an acute angle, zero cosine to a right angle, and negative cosine to an obtuse angle.

Remark

The dot product does not only help us compute angles. It helps us read the relationship between directions. Sometimes the sign of the dot product is already enough to understand the geometry of a problem.

Why this matters for lines and planes

The dot product will appear repeatedly in the next chapters. A line may be described by a direction vector. A plane may be described by a normal vector. Perpendicularity between a vector and a direction, or between a vector and a plane, can be expressed using the dot product.

For example, if n is a normal vector to a plane, then every vector lying inside that plane is perpendicular to n. Algebraically, this becomes an equation of the form

$$n \cdot v = 0.$$

This is the beginning of the standard equation of a plane.

Summary

When using the dot product, ask:

- Am I trying to measure length, angle, or perpendicularity?
- Is the result supposed to be a number rather than a vector?
- Does a zero dot product indicate a right angle?
- Does the sign of the dot product already tell me something about the angle?
- Can a geometric condition be written as a dot product equation?

The dot product is a bridge between algebra and geometry. It allows angles and perpendicularity to be handled with equations.

6.6 Cross product and normal vectors

The dot product produces a number and measures directional agreement. In three-dimensional space there is another important operation: the cross product. It takes two vectors and produces a new vector perpendicular to both of them.

This operation is specific to three-dimensional vector geometry in the form used here. It will provide the missing link between vectors and equations of planes.

Definition in coordinates

Let

$$u = (u_1, u_2, u_3), \qquad v = (v_1, v_2, v_3).$$

Their cross product is defined by

$$u \times v = \big(u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1\big).$$

Definition

For vectors $u, v \in \mathbb{R}^3$, the cross product $u \times v$ is the vector

$$u \times v = \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix}.$$

It is perpendicular to both u and v.

A common mnemonic writes this computation as

$$u \times v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix},$$

with the usual alternating signs in the expansion. The coordinate formula above is the definition; the determinant notation is only a compact memory aid.

Example

Let

$$u = (1, 2, 3), \qquad v = (2, -1, 1).$$

Then

$$u \times v = \big(2 \cdot 1 - 3(-1),\ 3 \cdot 2 - 1 \cdot 1,\ 1(-1) - 2 \cdot 2\big),$$

so

$$u \times v = (5, 5, -5).$$

We can check perpendicularity:

$$(5, 5, -5) \cdot (1, 2, 3) = 5 + 10 - 15 = 0,$$

and

$$(5, 5, -5) \cdot (2, -1, 1) = 10 - 5 - 5 = 0.$$

Thus the resulting vector is perpendicular to both original vectors.

Order and orientation

The order of the vectors matters:

$$v \times u = -(u \times v).$$

Reversing the order reverses the direction of the resulting normal vector, but it does not change the plane perpendicular to that normal direction.

Remark

The vectors $u \times v$ and $v \times u$ have opposite orientations. In plane equations, either one may be used as a normal vector because multiplying a normal vector by a non-zero scalar does not change the plane.

Parallel vectors and the zero cross product

If u and v are parallel, they do not determine two independent directions. In that case,

$$u \times v = 0.$$

Conversely, for non-zero vectors in $\mathbb{R}^3$,

$$u \times v = 0$$

implies that the vectors are parallel.

Theorem

For $u, v \in \mathbb{R}^3$,

$$u \times v = 0$$

if and only if u and v are linearly dependent.

This criterion will be useful when deciding whether three points are collinear and whether two direction vectors really span a plane.

Geometric size

If θ is the angle between non-zero vectors u and v, then

$$\|u \times v\| = \|u\|\|v\| \sin\theta.$$

Therefore $\|u \times v\|$ is the area of the parallelogram spanned by u and v.

This formula also explains why the cross product is zero for parallel vectors: their angle is 0 or π, so $\sin\theta = 0$.

Finding a normal vector to a plane

Suppose a plane contains two non-parallel direction vectors u and v. A normal vector must be perpendicular to both directions. Therefore

$$n = u \times v$$

is a natural choice.

Example

Consider the three points

$$A = (1, 0, 0), \qquad B = (0, 1, 0), \qquad C = (0, 0, 1).$$

Two direction vectors in their plane are

$$u = B - A = (-1, 1, 0),$$

and

$$v = C - A = (-1, 0, 1).$$

Their cross product is

$$u \times v = \big(1 \cdot 1 - 0 \cdot 0,\ 0(-1) - (-1) \cdot 1,\ (-1) \cdot 0 - 1(-1)\big) = (1, 1, 1).$$

Thus a normal vector is

$$n = (1, 1, 1).$$

Using the point A, the plane equation is

$$n \cdot (P - A) = 0,$$

which becomes

$$(x - 1) + y + z = 0,$$

or

$$x + y + z = 1.$$

The chain of ideas is now complete:

three points $\longrightarrow$ two direction vectors $\longrightarrow$ cross product $\longrightarrow$ normal vector $\longrightarrow$ plane equation.

Summary

When using the cross product, ask:

- Are the vectors in $\mathbb{R}^3$?
- Do I need a vector perpendicular to two given directions?
- Have I preserved the alternating signs in the coordinate formula?
- Does reversing the order only reverse orientation?
- Is the cross product zero, indicating dependent directions?
- Can the result be used as a normal vector to a plane?

The cross product turns two independent directions inside a plane into one direction perpendicular to the plane.

6.7 Coordinates as a language

The purpose of this chapter is not only to introduce several formulas. Its purpose is to build a language. Coordinates, points, vectors, lengths, distances, and dot products are the words of that language. Once we can read them correctly, we can use them to describe more complicated geometric objects.

The next chapters will study lines and planes. These objects will not appear from nowhere. They will be built from the ideas already introduced here.

Points give position

A point tells us where we are. In the plane, a point has the form

$$P = (x, y).$$

In space, it has the form

$$P = (x, y, z).$$

A point may be fixed, such as

$$A = (2, -1),$$

or variable, such as

$$P = (x, y).$$

When the coordinates are variable, the point is allowed to move through a set of possible positions.

This distinction is essential for equations. An equation in coordinates usually does not describe one number. It describes all points whose coordinates satisfy a condition.

Example

The condition

$$y = 3$$

describes all points (x, y) in the plane whose second coordinate is 3. The first coordinate

is free. Thus the equation describes a horizontal line.
The point $(2, 3)$ is one point on that line, but it is not the whole line.

A common mistake is to confuse one solution with the entire geometric object. A single point may satisfy an equation, but the equation may describe a whole set of points.

Vectors give displacement and direction

A vector tells us how to move. If

$$v = (v_1, v_2),$$

then v_1 and v_2 describe coordinate changes. The vector may be used as an actual displacement from one point to another, or it may be used only as a direction.

For example, the vector

$$v = (2, 1)$$

can mean: move two units in the x-direction and one unit in the y-direction. But it can also mean: consider the direction determined by this displacement.

This distinction will matter for lines. A line can be described by one point on the line and one non-zero direction vector. The point anchors the line. The vector gives the direction in which the line extends.

Remark

A point and a vector may have the same coordinates, but they play different roles. A point answers the question "where?" A vector answers the question "how far and in which direction?"

Equations describe conditions

An equation in analytic geometry is usually a condition on points. It says which points are allowed and which points are not.

For example, the equation

$$x^2 + y^2 = 1$$

describes all points whose distance from the origin is 1. This is a circle of radius 1 centered at the origin.

The formula is algebraic, but the object is geometric. The equation does not merely ask us to solve for x or y. It describes a set of points.

Example

Consider the condition

$$(x - 2)^2 + (y + 1)^2 = 9.$$

A point (x, y) satisfies this condition exactly when its distance from $(2, -1)$ is 3. Therefore the equation describes a circle with center $(2, -1)$ and radius 3.

This example shows the main power of analytic geometry. A geometric statement about distance becomes an algebraic equation. Conversely, an algebraic equation can be read as a geometric statement.

Fixed and variable quantities

Many mistakes in analytic geometry come from not distinguishing fixed objects from variable ones. A point may be fixed. A direction vector may be fixed. But the point (x, y) or (x, y, z)

often represents a variable point moving through a set.

When we write a line in the form

$$P = A + tv,$$

the point A is fixed, the vector v is fixed and non-zero, and the parameter t is variable. As t changes, the point P moves along the line.

This formula belongs to the next chapter, but the language is already available. A fixed point gives a starting position. A direction vector gives a possible movement. A parameter controls how far we move.

Remark

Whenever a formula contains letters, ask which letters represent fixed data and which letters are allowed to vary. This question often reveals the meaning of the formula.

From this chapter to lines and planes

The chapter has prepared three ideas that will be used repeatedly.

First, coordinates describe position. Second, vectors describe displacements and directions. Third, equations describe conditions on variable points.

Lines will be described by combining points and direction vectors. Planes will be described either by points and directions or by a point and a normal vector. The dot product will allow us to express perpendicularity, and therefore it will help us write equations of planes.

Summary

When working with coordinates and vectors, ask:

- Am I describing a point, a vector, or a set of points?
- Which quantities are fixed, and which are variable?
- Is a vector being used as a displacement, a direction, or a normal vector?
- Can a geometric condition be translated into an algebraic equation?
- After computing, can I interpret the result geometrically?

Analytic geometry is not a collection of unrelated formulas. It is a way of translating between geometric objects and algebraic statements. The translation is useful only when we understand both languages.

Chapter 7

Lines

7.1 What information defines a line

A line is one of the simplest geometric objects, but it is also one of the easiest objects to misunderstand. In elementary algebra a line is often introduced through an equation such as $y = mx + b$. That equation is useful, but it should not be confused with the object itself. A line is not a formula. A line is a set of points arranged in one direction.

Analytic geometry studies this set of points by translating geometric information into algebraic language. The first question is therefore not "Which formula should I use?" The first question is: what information is enough to determine the line?

A point and a direction

The most natural description of a line uses two pieces of information:

- one point on the line,
- one non-zero vector that gives the direction of the line.

If A is a point and $v \neq 0$ is a direction vector, then the line consists of all points obtained by starting at A and moving by arbitrary multiples of v.

Definition

Let A be a point and let $v \neq 0$ be a vector. The line through A in direction v is the set of points

$$P = A + tv,$$

where t runs through all real numbers.

The parameter t controls the movement along the line. When $t = 0$, we are at the point A. When $t = 1$, we move from A by exactly one copy of the vector v. When $t = -1$, we move by the same vector in the opposite direction.

Example

Let

$$A = (1, 2), \qquad v = (3, -1).$$

The line through A in direction v is

$$P = (1, 2) + t(3, -1).$$

For $t = 0$, we get

$$P = (1, 2).$$

For $t = 1$, we get

$$P = (4, 1).$$

For $t = -1$, we get

$$P = (-2, 3).$$

These are different points, but they all lie on the same line because they are obtained from the same starting point by moving in the same direction.

This description matches the language developed in the previous chapter. A point gives position. A vector gives direction. A parameter tells us how much of that direction to use.

Two distinct points

A second common way to determine a line is to give two distinct points. If the points are different, then the vector from one point to the other gives a direction.

Let

$$A = (a_1, a_2), \qquad B = (b_1, b_2), \qquad A \neq B.$$

Then

$$\overrightarrow{AB} = B - A = (b_1 - a_1,\, b_2 - a_2)$$

is a non-zero direction vector for the line through A and B. Therefore the line can be written as

$$P = A + t(B - A).$$

The condition $A \neq B$ is not a technical decoration. If $A = B$, then the vector $B - A$ is the zero vector, and the zero vector has no direction. One point alone does not determine a unique line.

Example

Let

$$A = (2, -1), \qquad B = (5, 3).$$

Then

$$B - A = (3, 4).$$

The line through A and B can be written as

$$P = (2, -1) + t(3, 4).$$

For $t = 1$, this gives $(5, 3)$, so the point B is indeed on the line.

A point and a normal vector in the plane

In the plane there is another important way to determine a line. Instead of giving a direction along the line, we may give a direction perpendicular to the line. Such a vector is called a normal vector.

If $n \neq 0$ is perpendicular to a line and A is a point on the line, then a point P lies on the line exactly when the displacement $P - A$ is perpendicular to n. Using the dot product, this becomes

$$n \cdot (P - A) = 0.$$

This form is the beginning of the normal description of a line. It is extremely useful because perpendicularity, distance from a point to a line, and the general equation of a line all become natural from it.

Remark

A direction vector points along the line. A normal vector points across the line. Both can describe the same line, but they reveal different information.

The same object, different data

A line may therefore be determined from different kinds of information. We may know a point and a direction. We may know two points. In the plane, we may know a point and a normal vector. We may also know an equation and need to read from it what the point, direction, or normal vector is.

This is why this chapter will not present only one formula for a line. A line has many algebraic descriptions. Each description is a way of reading the same geometric object.

Summary

When first meeting a line problem, ask:

- Am I given a point on the line?
- Am I given a direction vector?
- Am I given two points?
- Am I given a normal vector or an implicit equation?
- Is the line in the plane or in space?
- Which description makes the given information visible?

The goal is not to memorize all possible formulas at once. The goal is to understand what information each formula is carrying.

7.2 Many descriptions of the same line

A line is a single geometric object, but it can be described in many algebraic languages. This is not a defect of geometry. It is a strength. Different descriptions make different features visible.

A good student should not ask only: "What is the formula for a line?" A better question is: "Which description of the line is most useful for the question I am trying to answer?"

In this section we collect the most common descriptions of a line in the plane. Later sections will explain them more carefully and show how to move between them.

Vector form

The vector form is

$$P = A + tv,$$

where A is a point on the line, $v \neq 0$ is a direction vector, and $t \in \mathbb{R}$.

This form makes motion along the line visible. It says: start at a point, then move in a fixed direction by an arbitrary amount.

Coordinate parametric form

If

$$A = (x_0, y_0), \qquad v = (a, b),$$

then the vector form becomes

$$(x, y) = (x_0, y_0) + t(a, b).$$

Equivalently,

$$x = x_0 + ta, \qquad y = y_0 + tb.$$

This is the coordinate parametric form. It is useful when we want to generate points on the line, check membership, or describe a line in space later.

Two-point form

If a line passes through two distinct points A and B, then the vector $B - A$ gives a direction. The line can be written as

$$P = A + t(B - A).$$

This form is useful because two points often occur naturally in geometric problems.

Point-slope form

For a non-vertical line in the plane, if the line passes through (x_0, y_0) and has slope m, then

$$y - y_0 = m(x - x_0).$$

This form makes the slope and one point visible. It is useful in elementary coordinate geometry, but it does not describe vertical lines.

Slope-intercept form

The form

$$y = mx + b$$

shows the slope m and the intersection point $(0, b)$ with the y-axis. It is familiar and convenient, but it is not the most general way to describe a line. A vertical line cannot be written in this form.

Implicit or general form

The general form of a line in the plane is

$$Ax + By + C = 0,$$

where A and B are not both zero.

This form is very important. It describes vertical, horizontal, and oblique lines in one language. It is also useful for intersections, side-of-line tests, and distance from a point to a line.

Normal form

The normal form uses a point P_0 on the line and a normal vector $n \neq 0$, perpendicular to the line:

$$n \cdot (P - P_0) = 0.$$

If $n = (A, B)$, $P = (x, y)$, and $P_0 = (x_0, y_0)$, this becomes

$$A(x - x_0) + B(y - y_0) = 0.$$

After expanding, it becomes an equation of the form

$$Ax + By + C = 0.$$

Thus the general form hides a normal vector.

Why so many forms?

The forms are not separate topics to memorize independently. They are different ways of reading the same object.

Description	Visible information	Useful for
$P = A + tv$	point and direction	movement, parametrization
$P = A + t(B - A)$	two points	building a line from points
$y - y_0 = m(x - x_0)$	point and slope	non-vertical plane lines
$y = mx + b$	slope and intercept	quick graphing
$Ax + By + C = 0$	normal vector hidden in coefficients	intersections, distance
$n \cdot (P - P_0) = 0$	point and normal vector	perpendicularity, distance

This table should not be read as a list of unrelated formulas. It is a map. When a problem gives certain information, one description may be natural. When a problem asks for a certain property, another description may be more useful.

Remark

Changing the description of a line is like translating a sentence from one language into another. The meaning should remain the same, but different words may make different parts of the meaning easier to see.

Summary

When choosing a line description, ask:

- Do I need to see a direction vector?
- Do I need to use a normal vector?
- Do I need to check whether a point lies on the line?
- Do I need to compare this line with another line?
- Do I need to compute a distance?
- Is a special case, such as a vertical line, possible?

The rest of the chapter develops this map into a working method.

7.3 Parametric description of lines

The parametric description is one of the most natural ways to describe a line. It uses the idea that a line can be traced by a moving point.

We start with a fixed point and a fixed direction. Then we allow a parameter to vary. As the parameter changes, the point moves along the line.

Vector parametric form

Let A be a point and let $v \neq 0$ be a direction vector. The line through A in direction v is

$$P = A + tv, \qquad t \in \mathbb{R}.$$

This is a compact formula, but it contains three different roles:

- A is fixed and anchors the line,
- v is fixed and gives the direction,
- t varies and moves the point along the line.

If we do not distinguish these roles, the formula becomes only a string of symbols. If we do distinguish them, the formula becomes a description of motion.

Example

Consider the line

$$P = (2, 1) + t(3, 4).$$

For $t = 0$, we get

$$P = (2, 1).$$

For $t = 1$, we get

$$P = (5, 5).$$

For $t = -1$, we get

$$P = (-1, -3).$$

The same formula produces different points on the same line. The parameter controls how far we move in the direction $(3, 4)$.

The parameter is not something mysterious. It is a coordinate along the line, measured in copies of the direction vector.

Coordinate parametric form in the plane

If

$$A = (x_0, y_0), \qquad v = (a, b),$$

then

$$P = A + tv$$

means

$$(x, y) = (x_0, y_0) + t(a, b).$$

Therefore

$$x = x_0 + ta, \qquad y = y_0 + tb.$$

These are the parametric equations of the line in the plane.

Example

Let

$$A = (-1, 3), \qquad v = (2, -5).$$

Then the line is

$$(x, y) = (-1, 3) + t(2, -5),$$

or equivalently

$$x = -1 + 2t, \qquad y = 3 - 5t.$$

For $t = 2$, the point is

$$x = -1 + 4 = 3, \qquad y = 3 - 10 = -7.$$

Thus $(3, -7)$ lies on the line.

The coordinate equations should be read together. The same value of t must be used in both equations. We are not choosing one parameter for x and another parameter for y. One moving point has one parameter.

Checking whether a point lies on a parametric line

To check whether a point belongs to a parametric line, we ask whether there exists a value of the parameter that produces the point.

Example

Consider the line

$$x = 1 + 3t, \qquad y = 2 - t.$$

Does the point $(7, 0)$ lie on the line?
From the first equation,

$$7 = 1 + 3t,$$

so

$$t = 2.$$

Using the same value in the second equation gives

$$y = 2 - 2 = 0.$$

Therefore $(7, 0)$ lies on the line.
Now consider the point $(7, 1)$. The first coordinate again forces $t = 2$, but then the second coordinate would be 0, not 1. Therefore $(7, 1)$ does not lie on the line.

This example shows why the parameter must be shared. A point lies on the line only if all coordinate equations agree on the same parameter value.

Different parametrizations of the same line

A line does not have a unique parametric equation. We may choose a different point on the same line, or a different non-zero vector parallel to the same direction.

For example,

$$P = (2, 1) + t(3, 4)$$

and

$$P = (5, 5) + s(6, 8)$$

describe the same line. The point $(5, 5)$ lies on the first line, and the vector $(6, 8)$ is twice the direction vector $(3, 4)$. The second description moves along the same line, but uses a different starting point and a different scale for the parameter.

Remark

A parametric equation describes a line, but the parametrization itself is not unique. The geometric line remains the same even if we choose another point on it or replace the direction vector by a non-zero scalar multiple.

Why parametric form is useful

Parametric form is especially useful when the natural data are a point and a direction, or two points. It is also the most natural form for lines in space, where slope-intercept equations are no longer available.

Summary

When reading a parametric line, ask:

- What point is fixed on the line?
- What vector gives the direction?
- What does the parameter do geometrically?
- Does a given point arise from one common parameter value?
- Could a different point or parallel direction vector describe the same line?

The parametric form is not merely a formula. It is a way to watch a point move.

7.4 Two points and direction vectors

Many line problems begin with two points. This is natural: in Euclidean geometry, two distinct points determine exactly one line. In analytic geometry we translate this statement into coordinates and vectors.

The translation is simple. Two points give a displacement. That displacement gives a direction.

Building the direction vector

Let

$$A = (a_1, a_2), \qquad B = (b_1, b_2),$$

and suppose $A \neq B$. The vector from A to B is

$$\overrightarrow{AB} = B - A = (b_1 - a_1,\, b_2 - a_2).$$

This vector is non-zero because the points are distinct. It is therefore a valid direction vector for the line through A and B.

The line can then be written as

$$P = A + t(B - A), \qquad t \in \mathbb{R}.$$

Example

Find a parametric equation of the line through

$$A = (1, 2), \qquad B = (4, -1).$$

First compute the direction vector:

$$B - A = (4 - 1,\, -1 - 2) = (3, -3).$$

Therefore the line is

$$P = (1, 2) + t(3, -3).$$

In coordinate form,

$$x = 1 + 3t, \qquad y = 2 - 3t.$$

This procedure should be read as a chain of meanings. The two points determine a displacement. The displacement gives a direction. A point and a direction determine the line.

The order of the points

If we reverse the order of the two points, the direction vector changes sign:

$$A - B = -(B - A).$$

This does not change the line. It only reverses the orientation of the parametrization.

Using the previous example, we found

$$B - A = (3, -3).$$

If we start from B instead, then

$$A - B = (-3, 3).$$

The line may also be written as

$$P = (4, -1) + s(-3, 3).$$

This is the same line, but the parameter moves along it in the opposite orientation.

Remark

Changing the orientation of the direction vector does not change the geometric line. It changes how the line is traced by the parameter.

Non-unique direction vectors

A direction vector for a line is never unique. If v is a direction vector, then every non-zero scalar multiple λv is also a direction vector for the same line.

For instance, the vectors

$$(3, -3), \qquad (1, -1), \qquad (-2, 2)$$

all describe the same direction. They have different lengths and possibly different orientations, but they are parallel.

Example

The line

$$P = (1, 2) + t(3, -3)$$

can also be written as

$$P = (1, 2) + s(1, -1).$$

The second version uses a shorter direction vector. Both equations describe the same set of points, because $(3, -3) = 3(1, -1)$.

This is often useful. We may simplify a direction vector by dividing out a common factor. The line remains the same, but the notation becomes easier to read.

Finding a point from a parametrization

A parametric line already contains a point. In

$$P = A + tv,$$

the point A appears when $t = 0$. But every value of t gives another point on the same line. We may choose any of them as a new anchor point.

Example

Consider

$$P = (2, -1) + t(4, 2).$$

For $t = \frac{1}{2}$, we get

$$P = (2, -1) + \frac{1}{2}(4, 2) = (4, 0).$$

Thus $(4, 0)$ is also a point on the line. We could write the same line as

$$P = (4, 0) + s(4, 2).$$

This flexibility is not a problem. It is part of the geometry. A line has infinitely many points and infinitely many possible non-zero direction vectors.

When the two points are the same

If the two given points are equal, then they do not determine a unique line. The vector between them is the zero vector:

$$B - A = 0.$$

The zero vector cannot be a direction vector, because it gives no direction.

This is a common hidden issue in problems with parameters. If two points depend on a parameter, there may be special parameter values for which the points coincide. Those values must be checked separately.

Summary

When a line is given by two points, ask:

- Are the two points distinct?
- What is the vector from one point to the other?
- Can the direction vector be simplified?
- Does reversing the order only reverse orientation?
- Are there parameter values for which the two points might coincide?

The formula $P = A + t(B - A)$ is not just a shortcut. It expresses the idea that a line is obtained by moving from one point in the direction of another.

7.5 Normal and implicit forms in the plane

The parametric form describes a line by moving along it. The normal form describes a line in a different way: by using a vector perpendicular to it. This change of viewpoint is important. A direction vector tells us how to travel along the line. A normal vector tells us how to stand across the line.

Both viewpoints describe the same geometric object, but they make different properties visible.

Normal vector to a line

A non-zero vector n is called normal to a line if it is perpendicular to every direction vector of the line.

If a line has direction vector v, then a normal vector n satisfies

$$n \cdot v = 0.$$

In the plane, if

$$v = (a, b),$$

then one possible normal vector is

$$n = (-b, a),$$

because

$$(a, b) \cdot (-b, a) = -ab + ab = 0.$$

Another possible normal vector is $(b, -a)$. The normal vector is not unique, just as the direction vector is not unique.

Example

Let a line have direction vector

$$v = (3, 2).$$

Then

$$n = (-2, 3)$$

is a normal vector, because

$$(3, 2) \cdot (-2, 3) = -6 + 6 = 0.$$

Thus $(-2, 3)$ points perpendicular to the line.

Normal form from a point and a normal vector

Suppose a line passes through a fixed point $P_0 = (x_0, y_0)$, and suppose $n = (A, B) \neq (0, 0)$ is normal to the line. A variable point $P = (x, y)$ lies on the line exactly when the displacement from P_0 to P is perpendicular to n.

The displacement is

$$P - P_0 = (x - x_0,\ y - y_0).$$

The perpendicularity condition is

$$n \cdot (P - P_0) = 0.$$

Therefore

$$(A, B) \cdot (x - x_0,\ y - y_0) = 0,$$

so

$$A(x - x_0) + B(y - y_0) = 0.$$

Definition

A line in the plane with point $P_0 = (x_0, y_0)$ and normal vector $n = (A, B) \neq (0, 0)$ can be written in normal form as

$$A(x - x_0) + B(y - y_0) = 0.$$

Equivalently,

$$n \cdot (P - P_0) = 0.$$

This equation is not arbitrary. It says: the vector from the fixed point to the variable point must be perpendicular to the normal vector.

Example

Find an equation of the line through

$$P_0 = (1, -2)$$

with normal vector

$$n = (3, 4).$$

The normal form is

$$3(x - 1) + 4(y - (-2)) = 0.$$

Thus

$$3(x - 1) + 4(y + 2) = 0.$$

Expanding,

$$3x - 3 + 4y + 8 = 0,$$

so

$$3x + 4y + 5 = 0.$$

Implicit or general form

After expanding the normal form, every line in the plane can be written as

$$Ax + By + C = 0,$$

where A and B are not both zero.

This is called the implicit form or general form of a line. The word "implicit" means that the line is described as a condition. A point (x, y) lies on the line if its coordinates satisfy the equation.

Definition

The general form of a line in the plane is

$$Ax + By + C = 0,$$

where $(A, B) \neq (0, 0)$. The vector

$$n = (A, B)$$

is normal to the line.

The last sentence is essential. The coefficients of x and y are not just coefficients. Together they form a normal vector.

Example

Consider the line

$$2x - 5y + 7 = 0.$$

A normal vector is

$$n = (2, -5).$$

A direction vector must be perpendicular to n. One possible choice is

$$v = (5, 2),$$

because

$$(2, -5) \cdot (5, 2) = 10 - 10 = 0.$$

This example shows how information can be read from an equation. The equation does not display a direction vector directly, but it displays a normal vector.

Checking membership

The implicit form is often the easiest way to check whether a point lies on a line. We substitute the coordinates into the left-hand side.

Example

Let

$$\ell : 2x - 5y + 7 = 0.$$

Check whether $P = (4, 3)$ lies on ℓ. We compute

$$2 \cdot 4 - 5 \cdot 3 + 7 = 8 - 15 + 7 = 0.$$

Therefore $(4, 3)$ lies on the line.
For $Q = (1, 1)$, we get

$$2 \cdot 1 - 5 \cdot 1 + 7 = 4.$$

Since the result is not zero, $(1, 1)$ does not lie on the line.

The number obtained after substitution also has meaning. It measures, up to a scale factor, how far the point is from satisfying the line condition. This idea will later become the distance formula.

Why implicit form is useful

The general form is especially useful because it treats all lines in the plane uniformly. It includes vertical lines, horizontal lines, and oblique lines. It also makes normal vectors visible, which helps with perpendicularity and distance.

Summary

When reading an implicit equation $Ax + By + C = 0$, ask:

- Are A and B not both zero?
- What normal vector is visible in the equation?
- Can I find a direction vector perpendicular to that normal vector?
- Does a given point satisfy the equation?
- Would this form help with distance or intersection questions?

The implicit form is not merely another formula. It is the language of conditions, normal vectors, and perpendicularity.

7.6 Slope forms and special cases

Many students first meet lines through equations of the form $y = mx + b$. This form is useful, but it is only one description among several. It works well for non-vertical lines in the plane. It does not describe every line.

The purpose of this section is therefore not to reject slope forms, but to place them correctly.

They are convenient when slope is the natural information, but they should not replace the more general geometric understanding of a line.

Slope as a ratio of changes

For a non-vertical line, the slope measures the ratio of vertical change to horizontal change. If a line has direction vector

$$v = (a, b)$$

with $a \neq 0$, then its slope is

$$m = \frac{b}{a}.$$

This says that when the horizontal coordinate changes by a, the vertical coordinate changes by b.

Example

If a line has direction vector

$$v = (4, 6),$$

then its slope is

$$m = \frac{6}{4} = \frac{3}{2}.$$

The vector $(4, 6)$ and the vector $(2, 3)$ describe the same direction, and both give the same slope.

Slope is therefore not a new kind of direction. It is a way of encoding direction for non-vertical lines.

Point-slope form

Suppose a non-vertical line passes through (x_0, y_0) and has slope m. The point-slope form is

$$y - y_0 = m(x - x_0).$$

This equation says that every point (x, y) on the line has the same ratio of vertical change to horizontal change relative to (x_0, y_0).

Example

Find an equation of the line through $(2, -1)$ with slope 3.
Using point-slope form,

$$y - (-1) = 3(x - 2),$$

so

$$y + 1 = 3x - 6.$$

Thus

$$y = 3x - 7.$$

The computation is short, but the meaning is important. The equation does not simply contain the number 3. It says that vertical change is always three times horizontal change.

Slope-intercept form

If a non-vertical line crosses the y-axis at $(0, b)$ and has slope m, then its equation is

$$y = mx + b.$$

The number b is the value of y when $x = 0$. The number m describes the slope.

Example

The line

$$y = -2x + 5$$

has slope -2 and intersects the y-axis at

$$(0, 5).$$

A possible direction vector is

$$v = (1, -2),$$

because when x increases by 1, y changes by -2.

This form is good for quick graphing and for reading slope. It is not always the best form for distance, intersections, or vertical lines.

Horizontal lines

A horizontal line has slope 0. Its equation has the form

$$y = c.$$

Every point on the line has the same second coordinate.

For example,

$$y = 4$$

describes all points (x, y) with $y = 4$. The first coordinate is free.

In general form, this line is

$$y - 4 = 0.$$

A normal vector is $(0, 1)$, and a direction vector is $(1, 0)$.

Vertical lines

A vertical line has the form

$$x = c.$$

Every point on the line has the same first coordinate. Such a line does not have a finite slope and cannot be written as $y = mx + b$.

For example,

$$x = -2$$

describes all points (x, y) with first coordinate -2. The second coordinate is free.

In general form, this line is

$$x + 2 = 0.$$

A normal vector is $(1, 0)$, and a direction vector is $(0, 1)$.

Remark

The equation $y = mx + b$ is familiar, but it is not universal. Whenever vertical lines may appear, use a description that can include them, such as parametric form or general form.

From slope form to general form

Every non-vertical line written as

$$y = mx + b$$

can be rewritten in general form:

$$mx - y + b = 0.$$

If we prefer integer coefficients, we may multiply by a common denominator.

Example

Rewrite

$$y = \frac{2}{3}x - 5$$

in general form. Move all terms to one side:

$$\frac{2}{3}x - y - 5 = 0.$$

Multiplying by 3, we get

$$2x - 3y - 15 = 0.$$

A normal vector is therefore

$$n = (2, -3).$$

This example shows again that changing form changes what is visible. The slope form made the slope visible. The general form made the normal vector visible.

Summary

When using slope forms, ask:

- Is the line definitely non-vertical?
- Is slope the most useful information in this problem?
- Would a general form make a normal vector visible?
- Is the line horizontal or vertical?
- Which form avoids unnecessary special cases?

Slope forms are useful tools, but they should be used with awareness of their limits.

7.7 Converting between line descriptions

The same line can be written in many forms. Therefore one of the basic skills in analytic geometry is translation. We translate from the form that is given to the form that makes the next question easier.

This translation should not be mechanical. Each conversion has a meaning. When we convert a parametric equation into an implicit equation, we are changing from a direction-based description to a condition-based description. When we convert an implicit equation into a parametric equation, we are finding a point and a direction hidden inside a condition.

From parametric form to implicit form

Suppose a line is given in parametric form:

$$P = P_0 + tv.$$

In the plane, if

$$P_0 = (x_0, y_0), \qquad v = (a, b),$$

then a normal vector may be chosen as

$$n = (-b, a).$$

The implicit equation is then obtained from

$$n \cdot (P - P_0) = 0.$$

Example

Convert the line

$$(x, y) = (1, 2) + t(3, -1)$$

to implicit form.
The direction vector is

$$v = (3, -1).$$

A normal vector is

$$n = (1, 3),$$

because

$$(3, -1) \cdot (1, 3) = 3 - 3 = 0.$$

Using the point $(1, 2)$, the normal form is

$$1(x - 1) + 3(y - 2) = 0.$$

Thus

$$x - 1 + 3y - 6 = 0,$$

so

$$x + 3y - 7 = 0.$$

The direction vector was visible in the parametric form. The normal vector became visible after conversion to implicit form.

From implicit form to parametric form

Now suppose the line is given by

$$Ax + By + C = 0.$$

A normal vector is immediately visible:

$$n = (A, B).$$

A direction vector must be perpendicular to n. One convenient choice is

$$v = (-B, A),$$

because

$$(A, B) \cdot (-B, A) = -AB + AB = 0.$$

We also need one point on the line. This can be found by choosing one coordinate and solving for the other, whenever possible.

Example

Convert

$$2x - 5y + 10 = 0$$

to parametric form.
A normal vector is

$$n = (2, -5).$$

A direction vector is

$$v = (5, 2),$$

because

$$(2, -5) \cdot (5, 2) = 10 - 10 = 0.$$

To find a point, choose $x = 0$. Then

$$-5y + 10 = 0,$$

so

$$y = 2.$$

Thus $(0, 2)$ lies on the line. A parametric equation is

$$(x, y) = (0, 2) + t(5, 2).$$

There are many possible choices of the point. If we choose another point on the same line, we get another parametrization of the same geometric object.

From two points to implicit form

If a line is given by two distinct points, first build a direction vector. Then choose a normal vector perpendicular to that direction vector.

Example

Find an implicit equation of the line through

$$A = (1, -1), \qquad B = (4, 5).$$

The direction vector is

$$B - A = (3, 6).$$

We may simplify the direction to

$$v = (1, 2).$$

A normal vector is

$$n = (-2, 1),$$

because

$$(1, 2) \cdot (-2, 1) = -2 + 2 = 0.$$

Using the point $A = (1, -1)$, the normal form is

$$-2(x - 1) + 1(y - (-1)) = 0.$$

Thus

$$-2(x - 1) + y + 1 = 0.$$

Expanding,

$$-2x + 2 + y + 1 = 0,$$

so

$$-2x + y + 3 = 0.$$

Equivalently,

$$2x - y - 3 = 0.$$

Both final equations describe the same line. Multiplying every coefficient by a non-zero number does not change the set of solutions.

From slope form to direction and normal vectors

If a non-vertical line has equation

$$y = mx + b,$$

then a direction vector is

$$v = (1, m).$$

Indeed, when x increases by 1, the value of y changes by m.

After rewriting as

$$mx - y + b = 0,$$

we also see a normal vector:

$$n = (m, -1).$$

These two vectors are perpendicular, because

$$(1, m) \cdot (m, -1) = m - m = 0.$$

Example

For the line

$$y = -3x + 4,$$

a direction vector is

$$v = (1, -3).$$

Rewriting as

$$3x + y - 4 = 0,$$

we see a normal vector

$$n = (3, 1).$$

Indeed,

$$(1, -3) \cdot (3, 1) = 3 - 3 = 0.$$

Checking the translation

After converting one form into another, it is good practice to check that the descriptions agree. We can check that the chosen point satisfies the new equation. We can check that the direction vector is perpendicular to the normal vector. We can also test a second point generated by the parametrization.

Summary

When converting between descriptions, ask:

- What information is visible in the given form?
- Do I need a direction vector or a normal vector?
- Have I found at least one point on the line?
- Is the direction vector perpendicular to the normal vector?
- Does the converted equation describe the same set of points?

Conversion is not a ritual of rearranging symbols. It is a way of making hidden geometric information visible.

7.8 Parallel, perpendicular, and intersecting lines

Once we can describe one line, we can compare two lines. Direction vectors make parallelism natural. Dot products make perpendicularity natural. Implicit equations make intersections natural because they become systems of linear equations.

Parallel lines from direction vectors

Two lines are parallel when their non-zero direction vectors are scalar multiples of one another.

Definition

Let ℓ_1 and ℓ_2 have direction vectors v_1 and v_2. They are parallel if there exists a non-zero scalar λ such that

$$v_2 = \lambda v_1.$$

Example

For

$$\ell_1 : P = (1, 2) + t(2, -3)$$

and

$$\ell_2 : P = (-4, 0) + s(-4, 6),$$

we have $(-4, 6) = -2(2, -3)$, so the lines are parallel.

Parallel lines may be distinct or may describe the same line. If they share a point, they are the same line; otherwise they are distinct parallel lines.

Example

The lines

$$\ell_1 : P = (1, 2) + t(2, -3)$$

and

$$\ell_2 : P = (3, -1) + s(4, -6)$$

have parallel directions. Moreover, $(3, -1)$ is obtained from ℓ_1 when $t = 1$. Therefore the two parametrizations describe the same line.

Perpendicular lines

In the plane, two non-parallel lines automatically intersect. In space, position must also be checked: skew lines can have perpendicular direction vectors without meeting.

Definition

Let ℓ_1 and ℓ_2 be intersecting lines with non-zero direction vectors v_1 and v_2. The lines are perpendicular if

$$v_1 \cdot v_2 = 0.$$

For lines in the plane, perpendicular direction vectors already guarantee that the lines intersect.

Example

Let two lines in the plane have directions

$$v_1 = (3, 1), \qquad v_2 = (2, -6).$$

Then

$$v_1 \cdot v_2 = 3 \cdot 2 + 1 \cdot (-6) = 0.$$

The directions are perpendicular, and the plane lines therefore meet at a right angle.

Remark

For lines in $\mathbb{R}^3$, the equation $v_1 \cdot v_2 = 0$ describes perpendicular directions. To call the lines themselves perpendicular, one must additionally verify that they intersect.

Using normal vectors in the plane

For lines

$$\ell_1 : A_1x + B_1y + C_1 = 0, \qquad \ell_2 : A_2x + B_2y + C_2 = 0,$$

the normal vectors are

$$n_1 = (A_1, B_1), \qquad n_2 = (A_2, B_2).$$

The lines are parallel when their normal vectors are parallel. They are perpendicular when their normal vectors are perpendicular.

Example

For

$$\ell_1 : 2x - 3y + 1 = 0, \qquad \ell_2 : 4x - 6y - 5 = 0,$$

we have

$$(4, -6) = 2(2, -3),$$

so the lines are parallel.

Example

For

$$\ell_1 : x + 2y - 4 = 0, \qquad \ell_2 : 2x - y + 3 = 0,$$

the normals satisfy

$$(1, 2) \cdot (2, -1) = 0.$$

Therefore the lines are perpendicular.

Intersecting lines as a system of equations

Two non-parallel lines in the plane intersect in exactly one point. Their intersection is found by solving their equations simultaneously.

Example

For

$$\ell_1 : x + y - 3 = 0, \qquad \ell_2 : 2x - y = 0,$$

we solve

$$\begin{cases} x + y = 3, \\ 2x - y = 0. \end{cases}$$

Adding the equations gives $3x = 3$, so $x = 1$, and then $y = 2$. The intersection point is

$$(1, 2).$$

Summary

When comparing two lines, ask:

- Are the lines in the plane or in space?
- Are their direction vectors parallel or perpendicular?
- In space, do lines with perpendicular directions actually intersect?
- If the lines are parallel, are they identical or distinct?
- If plane lines are not parallel, what is their intersection point?

Direction, position, and dimension must all be read before the geometric relationship is named.

7.9 Distance from a point to a line

The distance from a point to a line is not the distance to an arbitrary point on the line. It is the shortest possible distance. Geometrically, the shortest path from a point to a line is perpendicular to the line.

This is why the normal form of a line is so useful. A normal vector already points in the direction in which this shortest distance is measured.

The geometric idea

Let a line be given in general form:

$$\ell : Ax + By + C = 0.$$

A normal vector to the line is

$$n = (A, B).$$

If a point $P_0 = (x_0, y_0)$ does not lie on the line, then the shortest segment from P_0 to the line is parallel to n. The distance should therefore be measured in the normal direction.

The expression

$$Ax_0 + By_0 + C$$

tells us how far the point is from satisfying the equation of the line, but it still depends on the length of the normal vector. To obtain the actual distance, we divide by the length of the normal vector:

$$\sqrt{A^2 + B^2}.$$

Theorem

Let

$$\ell : Ax + By + C = 0$$

be a line in the plane, with $(A, B) \neq (0, 0)$. The distance from the point $P_0 = (x_0, y_0)$ to ℓ is

$$d(P_0, \ell) = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

The absolute value appears because distance is non-negative. Without the absolute value, the sign tells us on which side of the oriented line the point lies.

Example of the distance formula

Example

Find the distance from the point

$$P_0 = (3, 1)$$

to the line

$$\ell : 2x - y - 4 = 0.$$

Here

$$A = 2, \qquad B = -1, \qquad C = -4.$$

Substitute $(x_0, y_0) = (3, 1)$:

$$Ax_0 + By_0 + C = 2 \cdot 3 + (-1) \cdot 1 - 4 = 6 - 1 - 4 = 1.$$

The length of the normal vector is

$$\sqrt{A^2 + B^2} = \sqrt{2^2 + (-1)^2} = \sqrt{5}.$$

Therefore

$$d(P_0, \ell) = \frac{|1|}{\sqrt{5}} = \frac{1}{\sqrt{5}}.$$

The computation is short, but it should be interpreted. The point does not satisfy the line equation because substitution gives 1, not 0. Dividing by $\sqrt{5}$ corrects for the length of the normal vector.

Why the denominator is needed

The equation of a line is not unique. For example,

$$2x - y - 4 = 0$$

and

$$4x - 2y - 8 = 0$$

describe the same line. The second equation is obtained by multiplying the first by 2.

If we used only $|Ax_0 + By_0 + C|$, the value would also be multiplied by 2. But the geometric distance should not change. The denominator $\sqrt{A^2 + B^2}$ is multiplied by the same factor, so the ratio remains the same.

Remark

The distance formula uses the equation of the line, but the result is geometric. It cannot depend on which equivalent equation we choose.

Signed distance and sides of a line

The expression

$$Ax_0 + By_0 + C$$

without absolute value has a sign. Points on one side of the line give one sign, and points on the other side give the opposite sign. Points on the line give zero.

A normalized signed quantity is

$$\frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}}.$$

Its absolute value is the distance to the line.

Example

For the line

$$\ell : x + y - 1 = 0,$$

consider the points

$$P = (2, 0), \qquad Q = (0, -2).$$

For P,

$$2 + 0 - 1 = 1.$$

For Q,

$$0 + (-2) - 1 = -3.$$

The signs are different, so the points lie on opposite sides of the oriented line.

This interpretation is useful in geometry and in applications. The equation of a line does more than describe the points on the line. It also separates the plane into two sides.

Distance and projection idea

There is another way to understand the formula. Choose any point A on the line. The vector $P_0 - A$ goes from the line to the point. The distance from P_0 to the line is the length of the projection of this vector onto the normal direction.

If $n = (A, B)$, then a unit normal vector is

$$\frac{n}{\|n\|}.$$

The signed length of the projection is

$$(P_0 - A) \cdot \frac{n}{\|n\|}.$$

This is the same idea hidden inside the distance formula.

Summary

When computing distance from a point to a line, ask:

- Is the line written in general form $Ax + By + C = 0$?
- What is the normal vector (A, B)?
- What value do I get after substituting the point?
- Did I divide by the length of the normal vector?
- Does the sign, before taking absolute value, carry side-of-line information?

The distance formula is not an isolated trick. It is the normal-vector interpretation of a line turned into a measurement.

7.10 Lines in space

Lines in space are best understood through points and direction vectors. The parametric form works in the plane and in space, while slope-intercept form belongs only to the plane. This is one reason why the vector description is so important.

In three-dimensional space, a line still needs the same kind of information: one point and one non-zero direction vector.

Parametric form in space

Let

$$A = (x_0, y_0, z_0)$$

be a point in space, and let

$$v = (a, b, c) \neq (0, 0, 0)$$

be a direction vector. The line through A in direction v is

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c), \qquad t \in \mathbb{R}.$$

Equivalently,

$$x = x_0 + ta, \qquad y = y_0 + tb, \qquad z = z_0 + tc.$$

Example

The line

$$(x, y, z) = (1, -2, 3) + t(4, 0, -1)$$

has point $(1, -2, 3)$ and direction vector $(4, 0, -1)$. Its coordinate equations are

$$x = 1 + 4t, \qquad y = -2, \qquad z = 3 - t.$$

The second coordinate is constant because the second component of the direction vector is zero.

This example is important. A zero component of a direction vector does not mean that the vector is invalid. It means that the line does not move in that coordinate direction.

A line through two points in space

Two distinct points in space determine a line in the same way as in the plane. If

$$A = (a_1, a_2, a_3), \qquad B = (b_1, b_2, b_3), \qquad A \neq B,$$

then

$$B - A = (b_1 - a_1,\, b_2 - a_2,\, b_3 - a_3)$$

is a direction vector. The line is

$$P = A + t(B - A).$$

Example

Find a parametric equation of the line through

$$A = (2, 1, -1), \qquad B = (5, 3, 4).$$

The direction vector is

$$B - A = (3, 2, 5).$$

Therefore the line is

$$(x, y, z) = (2, 1, -1) + t(3, 2, 5).$$

In coordinates,

$$x = 2 + 3t, \qquad y = 1 + 2t, \qquad z = -1 + 5t.$$

The idea is unchanged: two points give a displacement, and the displacement gives a direction.

Symmetric form

If all components of the direction vector are non-zero, then the parametric equations

$$x = x_0 + ta, \qquad y = y_0 + tb, \qquad z = z_0 + tc$$

can be rewritten as

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

This is called the symmetric form of a line in space.

Example

For the line

$$(x, y, z) = (1, 2, -3) + t(2, -1, 4),$$

all components of the direction vector are non-zero. Therefore

$$\frac{x - 1}{2} = \frac{y - 2}{-1} = \frac{z + 3}{4}.$$

The symmetric form is compact, but it must be used carefully. It comes from solving each parametric equation for the same parameter t. If a component of the direction vector is zero, we cannot divide by it.

Zero components in symmetric form

Suppose

$$(x, y, z) = (1, -2, 3) + t(4, 0, -1).$$

Then

$$x = 1 + 4t, \qquad y = -2, \qquad z = 3 - t.$$

The second component of the direction vector is zero, so the line has constant y-coordinate. We may write

$$\frac{x-1}{4} = \frac{z-3}{-1}, \qquad y = -2.$$

We should not write a fraction with denominator 0. The equation $y = -2$ must be kept separately.

Remark

Symmetric form is convenient only when used honestly. A zero component of the direction vector becomes a constant-coordinate condition, not a division by zero.

Why one equation is not enough in space

In the plane, a single linear equation

$$Ax + By + C = 0$$

describes a line. In space, a single linear equation

$$Ax + By + Cz + D = 0$$

usually describes a plane, not a line.

This is a fundamental difference. In the plane, one linear condition leaves one degree of freedom, which gives a line. In space, one linear condition leaves two degrees of freedom, which gives a plane.

A line in space can often be described as the intersection of two planes, that is, by a system

$$\begin{cases} A_1x + B_1y + C_1z + D_1 = 0, \\ A_2x + B_2y + C_2z + D_2 = 0. \end{cases}$$

However, the parametric form is usually the clearest first description of a line in space.

Example

The equation

$$z = 2$$

in space does not describe a line. It describes all points whose third coordinate is 2. This is a plane parallel to the xy-plane.
To describe a line in that plane, we need another independent condition or a parametric description.

Comparing lines in space

Lines in space may be parallel, intersecting, or skew. Skew lines are lines that do not intersect and are not parallel. This possibility does not occur for lines in the plane.

For now, the most important point is that direction vectors still detect parallelism. If two direction vectors are scalar multiples, the lines are parallel or the same line. If they are not scalar multiples, the lines may intersect or may be skew. To decide, one must solve the parametric equations together.

Summary

When working with lines in space, ask:

- What point lies on the line?
- What non-zero vector gives the direction?
- Are any direction-vector components zero?
- Is symmetric form safe to use?
- Am I mistakenly treating one plane equation as a line?
- Are two lines parallel, intersecting, or skew?

The vector-parametric description is the most reliable language for lines in space. It keeps the point, the direction, and the movement visible.

7.11 Choosing the right description

This chapter introduced many descriptions of a line. At first this may seem like too much notation. But the purpose is not to collect formulas. The purpose is to learn how to choose a description that fits the problem.

The same line can be written in several ways. The best form depends on what is known and what is being asked.

When a point and direction are given

If a problem gives a point and a direction vector, the parametric form is usually the most natural:

$$P = A + tv.$$

It keeps the given information visible. It also makes it easy to generate points and to understand movement along the line.

This form is also the safest first choice in space.

When two points are given

If a problem gives two distinct points A and B, first form the direction vector

$$B - A.$$

Then write

$$P = A + t(B - A).$$

From there, if needed, convert to another form.

The important point is that the two given points are not just coordinates to insert into a memorized formula. They define a displacement, and that displacement defines a direction.

When a normal vector is visible

If a problem gives an equation

$$Ax + By + C = 0,$$

then the normal vector

$$n = (A, B)$$

is immediately visible. This form is useful for distance questions, side-of-line questions, and comparisons using normal vectors.

If a point P_0 and a normal vector n are given, use

$$n \cdot (P - P_0) = 0.$$

This is often more meaningful than jumping directly to expanded coordinates.

When slope is useful

If the line is non-vertical and slope is central to the question, then slope forms may be convenient:

$$y - y_0 = m(x - x_0)$$

or

$$y = mx + b.$$

But these forms should be used with awareness. Vertical lines require a different description, such as

$$x = c,$$

or a parametric or general equation.

Which form helps which question?

The following guide is often useful.

Question	Useful description
Generate points on the line	parametric form
Line through two points	two-point form, then parametric form
Check whether a point lies on the line	implicit or parametric form
Find an intersection in the plane	general form as a system
Check parallelism	direction vectors or normal vectors
Check perpendicularity	dot product of directions or normals
Compute distance from a point	general form $Ax + By + C = 0$
Work in space	parametric form

This table should not be used mechanically. It is a guide for choosing what to read from the line.

A complete reading example

Consider the line

$$\ell : 2x - 3y + 6 = 0.$$

From this form we immediately read a normal vector:

$$n = (2, -3).$$

A direction vector can be chosen perpendicular to n:

$$v = (3, 2),$$

because

$$(2, -3) \cdot (3, 2) = 6 - 6 = 0.$$

To find a point on the line, set $x = 0$. Then

$$-3y + 6 = 0,$$

so

$$y = 2.$$

Thus $(0, 2)$ lies on the line. A parametric form is

$$P = (0, 2) + t(3, 2).$$

The same line can therefore be read in two ways:

$$2x - 3y + 6 = 0$$

shows the normal vector and is good for distance or side tests, while

$$P = (0, 2) + t(3, 2)$$

shows a point and direction and is good for motion along the line.

Remark

A good solution often changes the description of a line. This is not extra work. It is how we choose the language in which the problem becomes clear.

What to remember

A line is a set of points. Equations and parametrizations are descriptions of that set. The main descriptions are not competitors; they are tools.

Summary

When working with lines, ask:

- Is the line in the plane or in space?
- Am I given a point, two points, a direction vector, or a normal vector?
- Which description makes the given information visible?
- Which description makes the required property easiest to study?
- Am I studying membership, intersection, parallelism, perpendicularity, or distance?
- After computing, can I explain what the result means geometrically?

The most important lesson is not that lines have many formulas. The most important lesson is that each formula reveals a different aspect of the same geometric object.

Chapter 8

Planes

8.1 What information defines a plane

A plane is the three-dimensional analogue of a line in one important sense: it is a flat object. But a plane is not one-dimensional. A line has one direction. A plane has two independent directions.

This distinction changes how we describe planes. A line can be determined by a point and one non-zero direction vector. A plane can be determined by a point and two non-parallel direction vectors, or by a point and one normal vector.

A point and two directions

Imagine standing at a point on a flat surface. To move around the surface, one direction is not enough. You need two independent directions. If those directions are not parallel, then every point of the plane can be reached by combining them.

Let P_0 be a point in space and let u and v be two non-zero, non-parallel vectors. Then the plane through P_0 spanned by u and v is described by

$$P = P_0 + su + tv, \qquad s, t \in \mathbb{R}.$$

The two parameters s and t move independently in the two directions.

Example

Let

$$P_0 = (1, 2, 0), \qquad u = (1, 0, 1), \qquad v = (0, 1, 2).$$

Then a plane is given by

$$P = (1, 2, 0) + s(1, 0, 1) + t(0, 1, 2).$$

In coordinates,

$$x = 1 + s, \qquad y = 2 + t, \qquad z = s + 2t.$$

Changing s and t moves the point across the whole plane.

The condition that u and v are not parallel is essential. If they were parallel, they would describe only one direction, and the formula would trace a line, not a plane.

A point and a normal vector

There is another very powerful way to describe a plane. Instead of giving two directions inside the plane, we may give one direction perpendicular to the plane. Such a vector is called a normal vector.

If $n \neq 0$ is perpendicular to a plane and P_0 is a point on the plane, then a point P lies on the plane exactly when the displacement $P - P_0$ is perpendicular to n. Using the dot product, this condition is

$$n \cdot (P - P_0) = 0.$$

This is the normal description of a plane. It is especially useful for equations, distances, angles, and parallelism.

Remark

Two direction vectors describe movement inside the plane. A normal vector describes the direction across the plane. Both viewpoints describe the same geometric object.

Three non-collinear points

A plane can also be determined by three points, provided they are not collinear. If the points are A, B, and C, then the vectors

$$\overrightarrow{AB} = B - A, \qquad \overrightarrow{AC} = C - A$$

lie in the plane. If these vectors are not parallel, then they give the two independent directions needed to describe the plane.

Example

The points

$$A = (1, 0, 0), \qquad B = (0, 1, 0), \qquad C = (0, 0, 1)$$

are not collinear. The vectors

$$B - A = (-1, 1, 0), \qquad C - A = (-1, 0, 1)$$

are not parallel, so they determine a plane.

The same plane, different descriptions

As with lines, a plane is not the same thing as one formula. The same plane may be described parametrically, by a normal equation, by a general equation, or by three points.

Summary

A plane may be determined by:

- one point and two non-parallel direction vectors,
- one point and one non-zero normal vector,
- three non-collinear points,
- an equation of the form $Ax + By + Cz + D = 0$, where $(A, B, C) \neq (0, 0, 0)$.

The goal of this chapter is to learn how to move between these descriptions and how to use each one for the questions it answers best.

8.2 Normal and general equation of a plane

The normal form of a plane is one of the most useful descriptions in analytic geometry. It uses a point on the plane and a vector perpendicular to the plane.

Let

$$P_0 = (x_0, y_0, z_0)$$

be a point on the plane, and let

$$n = (A; B; C) \neq (0; 0; 0)$$

be a normal vector. A point

$$P = (x; y; z)$$

lies on the plane exactly when $P - P_0$ is perpendicular to n. Therefore

$$n \cdot (P - P_0) = 0:$$

Definition

A plane through $P_0 = (x_0; y_0; z_0)$ with normal vector $n = (A; B; C) \neq (0; 0; 0)$ can be written as

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0:$$

This is the normal equation of the plane.

The equation is not a memorized pattern. It says that the displacement from the fixed point to any point of the plane must be perpendicular to the normal vector.

From normal form to general form

Expanding the normal equation gives

$$Ax + By + Cz + D = 0;$$

where

$$D = -Ax_0 - By_0 - Cz_0:$$

This is called the general equation of a plane.

Definition

The general equation of a plane in space is

$$Ax + By + Cz + D = 0;$$

where $(A; B; C) \neq (0; 0; 0)$. The vector

$$n = (A; B; C)$$

is normal to the plane.

The coefficients of x, y, and z are not only numbers in an equation. Together they form a normal vector.

Example from a point and normal vector

Example

Find the equation of the plane passing through

$$A = (1; 0; 2)$$

with normal vector

$$n = (2; -1; 1):$$

Evaluating this integral exactly gives

$$A = \pi R^2.$$

Remark

The modelling step is elementary: vertical slices produce the integral. An exact evaluation of

$$\int \sqrt{R^2 - x^2}\, dx$$

usually uses a trigonometric substitution or another method beyond the integration techniques developed so far. The formula is stated here to show how the familiar area can be represented by accumulation, not as an exercise in an unintroduced technique.

Volumes of solids of revolution

If a non-negative function f is rotated around the x-axis, cross-sections perpendicular to the x-axis are disks of radius $f(x)$. Their area is

$$\pi[f(x)]^2,$$

so the volume is

$$V = \int_a^b \pi[f(x)]^2\, dx.$$

Example

A vessel is formed by rotating

$$y = \sqrt{x}$$

around the x-axis for $0 \le x \le 4$. Its volume is

$$V = \int_0^4 \pi(\sqrt{x})^2\, dx = \pi \int_0^4 x\, dx = 8\pi.$$

The formula is simple because each thin slice is approximately a disk.

Arc length

The length of a curve $y = f(x)$ on $[a, b]$ is given by

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx.$$

This formula combines derivatives and integrals. The derivative measures local slope; the integral accumulates small pieces of length.

Example

Suppose a hanging rope is approximated by

$$f(x) = \frac{1}{10}x^2, \qquad -10 \le x \le 10.$$

Then

$$f'(x) = \frac{x}{5},$$

and the rope length is

$$L = \int_{-10}^{10} \sqrt{1 + \frac{x^2}{25}}\, dx.$$

This integral may require a more advanced method to evaluate exactly, but the modelling step is already clear: slope determines local stretching, and the integral accumulates length.

Summary

For geometric integral problems, ask:

- What small pieces are being accumulated?
- What are the correct bounds of integration?
- Is the integral measuring area, volume, or length?
- Is the quantity signed or ordinary geometric size?
- Does exact evaluation require a method already introduced?

Geometric applications show that integration is a method for building whole quantities from infinitesimal pieces.

13.4 Motion, velocity, and acceleration

One of the clearest applications of calculus is motion along a line. If position depends on time, derivatives describe velocity and acceleration. Conversely, integrals can recover velocity and position from acceleration.

Let $s(t)$ be the position of a particle at time t. Then

$$v(t) = s'(t)$$

is velocity, and

$$a(t) = v'(t) = s''(t)$$

is acceleration.

Velocity and acceleration from position

Example

A particle moves along a line with position

$$s(t) = t^3 - 6t^2 + 9t + 2, \qquad t \geq 0.$$

Find velocity and acceleration.
Velocity is the derivative of position:

$$v(t) = s'(t) = 3t^2 - 12t + 9.$$

Acceleration is the derivative of velocity:

$$a(t) = v'(t) = 6t - 12.$$

The formulas are not only calculations. Velocity tells us how position changes. Acceleration tells us how velocity changes.

When does the particle stop?

A particle stops when its velocity is zero. In the example above,

$$v(t) = 3t^2 - 12t + 9 = 3(t^2 - 4t + 3) = 3(t-1)(t-3).$$

Thus

$$v(t) = 0$$

when

$$t = 1 \qquad \text{or} \qquad t = 3.$$

These are the times when the particle is instantaneously at rest.

Speeding up and slowing down

A particle speeds up when velocity and acceleration have the same sign. It slows down when they have opposite signs. This is because acceleration then changes velocity in the direction opposite to the motion.

For the example,

$$v(t) = 3(t-1)(t-3), \qquad a(t) = 6(t-2).$$

We inspect the intervals determined by $t = 1, 2, 3$, with $t \geq 0$:

$$[0,1), \quad (1,2), \quad (2,3), \quad (3,\infty).$$

On each interval we compare the signs of $v(t)$ and $a(t)$.

Interval	sign of v	sign of a	behavior
$[0,1)$	positive	negative	slowing down
$(1,2)$	negative	negative	speeding up
$(2,3)$	negative	positive	slowing down
$(3,\infty)$	positive	positive	speeding up

This is a good example of interpretation. The signs matter because velocity carries direction, while speed is the magnitude of velocity.

Recovering motion by integration

If acceleration is known, then velocity can be recovered by integration. If velocity is known, position can be recovered by integration.

For vertical motion with constant acceleration

$$a(t) = -g,$$

we integrate:

$$v(t) = \int -g \, dt = -gt + C.$$

If the initial velocity is $v(0) = v_0$, then $C = v_0$, so

$$v(t) = v_0 - gt.$$

Now integrate again:

$$h(t) = \int (v_0 - gt) \, dt = v_0 t - \frac{1}{2} g t^2 + C_2.$$

If $h(0) = h_0$, then $C_2 = h_0$, so

$$h(t) = h_0 + v_0 t - \frac{1}{2} g t^2.$$

The maximum height occurs when velocity becomes zero:

$$v_0 - gt = 0, \qquad t = \frac{v_0}{g}.$$

Substituting into $h(t)$ gives

$$h_{\max} = h_0 + \frac{v_0^2}{2g}.$$

Summary

For motion problems, ask:

- Is position, velocity, or acceleration given?
- Do I need to differentiate or integrate?
- What initial conditions determine constants of integration?
- Does velocity equal zero at stopping or turning points?
- Are velocity and acceleration signs the same or opposite?
- What does the answer mean physically?

Motion problems show both directions of calculus: derivatives move from position to velocity and acceleration, while integrals reconstruct motion from rates of change.

13.5 Work, force, and accumulation

Physics gives many situations where an integral measures accumulated effect. One of the most important examples is work. If a force acts along a displacement, then work measures the accumulated force over distance.

When the force is constant, work is simply

$$W = F \cdot d.$$

When the force changes with position, we replace this product by an integral.

Definition

If a force $F(x)$ acts along a line from $x = a$ to $x = b$, then the work done by the force is

$$W = \int_a^b F(x)\,dx.$$

This formula says that work is accumulated force over displacement.

Hooke's law

For a spring, Hooke's law says that the force needed to stretch the spring is proportional to its elongation:

$$F(x) = kx,$$

where $k > 0$ is the spring constant and x is the elongation from the natural length.

Example

Compute the work required to stretch a spring from $x = 0$ to $x = L$, assuming

$$F(x) = kx.$$

The work is

$$W = \int_0^L kx\,dx = k\left[\frac{x^2}{2}\right]_0^L = \frac{1}{2}kL^2.$$

The force is small near the natural length and larger when the spring is stretched more. The integral accumulates these changing force values.

Flow as accumulated rate

If water flows out of a tank at a non-negative rate $r(t)$, then the amount of water leaving between times a and b is

$$\int_a^b r(t)\,dt.$$

Example

Suppose water flows out of a tank at rate

$$r(t) = 10 - 2t$$

liters per minute for $0 \le t \le 5$. This rate is non-negative on the stated interval. The amount leaving during the first three minutes is

$$\int_0^3 (10 - 2t)\,dt = \left[10t - t^2\right]_0^3 = 30 - 9 = 21.$$

Therefore 21 liters leave the tank during the first three minutes.

If instead $r(t)$ is defined as a signed net flow rate, its sign must be tied explicitly to a chosen direction. A negative integral then represents flow opposite to that positive direction, not a negative physical amount of water.

Average value

The average value of a continuous function on $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

This is the continuous analogue of averaging finitely many numbers.

For example, the root mean square value of a voltage $U(t)$ over one period T is

$$U_{\text{RMS}} = \sqrt{\frac{1}{T}\int_0^T [U(t)]^2\,dt}.$$

The square makes positive and negative oscillations contribute to energy-like effect rather than cancel.

Summary

For physical accumulation problems, ask:

- What rate or force is being accumulated?
- What is the variable of accumulation: time, position, or another quantity?
- What are the correct bounds?
- Is the rate an unsigned amount or a signed directional quantity?
- What units should the final answer have?

Integrals are natural in physics because many physical quantities are totals built from continuously changing contributions.

13.6 Growth, decay, and simple differential models

Some applications do not give the function directly. Instead, they describe how the function changes. This leads to differential equations: equations involving an unknown function and its derivative.

In this introductory chapter, we only need the simplest idea. If the rate of change is known, integration can recover the function. If the rate of change is proportional to the current value, exponential functions appear naturally.

Rate proportional to the quantity

A common model says that a quantity grows at a rate proportional to its current size:

$$N'(t) = kN(t).$$

Here $N(t)$ might represent a population, and k is a constant. If $k > 0$, the population grows. If $k < 0$, it decays.

A function satisfying this equation is

$$N(t) = N_0 e^{kt},$$

where $N_0 = N(0)$ is the initial value.

Example

Suppose a bacterial colony grows according to

$$N'(t) = kN(t), \qquad N(0) = N_0.$$

Then

$$N(t) = N_0 e^{kt}.$$

To find the doubling time, solve

$$N(t) = 2N_0.$$

Thus

$$N_0 e^{kt} = 2N_0.$$

Assuming $N_0 > 0$, divide by N_0:

$$e^{kt} = 2.$$

Take logarithms:

$$kt = \ln 2.$$

Therefore the doubling time is

$$t = \frac{\ln 2}{k}.$$

The result depends only on k, not on the initial population size. This is a feature of pure exponential growth.

Recovering a function from its derivative

If a rate of change is given directly, integration recovers the original quantity up to a constant. An initial condition determines that constant.

Example

Suppose the velocity of a particle is

$$v(t) = 3t^2 - 4t.$$

Find the position $s(t)$, assuming $s(0) = 5$.
Since velocity is the derivative of position,

$$s'(t) = v(t) = 3t^2 - 4t.$$

Integrate:

$$s(t) = \int (3t^2 - 4t)\, dt = t^3 - 2t^2 + C.$$

Use $s(0) = 5$:

$$5 = 0 - 0 + C,$$

so $C = 5$. Therefore

$$s(t) = t^3 - 2t^2 + 5.$$

The constant of integration is not optional. In applications it is determined by initial data.

Reading the model

A differential equation must be read as a relationship. For example,

$$N'(t) = kN(t)$$

does not give $N(t)$ immediately. It says that the rate of change of N at time t is proportional to the current value of $N(t)$.

This kind of statement is common in modelling. The derivative expresses a law of change.

Summary

When reading a simple differential model, ask:

- What quantity is unknown?
- What does its derivative represent?
- Is the rate given directly or in terms of the quantity itself?
- What initial condition determines the constant?

- Does the solution grow, decay, or remain constant?
- What does the solution mean in the original context?

Differential models show why derivatives and integrals belong together: derivatives describe laws of change, while integrals and antiderivatives recover quantities from those laws.

13.7 Mixed optimisation models

Some optimisation problems combine geometry, physics, and cost. The calculus step may be short, but the modelling step can be substantial. These problems are important because they show calculus acting as a reasoning tool rather than as a mechanical procedure.

Minimising material for a cylindrical can

Example

A cylindrical can must have fixed volume V. Find the relation between radius and height that minimises the surface area of the closed can.
Let r be the radius and h the height. The volume constraint is

$$\pi r^2 h = V.$$

The surface area of a closed cylinder is

$$S = 2\pi r^2 + 2\pi r h.$$

Use the constraint to express h in terms of r:

$$h = \frac{V}{\pi r^2}.$$

Then

$$S(r) = 2\pi r^2 + 2\pi r \cdot \frac{V}{\pi r^2} = 2\pi r^2 + \frac{2V}{r}.$$

The domain is $r > 0$. Differentiate:

$$S'(r) = 4\pi r - \frac{2V}{r^2}.$$

Set $S'(r) = 0$:

$$4\pi r = \frac{2V}{r^2}.$$

Thus

$$4\pi r^3 = 2V, \qquad 2\pi r^3 = V.$$

Using $V = \pi r^2 h$, this becomes

$$2\pi r^3 = \pi r^2 h,$$

so

$$h = 2r.$$

Therefore the material is minimised when the height equals the diameter.

The final answer is more meaningful as a design relation than as an isolated formula.

A cable cost problem

A power station is on one side of a river, and a factory is downstream on the opposite side. Cable laid underwater is more expensive than cable laid on land. The question is where the cable should leave the water.

The modelling idea is to choose a variable x for the horizontal distance from the point directly opposite the station to the landing point on the far bank. Then the underwater length is found by the Pythagorean theorem, while the land length is the remaining downstream distance. A cost function is built from these lengths.

If the river has width 100 m and the factory is 500 m downstream, and if underwater cable costs twice as much as land cable, then a simplified cost model is

$$C(x) = 2\sqrt{x^2 + 100^2} + (500 - x), \qquad 0 \le x \le 500.$$

The factor 2 represents the higher underwater cost, while $500 - x$ is the remaining land distance.

To minimise cost, differentiate:

$$C'(x) = 2 \cdot \frac{x}{\sqrt{x^2 + 100^2}} - 1.$$

Set $C'(x) = 0$:

$$\frac{2x}{\sqrt{x^2 + 10000}} = 1.$$

Thus

$$2x = \sqrt{x^2 + 10000}.$$

Squaring gives

$$4x^2 = x^2 + 10000,$$

so

$$3x^2 = 10000, \qquad x = \frac{100}{\sqrt{3}}.$$

This value lies in the allowed interval, so it is the optimal landing distance from the point directly opposite the station.

Remark

The derivative calculation is short. The main work is choosing the variable and expressing the physical cost as a function.

What mixed models teach

Mixed optimisation problems require careful translation. Geometry may provide distances or areas. Physics may provide rates or costs. Calculus then identifies the optimum.

Summary

For mixed optimisation models, ask:

- What is the decision variable?
- What quantity is being minimised or maximised?
- Which geometric formulas are needed?
- What constraints reduce the model to one variable?
- What interval of values is meaningful?

- Does the critical point satisfy the original physical restrictions?

The value of calculus in applications is that it connects a model to a decision.

13.8 Choosing calculus tools

The final skill of this chapter is choosing the right tool. In real problems, no one announces in advance whether the solution requires a derivative, an integral, optimisation, or a differential equation. The wording of the problem must be read carefully.

Questions that suggest derivatives

Use derivatives when the problem asks about local change, slope, rate, growth, decrease, velocity, acceleration, or sensitivity.

Typical phrases include:

- instantaneous velocity,
- rate of change,
- increasing or decreasing,
- maximum or minimum,
- fastest increase or decrease,
- tangent slope.

For example, if a concentration is given by $C(t)$, then the time of maximum concentration is found by studying $C'(t)$. If the question asks when the concentration decreases most rapidly, then the derivative and second derivative may both be relevant.

Questions that suggest integrals

Use integrals when the problem asks for accumulation, total amount, area, volume, work, distance from velocity, or average value.

Typical phrases include:

- total distance,
- accumulated amount,
- area under a graph,
- work done by a variable force,
- volume of a solid of revolution,
- average value over an interval.

For example, if a force depends on position, work is computed by integrating force over displacement. If a flow rate depends on time, total flow is computed by integrating the rate over time.

Questions that suggest optimisation

Optimisation problems ask for a best possible value under constraints. They often use derivatives, but the model-building step comes first.

Typical phrases include:

- maximise area,
- minimise cost,
- use the least material,
- find the best dimensions,
- largest volume,
- shortest time or distance.

The important step is to build an objective function and identify the allowed domain.

Questions that suggest differential models

A differential model appears when the problem describes a law of change rather than the quantity itself.

Typical phrases include:

- the rate is proportional to the current amount,
- acceleration is constant,
- velocity is the derivative of position,
- the quantity changes according to its present value.

Such problems often require integration or solving a simple differential equation with an initial condition.

A final comparison

Problem type	Mathematical object	Common tool
Local change	slope or rate	derivative
Optimisation	best value of a function	derivative and domain check
Accumulation	total from small contributions	definite integral
Motion	position, velocity, acceleration	derivatives and integrals
Work	force over distance	definite integral
Growth model	rate law	differential equation
Geometry	area, volume, length	definite integral

This table should be read as a guide. Some problems use more than one tool. For example, an optimisation problem may require geometry to build a function, derivatives to find a critical point, and endpoint checks to confirm the result.

Summary

When facing an application problem, ask:

- What is being asked: rate, total, best value, or model?

- What quantity should become a function?
- What is the independent variable?
- What restrictions or units are present?
- Which calculus tool matches the question?
- How should the mathematical result be interpreted in context?

This chapter closes the calculus part of the course by showing that calculus is a language for reasoning. Derivatives and integrals are not only techniques. They are tools for turning change and accumulation into mathematical understanding.

Chapter 14

Weekly Problem Sets

This chapter contains one problem set for each substantive chapter of the course. Each set is intended to support one week of study and contains twenty problems arranged approximately from direct practice to interpretation and synthesis.

Unless a problem explicitly asks only for a numerical answer, show the essential reasoning. When a problem asks for an interpretation, explain the meaning of the result in words rather than giving only a formula.

14.1 Problem Set 1: Matrices

1. For
$$A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix},$$
state the dimensions of A and identify a_{12}, a_{21}, and a_{23}.

2. Construct the 3×2 matrix $B = (b_{ij})$ defined by
$$b_{ij} = 2i - j.$$

3. Let
$$A = \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & -1 \\ 2 & 0 \end{pmatrix}.$$
Compute $A + B$, $A - B$, $3A$, and $2A - B$.

4. Compute the transpose of
$$C = \begin{pmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{pmatrix}.$$
Then state the dimensions of C^{T}.

5. Determine which of the following matrices are symmetric:
$$A = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

6. Let A be 2×3, B be 3×4, and C be 4×2. Determine which of the products AB, BA, BC, CB, AC, and CA are defined. State the dimensions of every defined product.

7. Compute
$$AB \quad \text{for} \quad A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & -2 \end{pmatrix}.$$

8. For
$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix},$$
compute both AB and BA. Explain what the result shows about matrix multiplication.

9. Let
$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad x = \begin{pmatrix} 5 \\ -2 \end{pmatrix}.$$
Compute Ax and interpret the result as a new column vector.

10. Verify directly that
$$AI_2 = I_2 A = A$$
for
$$A = \begin{pmatrix} 3 & -2 \\ 1 & 5 \end{pmatrix}.$$

11. Let
$$D = \operatorname{diag}(2, -1, 3).$$
Compute D^2 and D^3. Describe the pattern.

12. Starting with
$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \\ -2 & 1 & 5 \end{pmatrix},$$
perform the row operation $R_2 \leftarrow R_2 - 3R_1$.

13. Starting with
$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$
first interchange R_1 and R_2, and then perform $R_2 \leftarrow R_2 - \frac{1}{3}R_1$. Write the matrix after each step.

14. Find x, y, z, w if
$$\begin{pmatrix} x & 2y \\ z-1 & w+3 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 2 & 1 \end{pmatrix}.$$

15. Find the matrix X satisfying
$$2X + \begin{pmatrix} 1 & -1 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 4 & -1 \end{pmatrix}.$$

16. Verify the distributive law $A(B + C) = AB + AC$ for
$$A = \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} -1 & 4 \\ 2 & 0 \end{pmatrix}.$$

17. The columns of
$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ -1 & 3 \end{pmatrix}$$
are vectors u and v. Write u and v, and express
$$A \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$
as a linear combination of these columns.

18. The matrix
$$T = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$
acts on points in the plane by matrix multiplication. Find the images of
$$(1,2)^{\mathsf{T}}, \quad (-3,1)^{\mathsf{T}}, \quad (0,4)^{\mathsf{T}}.$$
Describe what T does geometrically.

19. A shop records sales of two products on three days in the matrix
$$S = \begin{pmatrix} 12 & 8 \\ 15 & 11 \\ 9 & 14 \end{pmatrix}.$$
Explain what the rows and columns could represent. Compute the total sales of each product using matrix or vector notation.

20. For each expression below, decide whether it is defined. If it is defined, compute it; if it is not, explain the dimension mismatch:
$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 4 \\ -2 & 3 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$
Consider AB, BA, Au, Bu, and $u^{\mathsf{T}}B$.

14.2 Problem Set 2: Determinants

1. Compute
$$\det \begin{pmatrix} 4 & -3 \\ 2 & 5 \end{pmatrix}.$$

2. Find the determinant of
$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}$$
by expanding along the first row.

3. Compute the determinant of
$$B = \begin{pmatrix} 2 & 0 & 0 \\ -1 & 3 & 0 \\ 4 & 5 & -2 \end{pmatrix}$$
without a Laplace expansion. Explain why your method works.

4. Expand
$$\det \begin{pmatrix} 1 & 0 & 2 \\ 3 & -1 & 4 \\ 0 & 5 & 2 \end{pmatrix}$$
along the row or column that gives the shortest calculation.

5. A matrix B is obtained from a 3×3 matrix A by interchanging two rows. If $\det A = -7$, find $\det B$.

6. A matrix C is obtained from a 4×4 matrix A by multiplying one row by 5. If $\det A = 3$, find $\det C$.

7. A matrix D is obtained from A by adding 4 times the first row to the third row. If $\det A = -2$, determine $\det D$.

8. Suppose $\det A = 3$ and $\det B = -4$ for two 3×3 matrices. Find $\det(AB)$, $\det(A^{\mathsf{T}})$, and $\det(2A)$.

9. Determine whether
$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}$$
is singular without carrying out a full determinant expansion.

10. Find all values of k for which
$$A(k) = \begin{pmatrix} k & 2 \\ 3 & k-1 \end{pmatrix}$$
is singular.

11. Find a if
$$\det \begin{pmatrix} a & 1 \\ 2 & 3 \end{pmatrix} = 10.$$

12. Compute
$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$$
and explain the general rule suggested by the result.

13. Let
$$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 4 \\ 2 & -1 & 5 \end{pmatrix}.$$
Use row operations to transform A into an upper triangular matrix, keeping track of every change to the determinant. Then compute $\det A$.

14. A 3×3 matrix has determinant 6. What happens to its determinant if:
    (a) the first and second rows are interchanged;
    (b) the third row is multiplied by -2;
    (c) twice the first row is added to the second row?

15. The columns of a 3×3 matrix satisfy $c_3 = 2c_1 - c_2$. What is the determinant? Explain the structural reason.

16. The linear transformation
$$T(x) = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} x$$
acts on the plane. By what factor does T scale areas? Does it preserve orientation?

17. Find the area of the parallelogram spanned by
$$u = (3, 1), \qquad v = (-2, 4)$$
using a determinant.

18. Compute the Vandermonde determinant

$$\det \begin{pmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ x_1^2 & x_2^2 & x_3^2 \end{pmatrix}$$

for $x_1 = 1$, $x_2 = 2$, and $x_3 = 4$, first directly and then using the product formula.

19. Consider

$$A(t) = \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & t \\ t & 0 & 1 \end{pmatrix}.$$

Compute $\det A(t)$ and determine all real values of t for which the matrix is singular.

20. For

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 0 & 1 & 4 \end{pmatrix},$$

compute the determinant in two different ways: by a Laplace expansion and by row reduction. Compare the amount of work and explain which method is more efficient.

14.3 Problem Set 3: Matrix Inversion

1. Decide whether each matrix is invertible:

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

2. Find the inverse of

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$$

using the 2×2 inverse formula.

3. Verify your answer to the previous problem by computing both AA^{-1} and $A^{-1}A$.

4. Find the inverse of

$$D = \operatorname{diag}(2, -3, 5).$$

5. Find all values of k for which

$$A(k) = \begin{pmatrix} k & 1 \\ 2 & k \end{pmatrix}$$

has an inverse.

6. Use Gauss–Jordan elimination to find the inverse of

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}.$$

7. Use Gauss–Jordan elimination to find the inverse of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

8. Find the adjugate of

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

and use it to compute A^{-1}.

9. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Solve $Ax = b$ using A^{-1}.

10. Solve

$$XA = B$$

for X, where

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}.$$

Pay attention to the side on which A^{-1} must be multiplied.

11. Solve

$$2X - A = B$$

for X, where

$$A = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}.$$

12. If

$$A^{-1} = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} \quad \text{and} \quad B^{-1} = \begin{pmatrix} 1 & 0 \\ 4 & 2 \end{pmatrix},$$

find $(AB)^{-1}$.

13. Prove by direct multiplication that

$$(A^{\mathsf{T}})^{-1} = (A^{-1})^{\mathsf{T}}$$

for

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}.$$

14. Find the inverse of the elementary matrix

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{pmatrix}$$

and explain which row operation E represents.

15. A proposed inverse of

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$$

is

$$C = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}.$$

Check whether the proposal is correct.

16. Explain why a matrix with two identical rows cannot have an inverse. Use both a determinant argument and an information-loss interpretation.

17. The transformation
$$y = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} x$$
maps an unknown vector x to
$$y = \begin{pmatrix} 8 \\ 5 \end{pmatrix}.$$
Recover x.

18. Choose the most efficient method for finding the inverse of each matrix and justify the choice:
$$A = \begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$
Then compute the inverses.

19. Let
$$A = \begin{pmatrix} 1 & a \\ 2 & 2a \end{pmatrix}.$$
Explain, without attempting Gauss–Jordan elimination, why A^{-1} does not exist for any real a.

20. For
$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$
find A^{-1}, solve $Ax = b$ for
$$b = \begin{pmatrix} 5 \\ 11 \end{pmatrix},$$
and verify the solution in the original equation. Finally, explain what the inverse accomplishes conceptually.

14.4 Problem Set 4: Systems of Linear Equations

1. Solve by substitution:
$$\begin{cases} x + y = 7, \\ 2x - y = 2. \end{cases}$$

2. Write the augmented matrix for
$$\begin{cases} 2x - y + 3z = 5, \\ x + 4y - z = 2, \\ 3x + 2z = 7. \end{cases}$$

3. Solve by Gaussian elimination:
$$\begin{cases} x + 2y = 5, \\ 3x - y = 4. \end{cases}$$

4. Solve by Gaussian elimination:

$$\begin{cases} x + y + z = 6, \\ 2x - y + z = 3, \\ x + 3y - 2z = 1. \end{cases}$$

5. Reduce the augmented matrix

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 8 \end{array}\right)$$

to echelon form and classify the system.

6. Reduce

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 0 & 1 & 1 & 2 \end{array}\right)$$

and express all solutions using a parameter.

7. Determine whether

$$\begin{cases} x - 2y = 1, \\ 2x - 4y = 5 \end{cases}$$

has no solution, one solution, or infinitely many solutions. Interpret the result geometrically.

8. Solve

$$\begin{cases} 2x + y = 4, \\ x - 3y = -5 \end{cases}$$

using Cramer's rule.

9. Solve the system in the previous problem using the inverse matrix method. Compare the two methods.

10. For which values of k does

$$\begin{cases} kx + y = 2, \\ 2x + ky = 3 \end{cases}$$

have a unique solution?

11. Classify the system according to the value of a:

$$\begin{cases} x + y = 2, \\ 2x + 2y = a. \end{cases}$$

12. Find all values of k for which

$$\begin{cases} x + y + z = 1, \\ 2x + 2y + 2z = 2, \\ x + y + kz = 3 \end{cases}$$

is consistent. Describe the solution set in each consistent case.

13. The row-reduced augmented matrix of a system is

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & -1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right).$$

Identify the pivot and free variables and write the complete solution parametrically.

14. Determine whether the point $(2, -1)$ is the solution of

$$\begin{cases} 3x + y = 5, \\ x - 2y = 4. \end{cases}$$

If it is not, find the correct solution.

15. Find the intersection point of the lines

$$2x - y - 1 = 0, \qquad x + 3y - 9 = 0.$$

16. Find the intersection of the three planes

$$\begin{cases} x + y + z = 3, \\ x - y + z = 1, \\ 2x + z = 2. \end{cases}$$

17. A cinema sells adult tickets for 12 units and student tickets for 8 units. In one evening it sells 180 tickets for a total of 1840 units. Form and solve a linear system to find the number of each type of ticket.

18. A system reduces to a row

$$\left(\begin{array}{ccc|c} 0 & 0 & 0 & 6 \end{array}\right).$$

What does this row mean, and what can be concluded about the original system?

19. Construct a system of two linear equations in two variables that has:

    (a) exactly one solution;

    (b) no solution;

    (c) infinitely many solutions.

    Explain how the coefficients produce each case.

20. Consider

$$\begin{cases} x + ay = 1, \\ ax + y = 1. \end{cases}$$

Classify the system for every real value of a. For each case, determine whether the solution is unique, nonexistent, or non-unique, and find the solution set where possible.

14.5 Problem Set 5: Vectors and Coordinates

1. Explain the difference between the point $P = (3, -2)$ and the vector $v = (3, -2)$. What does each object describe?

2. For
$$A = (1, 4, -2), \qquad B = (5, -1, 3),$$
find $\overrightarrow{AB}$ and $\overrightarrow{BA}$.

3. Let
$$u = (2, -1, 3), \qquad v = (-4, 5, 1).$$
Compute $u + v$, $u - v$, $3u$, and $2u - v$.

4. Find the position vector of the point $P = (-2, 5, 1)$. Then find the endpoint of this vector if it begins at $A = (3, -1, 4)$ instead of the origin.

5. Find the length of
$$v = (3, -4, 12).$$

6. Find the distance between
$$A = (1, -2, 3) \quad \text{and} \quad B = (5, 1, -1).$$

7. Find the midpoint of the segment joining
$$A = (-3, 4, 2) \quad \text{and} \quad B = (5, -2, 8).$$

8. Find a unit vector in the direction of
$$v = (2, -2, 1).$$

9. For
$$u = (1, 2, -1), \qquad v = (3, 0, 4),$$
compute $u \cdot v$.

10. Find the angle between
$$u = (1, 0, 1), \qquad v = (1, 1, 0).$$

11. Find the value of k for which
$$u = (2, k, -1) \quad \text{and} \quad v = (1, 3, 5)$$
are perpendicular.

12. Determine whether the triangle with vertices
$$A = (0, 0), \quad B = (4, 0), \quad C = (4, 3)$$
is right-angled. Use vectors and the dot product.

13. Compute
$$u \times v \quad \text{for} \quad u = (1, 2, 0), \qquad v = (3, -1, 4).$$

14. Verify that the vector found in the previous problem is perpendicular to both u and v.

15. Find the area of the parallelogram spanned by

$$u = (2, 1, 0), \qquad v = (1, -1, 3).$$

Then find the area of the triangle spanned by the same vectors.

16. Find a normal vector to the plane containing the direction vectors

$$u = (1, 2, -1), \qquad v = (2, 0, 3).$$

17. Determine whether the points

$$A = (1, 2, 3), \quad B = (3, 6, 5), \quad C = (5, 10, 7)$$

are collinear.

18. A particle moves from $A = (2, -1)$ by the displacement $v = (5, 3)$, and then by $w = (-2, 4)$. Find its final position and its total displacement from A.

19. The midpoint of AB is $M = (2, 1, -3)$, and $A = (-1, 4, 2)$. Find B.

20. For the points

$$A = (1, 0, 2), \quad B = (4, 2, 1), \quad C = (2, 5, 3),$$

find $\overrightarrow{AB}$ and $\overrightarrow{AC}$, compute their dot product and cross product, determine the angle at A, and find the area of triangle ABC.

14.6 Problem Set 6: Lines

1. Write a vector-parametric equation of the line through

$$A = (2, -1)$$

with direction vector

$$v = (3, 4).$$

2. Find a parametric equation of the line through

$$A = (-1, 2) \quad \text{and} \quad B = (5, -4).$$

3. Determine whether the point $P = (7, 3)$ lies on the line

$$(x, y) = (1, -1) + t(2, 1).$$

4. Write the coordinate parametric equations of

$$P = (2, 3) + t(-1, 4).$$

Find the points corresponding to $t = -2$, $t = 0$, and $t = 3$.

5. Find the point-slope equation of the line through $(3, -2)$ with slope 5.

6. Convert

$$y - 4 = -2(x + 1)$$

to slope-intercept form and general form.

7. Write equations for:

(a) the vertical line through $(3, -1)$;

(b) the horizontal line through $(-2, 5)$.

8. Convert the line
$$(x, y) = (1, 2) + t(3, -1)$$
to general form.

9. Convert
$$2x - 5y + 10 = 0$$
to a parametric equation.

10. Determine whether the lines
$$\ell_1 : (x, y) = (1, 0) + t(2, 3), \qquad \ell_2 : (x, y) = (-1, 4) + s(4, 6)$$
are distinct parallel lines or the same line.

11. Determine whether
$$\ell_1 : 2x - y + 3 = 0 \quad \text{and} \quad \ell_2 : x + 2y - 4 = 0$$
are perpendicular.

12. Find the intersection point of
$$3x - y - 5 = 0, \qquad x + 2y - 8 = 0.$$

13. Find the distance from $P = (4, -1)$ to the line
$$2x + 3y - 6 = 0.$$

14. For the oriented line
$$x - 2y + 1 = 0,$$
determine on which side of the line the points $P = (0, 0)$ and $Q = (4, 1)$ lie. Are they on the same side?

15. Write a parametric equation of the line in space through
$$A = (1, -2, 3)$$
with direction vector
$$v = (2, 0, -1).$$

16. Convert
$$(x, y, z) = (2, 1, -3) + t(4, -2, 5)$$
to symmetric form.

17. Write an appropriate symmetric description of
$$(x, y, z) = (1, -2, 3) + t(4, 0, -1),$$
taking proper account of the zero component.

18. Classify the following pair of lines in space as parallel, intersecting, or skew:

$$\ell_1 : (x, y, z) = (0, 0, 0) + t(1, 1, 0),$$

$$\ell_2 : (x, y, z) = (1, 0, 1) + s(0, 1, 1).$$

19. A line in the plane is given by a point $A = (2, 1)$ and a normal vector $n = (3, -4)$. Write its normal equation, general equation, and one direction vector.

20. The line passes through $A = (-1, 3)$ and $B = (5, -1)$. Find:

    (a) a direction vector;
    (b) a parametric equation;
    (c) a normal vector;
    (d) a general equation;
    (e) its slope, if defined;
    (f) the distance from the origin to the line.

14.7 Problem Set 7: Planes

1. Write the normal equation and general equation of the plane through

$$P_0 = (1, -2, 3)$$

with normal vector

$$n = (2, 1, -1).$$

2. For the plane

$$3x - 2y + 4z - 7 = 0,$$

state a normal vector and find one convenient point on the plane.

3. Determine whether $P = (2, 1, -1)$ lies on

$$x + 2y - z - 5 = 0.$$

4. Write a parametric equation of the plane through

$$P_0 = (1, 0, 2)$$

with direction vectors

$$u = (1, 2, 0), \qquad v = (0, 1, 3).$$

5. Find a parametric equation of the plane through

$$A = (1, 0, 0), \quad B = (0, 2, 0), \quad C = (0, 0, 3).$$

6. Find a normal vector to the plane through

$$A = (1, 0, 1), \quad B = (3, 1, 0), \quad C = (0, 2, 4).$$

7. Find the general equation of the plane through the three points in the previous problem.

8. Convert the parametric plane

$$P = (1, 2, 0) + s(1, 0, 1) + t(0, 2, -1)$$

to general form.

9. Convert

$$2x - y + z - 4 = 0$$

to one possible parametric form.

10. Determine whether the planes

$$\pi_1 : x - 2y + 3z - 1 = 0, \qquad \pi_2 : 2x - 4y + 6z + 5 = 0$$

are distinct parallel planes or the same plane.

11. Determine whether

$$\pi_1 : x + y - z = 0, \qquad \pi_2 : 2x - y + z - 3 = 0$$

are perpendicular.

12. Find the acute angle between

$$\pi_1 : x + 2y + 2z = 0 \quad \text{and} \quad \pi_2 : 2x - y + 2z - 4 = 0.$$

13. Find the distance from

$$P = (3, -1, 2)$$

to the plane

$$2x - y + 2z - 7 = 0.$$

14. Find the intersection point of the line

$$(x, y, z) = (1, 0, 2) + t(2, -1, 1)$$

with the plane

$$x + y + z - 6 = 0.$$

15. Determine whether the line

$$(x, y, z) = (1, 2, 0) + t(1, -1, 2)$$

lies in, is parallel to, or intersects the plane

$$2x + 2y - z - 6 = 0.$$

16. Find the equation of the plane through $P = (2, -1, 4)$ parallel to

$$3x - y + 2z + 5 = 0.$$

17. Find a plane through $P = (1, 2, 3)$ that is perpendicular to the plane

$$x - y + z = 0.$$

Give one valid answer and explain why it works.

18. Find a parametric equation of the line of intersection of

$$x + y + z = 3, \qquad x - y + z = 1.$$

19. A plane contains the line

$$\ell : (x, y, z) = (1, 0, 2) + t(2, 1, -1)$$

and the point $Q = (0, 3, 1)$. Find a general equation of the plane.

20. For

$$A = (1, 0, 0), \quad B = (0, 1, 0), \quad C = (0, 0, 1),$$

find a parametric and a general equation of the plane through the points, verify that all three points satisfy the equation, find the distance from the origin to the plane, and determine where the line

$$(x, y, z) = t(1, 1, 1)$$

intersects it.

14.8 Problem Set 8: Sequences and Functions

1. Decide which of the following assignments define functions:

   (a) each real number x is assigned x^2;
   (b) each positive number x is assigned both $\sqrt{x}$ and $-\sqrt{x}$;
   (c) each student in a class is assigned the student's date of birth.

   Explain each decision.

2. For

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2 - 4,$$

state the domain and codomain, and determine the range.

3. Let

$$f(x) = 2x^2 - 3x + 1.$$

Compute $f(0)$, $f(-2)$, $f(a)$, and $f(x + h)$.

4. If

$$g(x) = \frac{1}{x - 3},$$

determine the natural real domain of g and compute $g(1)$.

5. The graph of a function contains the points

$$(-2, 3), \quad (0, -1), \quad (4, 5).$$

State the corresponding function values.

6. For

$$f(x) = x^2 - 5x + 6,$$

find the y-intercept and all x-intercepts.

7. Determine the natural real domain of each function:

$$f(x) = \sqrt{x + 2}, \qquad g(x) = \frac{x + 1}{x^2 - 9}, \qquad h(x) = \ln(4 - x).$$

8. Find the range of each function:
$$f(x) = x^2 + 2, \qquad g(x) = -3x + 1, \qquad h(x) = \sqrt{x}.$$

9. Classify each function as constant, affine, quadratic, polynomial, rational, exponential, logarithmic, or trigonometric:
$$3, \quad 2x - 5, \quad x^2 + 1, \quad \frac{x+1}{x-2}, \quad 4^x, \quad \ln x, \quad \sin x.$$

10. Determine which of the functions
$$f(x) = 3x, \qquad g(x) = 3x + 2, \qquad h(x) = -x$$
are linear in the sense $f(x + y) = f(x) + f(y)$. Explain why an affine function need not be linear.

11. For
$$f(x) = 2^x \quad \text{and} \quad g(x) = \log_2 x,$$
compute $f(-2)$, $f(3)$, $g(1)$, $g(8)$, and explain the relationship between the two functions.

12. Let
$$a_n = 3n - 1, \qquad n \geq 1.$$
Write the first five terms and compute a_{20}.

13. A sequence is defined by
$$a_1 = 4, \qquad a_{n+1} = a_n - 3.$$
Write the first six terms.

14. Find an explicit formula for the arithmetic sequence
$$5,\ 9,\ 13,\ 17, \ldots$$

15. Find an explicit formula for the geometric sequence
$$3,\ 6,\ 12,\ 24, \ldots$$

16. Define recursively the sequence
$$2,\ 6,\ 18,\ 54, \ldots$$
and state the starting index.

17. For
$$a_n = \frac{1}{n},$$
list the first five graph points (n, a_n). Explain why the graph should normally be drawn as separated points.

18. Compare the domains of
$$a_n = n^2, \qquad n \in \mathbb{N},$$
and
$$f(x) = x^2, \qquad x \in \mathbb{R}.$$
Which expressions $a_{3/2}$ and $f(3/2)$ are meaningful?

19. A graph rises on $(-3, 1)$, falls on $(1, 4)$, crosses the x-axis at -2 and 3, and has a local maximum at $(1, 5)$. Translate this information into statements about the function.

20. A taxi fare consists of a fixed fee of 6 units and 2.5 units for every kilometre travelled. Define a function $C(d)$, state a meaningful domain, compute $C(8)$, and explain whether the function is linear or affine.

14.9 Problem Set 9: Limits and Continuity

1. Explain in words the difference between
$$f(a) \quad \text{and} \quad \lim_{x\to a} f(x).$$

2. Compute
$$\lim_{n\to\infty} \frac{1}{n}, \qquad \lim_{n\to\infty} \frac{5n+2}{2n-1}.$$

3. Compute
$$\lim_{n\to\infty} \frac{3n^2 - n + 1}{2 - n^2}.$$

4. Determine whether each sequence converges:
$$a_n = (-1)^n, \qquad b_n = \frac{(-1)^n}{n}, \qquad c_n = 2 + \frac{3}{n}.$$

5. Compute
$$\lim_{n\to\infty} \left(\sqrt{n^2 + 4n} - n\right)$$
by rationalizing.

6. Compute by direct substitution:
$$\lim_{x\to 2} (3x^2 - x + 4).$$

7. Compute
$$\lim_{x\to 3} \frac{x^2 - 9}{x - 3}.$$

8. Compute
$$\lim_{x\to 9} \frac{\sqrt{x} - 3}{x - 9}.$$

9. Let
$$f(x) = \begin{cases} 2x + 1, & x < 2, \\ x^2 - 1, & x \geq 2. \end{cases}$$
Find the left-hand limit, right-hand limit, two-sided limit, and value $f(2)$.

10. Compute
$$\lim_{x\to 0^-} \frac{|x|}{x}, \qquad \lim_{x\to 0^+} \frac{|x|}{x},$$
and decide whether the two-sided limit exists.

11. Compute
$$\lim_{x\to\infty} \frac{4x^2 - 3x + 1}{2x^2 + 5}.$$

12. Find all horizontal asymptotes of
$$f(x) = \frac{3x - 1}{x + 4}.$$

13. Determine the one-sided infinite limits of

$$f(x) = \frac{1}{x-2}$$

as $x \to 2^-$ and $x \to 2^+$. State the vertical asymptote.

14. Check whether

$$f(x) = \frac{x^2-4}{x-2}, \qquad x \neq 2,$$

is continuous at $x = 2$. If not, give a value of $f(2)$ that makes it continuous.

15. Find k so that

$$f(x) = \begin{cases} kx + 1, & x < 3, \\ x^2 - 2, & x \geq 3 \end{cases}$$

is continuous at $x = 3$.

16. Classify the discontinuity at the indicated point:

(a) $\dfrac{x^2-1}{x-1}$ at $x = 1$;

(b) $\dfrac{1}{x+2}$ at $x = -2$;

(c) $\dfrac{|x|}{x}$ at $x = 0$;

(d) $\sin(1/x)$ at $x = 0$.

17. Compute

$$\lim_{x \to 0} \frac{\sin 5x}{2x}.$$

18. Compute

$$\lim_{x \to 0} \frac{\tan 3x}{\sin 2x}.$$

19. Compute

$$\lim_{x \to 0} (1 + 4x)^{1/x}.$$

20. For each limit, identify the most appropriate first method and then compute:

$$\lim_{x \to 1} \frac{x^3 - 1}{x - 1}, \qquad \lim_{x \to \infty} \frac{2x + 5}{x^2 + 1}, \qquad \lim_{x \to 0} \frac{\sin x}{x}, \qquad \lim_{x \to 4} \frac{x - 4}{\sqrt{x} - 2}.$$

14.10 Problem Set 10: Derivatives and Function Analysis

1. For

$$f(x) = x^2 + 1,$$

form and simplify the difference quotient

$$\frac{f(a+h) - f(a)}{h}.$$

2. Use the limit definition to find the derivative of

$$f(x) = 3x + 2.$$

3. Find the tangent line to
$$f(x) = x^2 - 1$$
at $x = 2$.

4. Differentiate, stating the relevant domain:
$$f(x) = x^7, \qquad g(x) = x^{-3}, \qquad h(x) = \sqrt{x}.$$

5. Differentiate:
$$f(x) = 4e^x - 3\ln x + 2\sin x - \cos x.$$

6. Differentiate using the product rule:
$$f(x) = x^3 e^x.$$

7. Differentiate using the quotient rule:
$$f(x) = \frac{x^2 + 1}{x + 2}.$$

8. Differentiate using the chain rule:
$$f(x) = (2x - 3)^6.$$

9. Differentiate:
$$f(x) = e^{\sin(x^2)}.$$
Identify all layers of composition.

10. Find all critical points of
$$f(x) = x^3 - 6x^2 + 9x.$$

11. Study the monotonicity of
$$f(x) = x^2 - 4x + 1$$
using a sign chart for f'.

12. Find and classify the local extrema of
$$f(x) = x^3 - 3x.$$

13. Find the absolute maximum and minimum of
$$f(x) = x^2 - 2x$$
on $[-1, 3]$.

14. Determine the intervals of convexity and concavity of
$$f(x) = x^3 - 6x.$$

15. Find all inflection points of
$$f(x) = x^4 - 4x^3.$$
Remember that $f''(a) = 0$ gives only a candidate.

16. Use the second derivative test to classify the critical points of

$$f(x) = x^4 - 4x^2.$$

For any inconclusive point, use another method.

17. Find all vertical and horizontal asymptotes of

$$f(x) = \frac{2x+1}{x-3}.$$

18. Perform a complete analysis of

$$f(x) = x^3 - 3x^2.$$

Include domain, zeros, behavior at infinity, monotonicity, extrema, convexity, and inflection points.

19. Perform a complete analysis of

$$f(x) = \frac{x}{x-1}.$$

Include domain, asymptotes, monotonicity, and a coherent description of the graph.

20. A quantity is modelled by

$$Q(t) = t^3 - 6t^2 + 9t + 4, \qquad t \geq 0.$$

Find $Q'(t)$, determine when the quantity is increasing or decreasing, identify local extrema, and explain what the sign of $Q'(t)$ means in a modelling context.

14.11 Problem Set 11: Integrals

1. Find all antiderivatives of

$$f(x) = 6x^2 - 4.$$

2. Compute, on appropriate intervals,

$$\int x^5\,dx, \qquad \int x^{-4}\,dx, \qquad \int \sqrt{x}\,dx.$$

3. Compute

$$\int \left(3e^x - \frac{2}{x} + 4\cos x\right)\,dx.$$

4. Rewrite and integrate:

$$\int \frac{\sqrt{x}}{x^3}\,dx.$$

5. Use substitution to compute

$$\int 2x(1+x^2)^5\,dx.$$

6. Use substitution to compute

$$\int \frac{e^x}{1+e^x}\,dx.$$

7. Use integration by parts to compute

$$\int xe^x\,dx.$$

8. Compute
$$\int x^2 \cos x \, dx$$
using integration by parts more than once.

9. Compute
$$\int \frac{2x+3}{x^2+3x+5} \, dx.$$

10. Compute
$$\int \frac{x}{x^2-2x+5} \, dx$$
by splitting the numerator into a derivative part and a remainder.

11. Compute
$$\int \frac{x^2+2}{x-1} \, dx$$
by polynomial division.

12. Compute
$$\int \sin^4 x \cos x \, dx.$$

13. Compute
$$\int \cos^2 x \, dx$$
using a trigonometric identity.

14. Evaluate
$$\int_0^2 (3x+1) \, dx.$$

15. Find the signed area represented by
$$\int_{-2}^{2} x \, dx$$
and compare it with the ordinary geometric area between $y = x$ and the x-axis on $[-2, 2]$.

16. Use properties of definite integrals to simplify:
$$\int_0^5 f(x) \, dx - \int_2^5 f(x) \, dx,$$
and
$$\int_3^1 g(x) \, dx.$$

17. Use the fundamental theorem of calculus to evaluate
$$\int_1^4 \frac{1}{x} \, dx.$$

18. Let
$$A(x) = \int_0^x (t^2+1) \, dt.$$
Find $A'(x)$ and compute $A(2)$.

19. Choose an appropriate integration method for each integral and justify the choice before computing:
$$\int x\sin(x^2)\,dx, \qquad \int x\ln x\,dx, \qquad \int \frac{3x^2}{x^3+1}\,dx.$$

20. Compute and verify by differentiation:
$$\int \left(\frac{2x}{1+x^2} + xe^{x^2} + \sin x\cos x\right) dx.$$

14.12 Problem Set 12: Applications of Calculus

1. For each question, state whether the main tool is a derivative, a definite integral, optimisation, or a differential model:
   (a) find instantaneous velocity;
   (b) find total water discharged from a known flow rate;
   (c) choose dimensions giving the largest area;
   (d) model a population whose growth rate is proportional to its size.

2. A rectangle has perimeter 40. Express its area as a function of one side and find the dimensions that maximise the area.

3. A rectangular enclosure is built beside a straight wall, so only three sides require fencing. If 60 metres of fencing are available, find the dimensions that maximise the area.

4. Squares of side x are cut from the corners of a 20 cm $\times$ 30 cm sheet to form an open box. Build the volume function, determine the meaningful domain, and find the value of x that maximises the volume.

5. Find the area enclosed by
$$y = 4 - x^2$$
and the x-axis.

6. The curve
$$y = \sqrt{x}, \qquad 0 \le x \le 9,$$
is rotated about the x-axis. Find the resulting volume.

7. Set up, but do not necessarily evaluate, the arc-length integral for
$$y = x^2, \qquad 0 \le x \le 2.$$

8. A particle has position
$$s(t) = t^3 - 6t^2 + 9t, \qquad t \ge 0.$$
Find its velocity and acceleration.

9. For the particle in the previous problem, determine when it is at rest and on which intervals it is speeding up or slowing down.

10. A particle has acceleration
$$a(t) = 6t - 4,$$
initial velocity $v(0) = 3$, and initial position $s(0) = 2$. Find $v(t)$ and $s(t)$.

11. A spring obeys Hooke's law $F(x) = 40x$ newtons. Find the work required to stretch it from $x = 0.1$ m to $x = 0.25$ m.

12. Water flows into a tank at the non-negative rate

$$r(t) = 12 - 2t$$

litres per minute for $0 \le t \le 6$. How much water enters during the first four minutes?

13. The temperature over a six-hour interval is modelled by

$$T(t) = 18 + 2t - \frac{t^2}{3}, \qquad 0 \le t \le 6.$$

Find its average value on the interval.

14. A population satisfies

$$N'(t) = 0.04N(t), \qquad N(0) = 500.$$

Find $N(t)$ and the doubling time.

15. A radioactive substance satisfies

$$M'(t) = -0.08M(t), \qquad M(0) = 120.$$

Find $M(t)$ and its half-life.

16. A quantity $Q(t)$ has rate of change

$$Q'(t) = 3t^2 - 4t + 1$$

and $Q(0) = 7$. Find $Q(t)$ and explain the role of the initial condition.

17. A closed cylindrical can must have fixed volume V. Derive the relation between radius and height that minimises its surface area.

18. A factory is 400 m downstream on the opposite side of a river 80 m wide. Underwater cable costs three times as much per metre as land cable. Let x be the downstream distance from the point directly opposite the station to the landing point. Build the cost function and determine the optimal x.

19. A car's velocity is

$$v(t) = t^2 - 6t + 8, \qquad 0 \le t \le 6.$$

Find the signed displacement and the total distance travelled on the interval.

20. A company models daily profit by

$$P(x) = -2x^3 + 30x^2 - 96x - 50, \qquad 0 \le x \le 10,$$

where x is production in hundreds of units. Determine the production levels at which profit is locally maximal or minimal, compare all relevant endpoint and critical values, and interpret the economically best choice within the model.